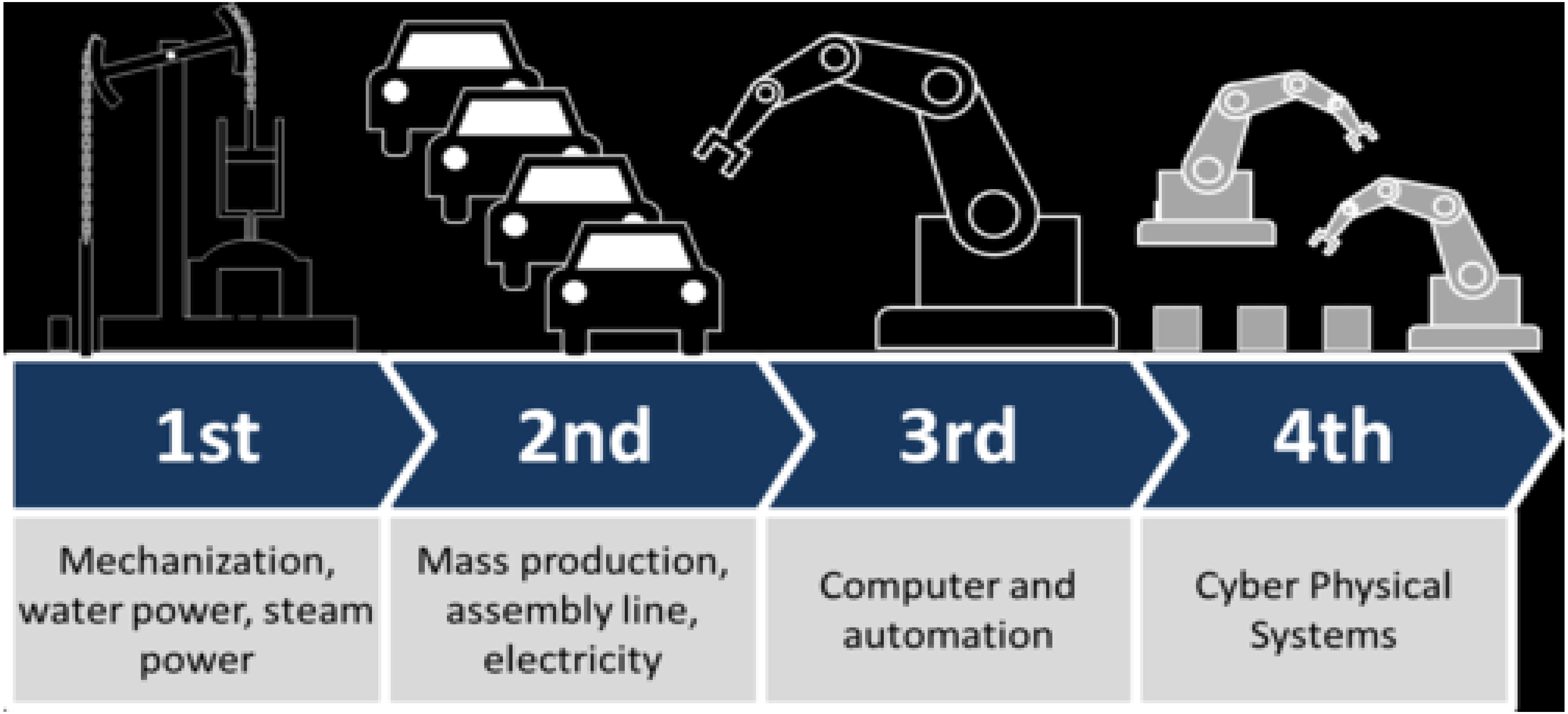
CEN520

Part 2

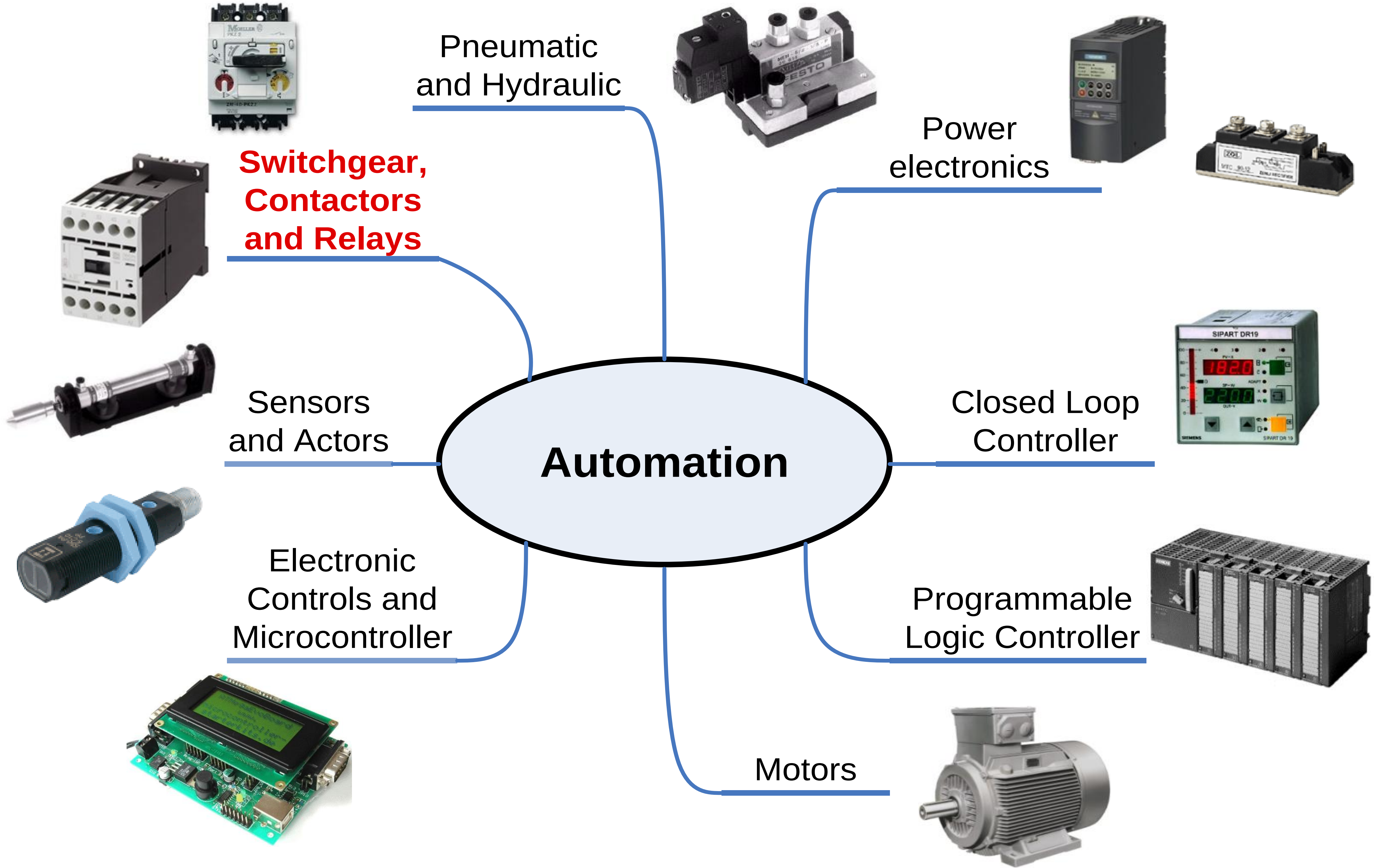
automation

Term paper: Future automated factory

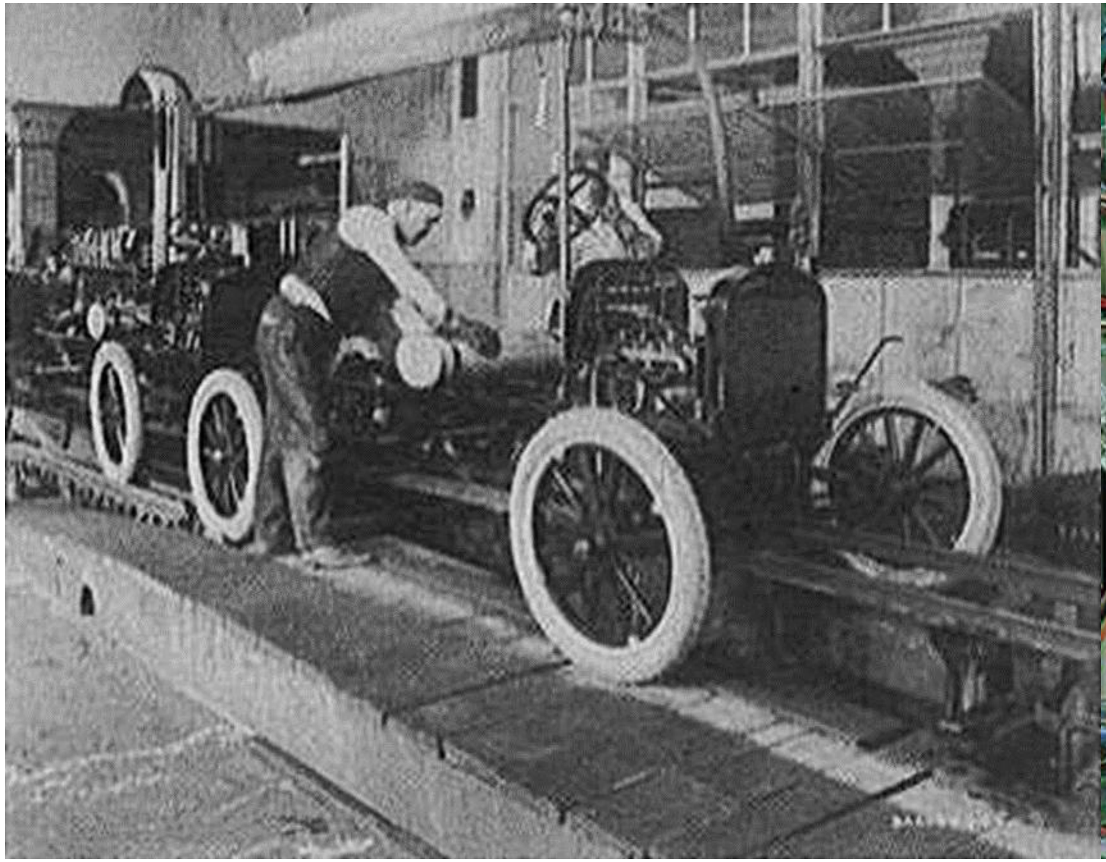


Definition - What does *Automation* mean?

- Automation is the creation of technology and its application in order to control and monitor the production and delivery of various goods and services.
- It performs tasks that were previously performed by humans.


Automation
Pneumatic and Hydraulic
Power electronics
Switchgear, Contactors and Relays
Sensors and Actors
Closed Loop Controller
Electronic Controls and Microcontroller
Programmable Logic Controller
Motors

Class discussion : compare and contrast the two plants



Ford motors plant

Mercedes plant

Advantages of automation

Advantages commonly attributed to automation include

1. To increase labour productivity
2. To reduce labour cost
3. To mitigate the effects of labour shortages
4. To reduce or remove routine manual and clerical tasks
5. To improve worker safety
6. To improve product quality
7. To reduce manufacturing lead time
8. To accomplish what cannot be done manually
9. To avoid the high cost of not automating

Assignments (3-5 pages)

- Assignment 1

List and briefly explain seven (7) disadvantages of automation

- Assignment 2

Is there a place for manual labor in the modern production system?

(hand written)

Applications of automated processes

Automation is being used in a number of areas such as ---

- manufacturing,
- transport,
- utilities,
- defense,
- facilities,
- operations
- information technology

Automated Manufacturing Systems

Examples:

- Automated machine tools
- Transfer lines
- Automated assembly systems
- Industrial robots that perform processing or assembly operations
- Automated material handling and storage systems to integrate manufacturing operations
- Automatic inspection systems for quality control

Motivations for automation

- *Automation* - automated equipment instead of labor
- *Material handling technologies* - because manufacturing usually involves a sequence of activities
- *Manufacturing systems* - integration and coordination of multiple automated or manual workstations
- *Flexible manufacturing* - to compete in the low-volume/high-mix product categories
- *Quality programs* - to achieve the high quality expected by today's customers
- *CIM* - to integrate design, production, and logistics
- *Lean production* - more work with fewer resources

Ten Strategies for Automation and Process Improvement

1. Specialization of operations
2. Combined operations
3. Simultaneous operations
4. Integration of operations
5. Increased flexibility
6. Improved material handling and storage
7. On-line inspection
8. Process control and optimization
9. Plant operations control
10. Computer-integrated manufacturing

Strategies of Automation

There are certain fundamental strategies that can be employed to improve productivity using automation . These are referred as automation strategies.

- ***Specialization of operations:***

The first strategy involves the use of special purpose equipment designed to perform one operation with the greatest possible efficiency. This is analogous to the concept of labor specializations, which has been employed to improve labor productivity.

Strategies of Automation

- ***Combined operations:***

Production occurs as a sequence of operations. Complex parts may require dozens, or even hundreds, of processing steps. The strategy of combined operations involves reducing the number of distinct production machines or workstations through which the part must be routed. This is accomplished by performing more than one operation at a given machine, thereby reducing the number of separate machines needed. Since each machine typically involves a setup, setup time can be saved as a consequence of this strategy. Material handling effort and non-operation time are also reduced.

- ***Simultaneous operations:***

A logical extension of the combined operations strategy is to perform at the same time the operations that are combined at one workstation. In effect, two or more processing (or assembly) operations are being performed simultaneously on the same workpart, thus reducing total processing time.

Strategies of Automation

- ***Integration of operations:***

Another strategy is to link several workstations into a single integrated mechanism using automated work handling devices to transfer parts between stations. In effect, this reduces the number of separate machines though which the product must be scheduled. With more than one workstation, several parts can be processed simultaneously, thereby increasing the overall output of the system.

- ***Increased flexibility:***

This strategy attempts to achieve maximum utilization of equipment for job shop and medium volume situations by using the same equipment for a variety of products. It involves the use of the flexible automation concepts. Prime objectives are to reduce setup time and programming time for the production machine. This normally translates into lower manufacturing lead time and lower work-in-process.

Strategies of Automation

- ***Improved material handling and storage systems:***

A great opportunity for reducing non-productive time exists in the use of automated material handling and storage systems. Typical benefits included reduced work-in-process and shorter manufacturing lead times.

- ***On-line inspection:***

Inspection for quality of work is traditionally performed after the process. This means that any poor quality product has already been produced by the time it is inspected. Incorporating inspection into the manufacturing process permits corrections to the process as product is being made. This reduces scrap and brings the overall quality of product closer to the nominal specifications intended by the designer.

Strategies of Automation

- ***Process control and optimization:***

This includes a wide range of control schemes intended to operate the individual process and associated equipment more efficiency. By this strategy, the individual process times can be reduced and product quality improved.

- ***Plant operations control:***

Whereas the previous strategy was concerned with the control of the individual manufacturing process, this strategy is concerned with control at the plant level of computer networking within the factory.

- ***Computer integrated manufacturing (CIM):***

Taking the previous strategy one step further, the integration of factory operations with engineering design and many of the other business functions of the firm. CIM involves extensive use of computer applications, computer data bases, and computer networking in the company

Automated Manufacturing Systems

Three basic types:

1. Fixed automation
2. Programmable automation
3. Flexible automation

Fixed Automation

A manufacturing system in which the sequence of processing (or assembly) operations is fixed by the equipment configuration

Typical features:

- Suited to high production quantities
- High initial investment for custom-engineered equipment
- High production rates
- Relatively inflexible in accommodating product variety

Programmable Automation

A manufacturing system designed with the capability to change the sequence of operations to accommodate different product configurations

Typical features:

- High investment in general purpose equipment
- Lower production rates than fixed automation
- Flexibility to deal with variations and changes in product configuration
- Most suitable for batch production
- Physical setup and part program must be changed between jobs (batches)

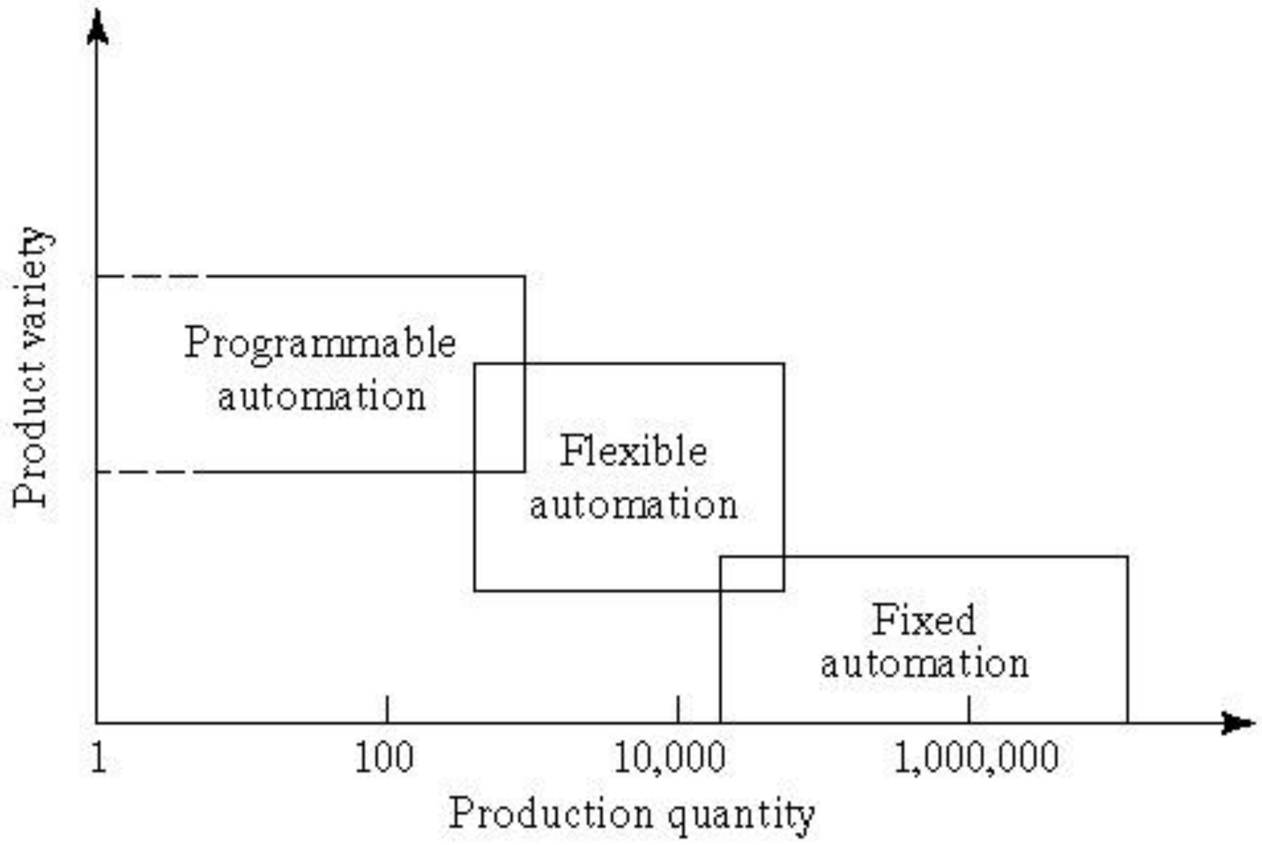
Flexible Automation

An extension of programmable automation in which the system is capable of changing over from one job to the next with no lost time between jobs

Typical features:

- High investment for custom-engineered system
- Continuous production of variable mixes of products
- Medium production rates
- Flexibility to deal with soft product variety

Product Variety and Production Quantity for Three Automation Types




Vince Penman

Term paper

The fourth industrial revolution:

Future for business, workplace and communities – the Computer Engineer's perspective.

CEN520

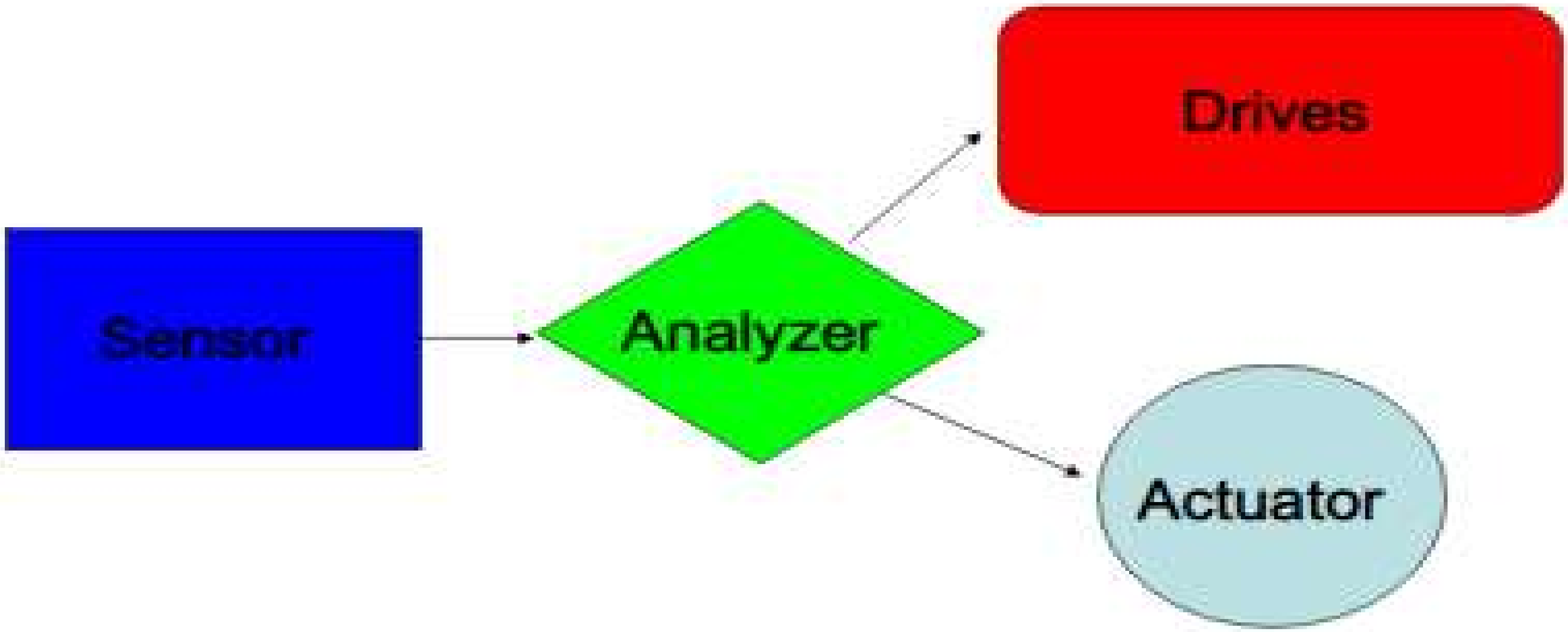
Part 2

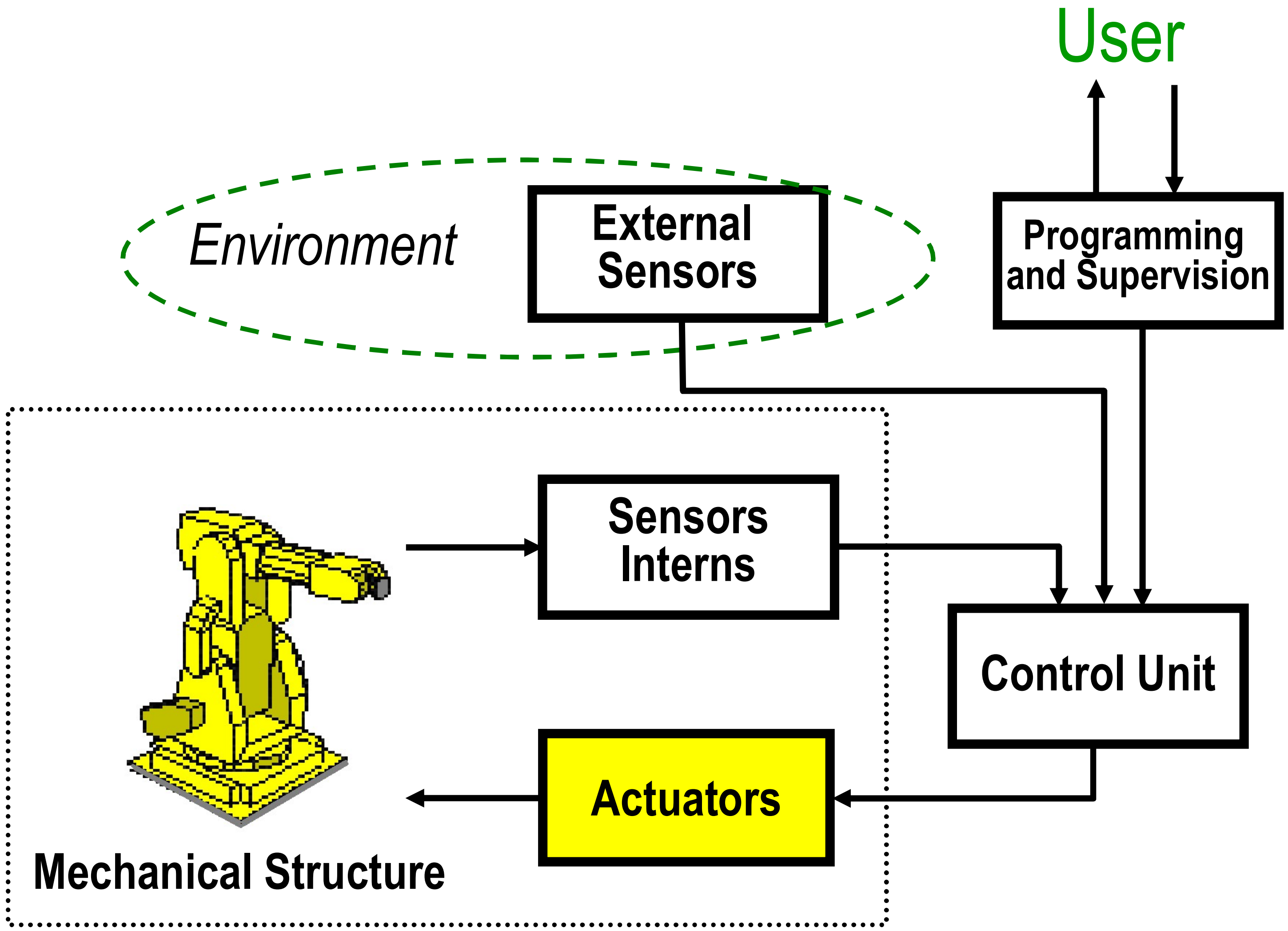
automation

Topics

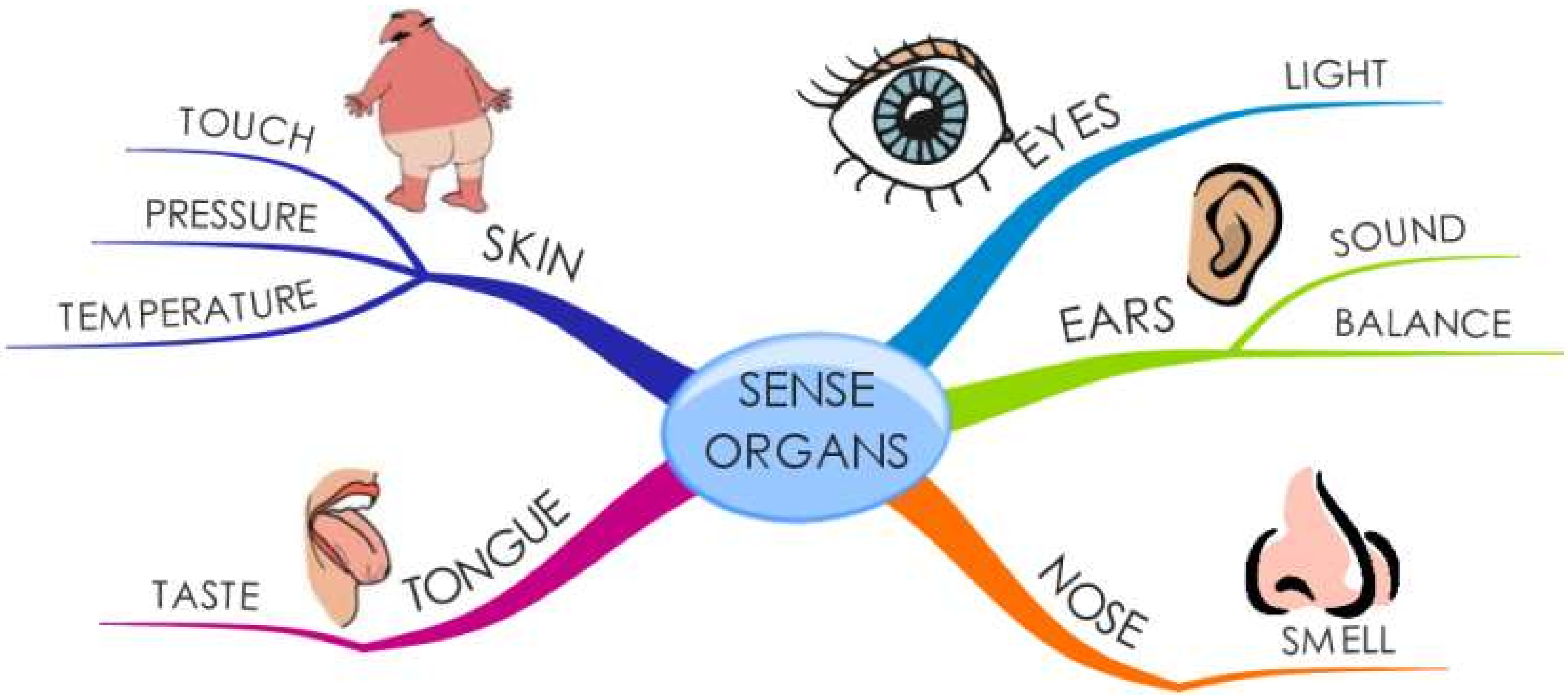
- Analysis of automated flow lines.
- Assembly systems and line balancing.
- Automated assembly systems

Building Blocks of Automation




User
Environment
External Sensors
Programming and Supervision
Sensors Interns
Control Unit
Actuators
Mechanical Structure

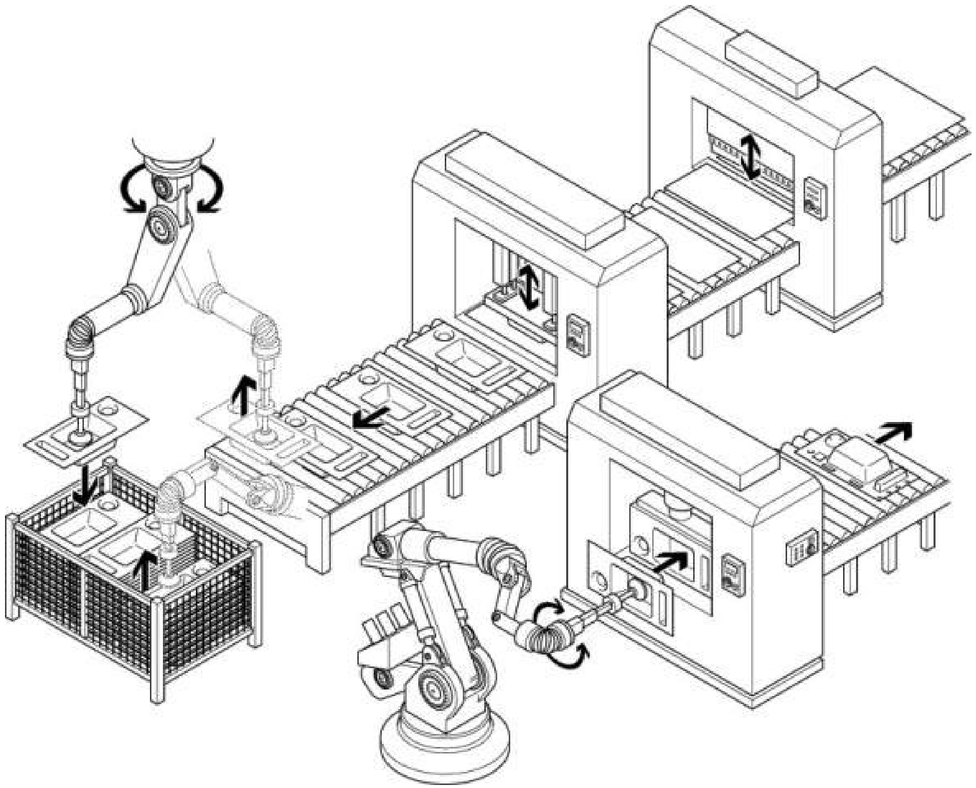
Sensors


SENSE ORGANS
SKIN
TOUCH
PRESSURE
TEMPERATURE
EYES
LIGHT
EARS
SOUND
BALANCE
TONGUE
TASTE
NOSE
SMELL

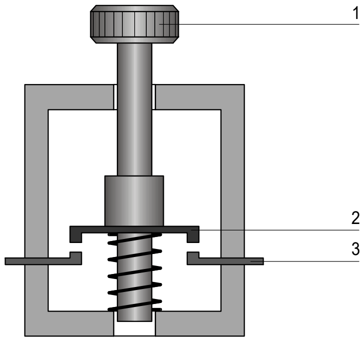
Sensor

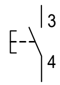
- A **Sensor** converts the physical parameter (for example: temperature, blood pressure, humidity, speed, etc.) into a signal which can be measured electrically

Automated Assembly Line

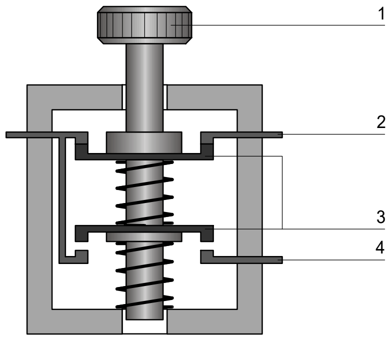


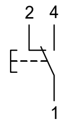
Electrical Switches



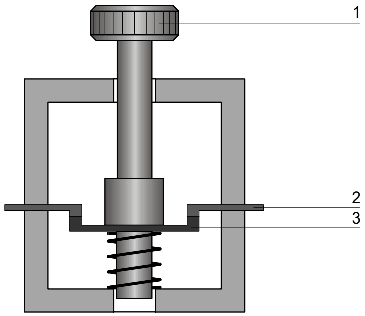


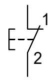
Normally open contact





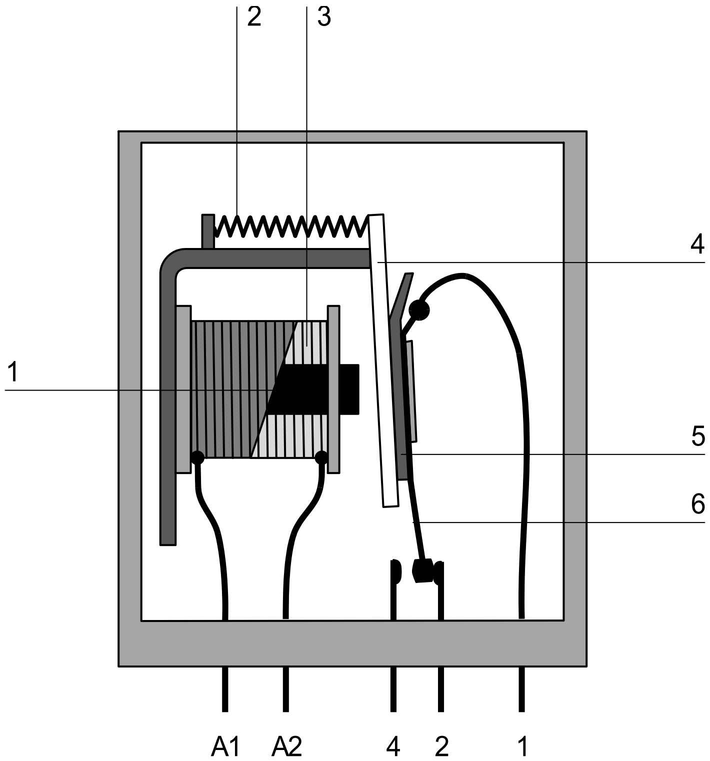
toggle switch

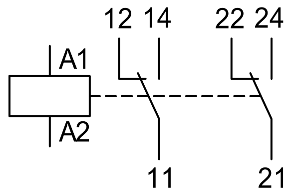


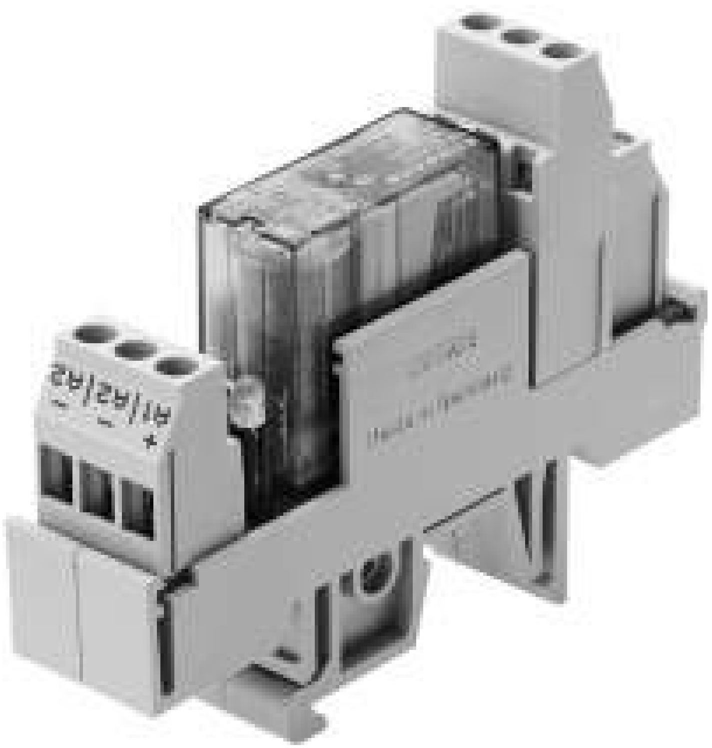


Normally closed contact

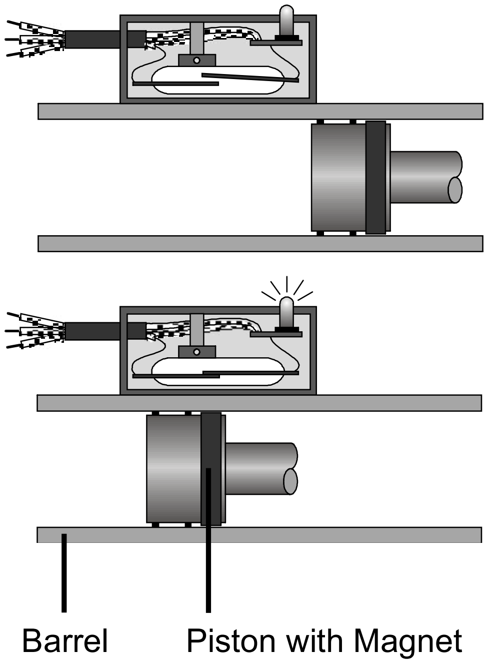
Relays

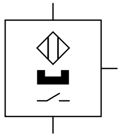




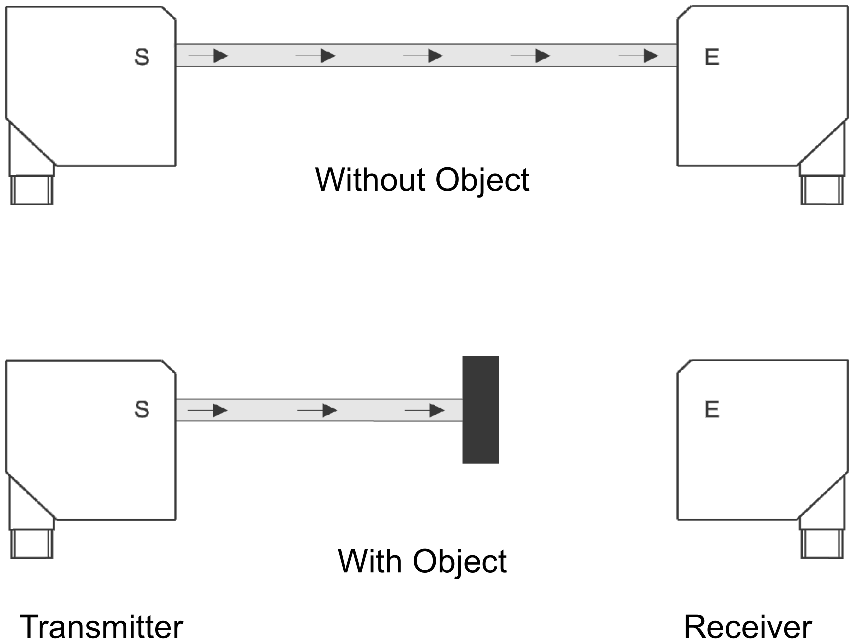


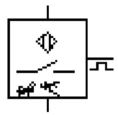
Proximity Switches for Pneumatical Cylinders



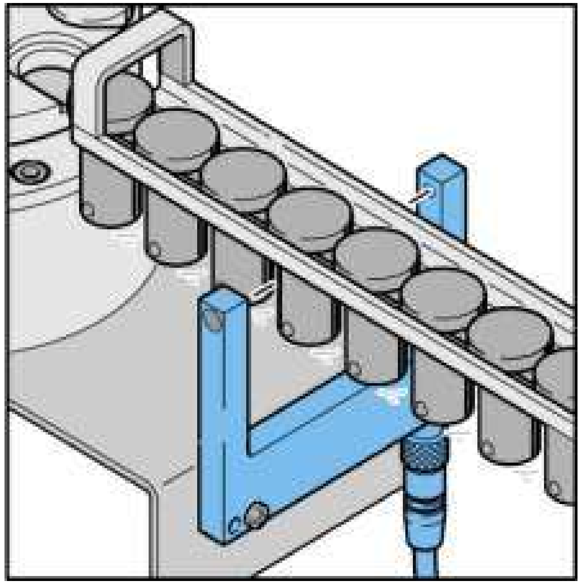


Optical Proximity Switches (Thru Beam)

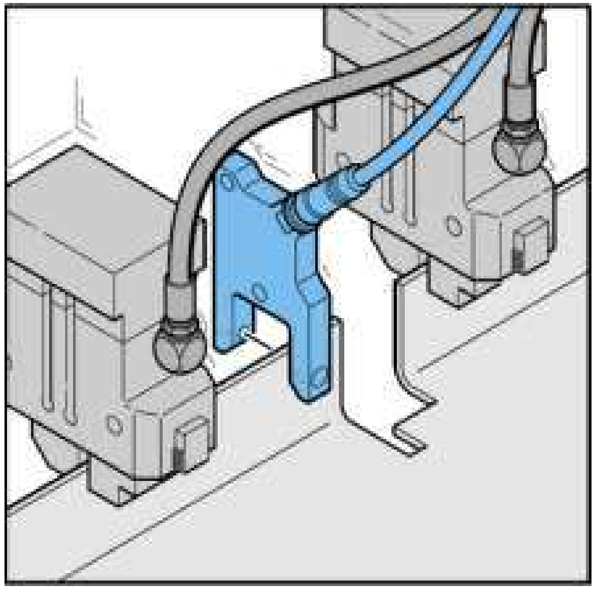




Thru Beam Sensor - Applications

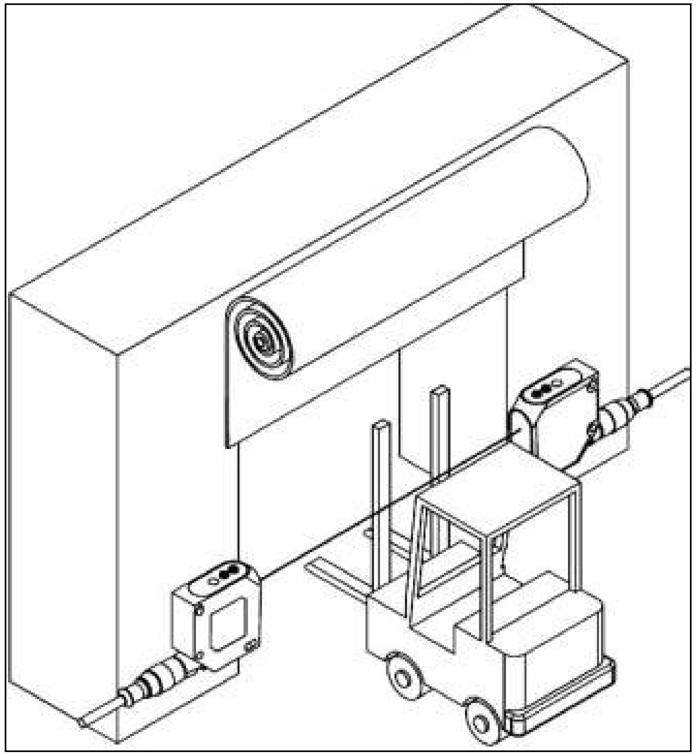


Standstill control in feeding units

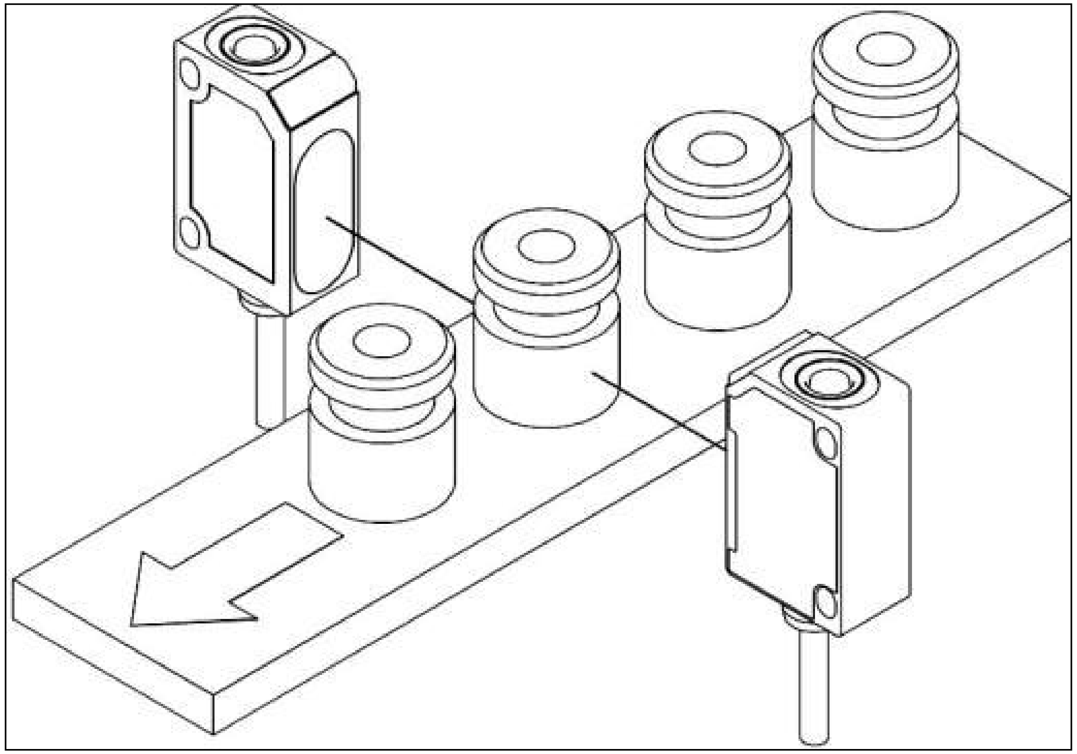


Parts detection in handling devices and transfer presses

Thru Beam Sensor – Applications (II)

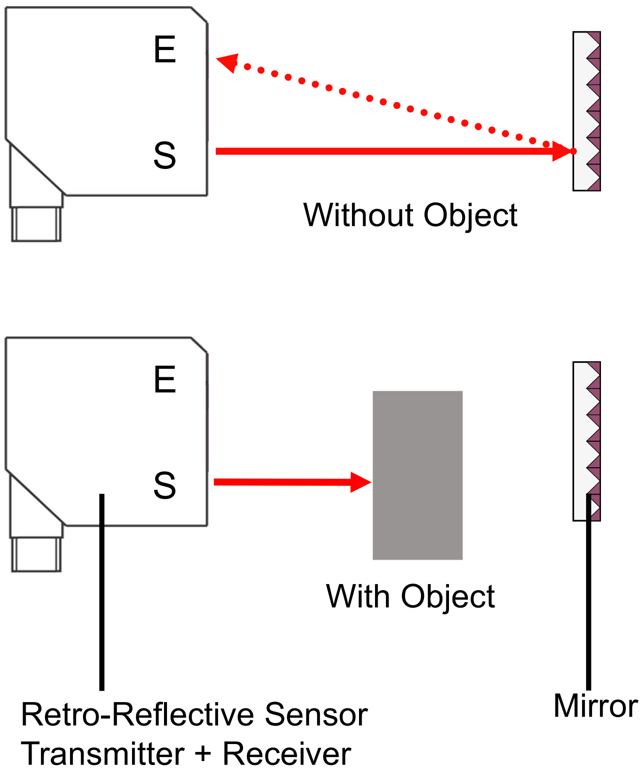


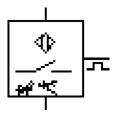
Object detection at long distances



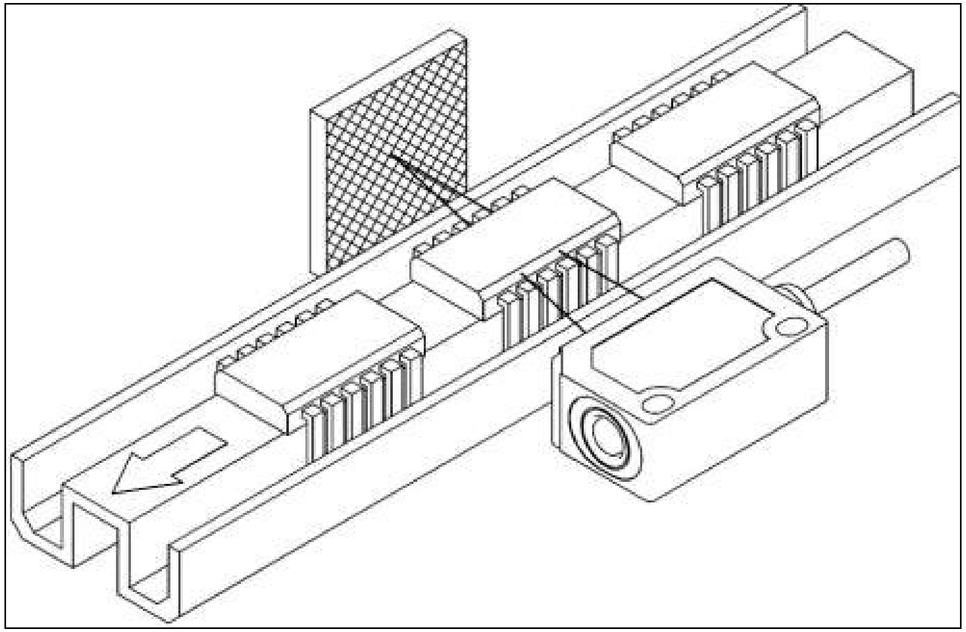
Object detection in harsh environments

Optical Proximity Switches – Retro Reflective Sensor

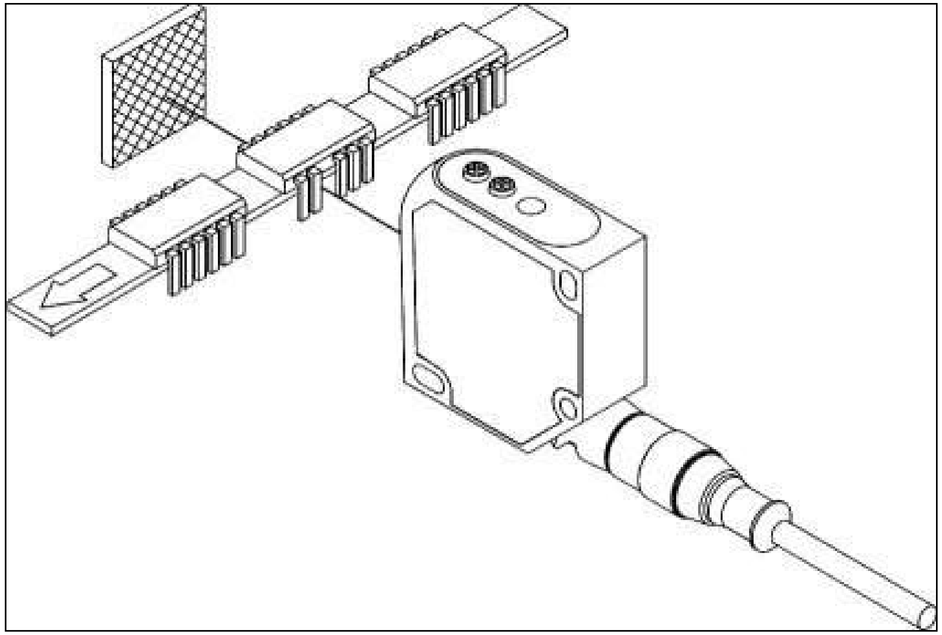




Retro Reflective Sensor - Applications

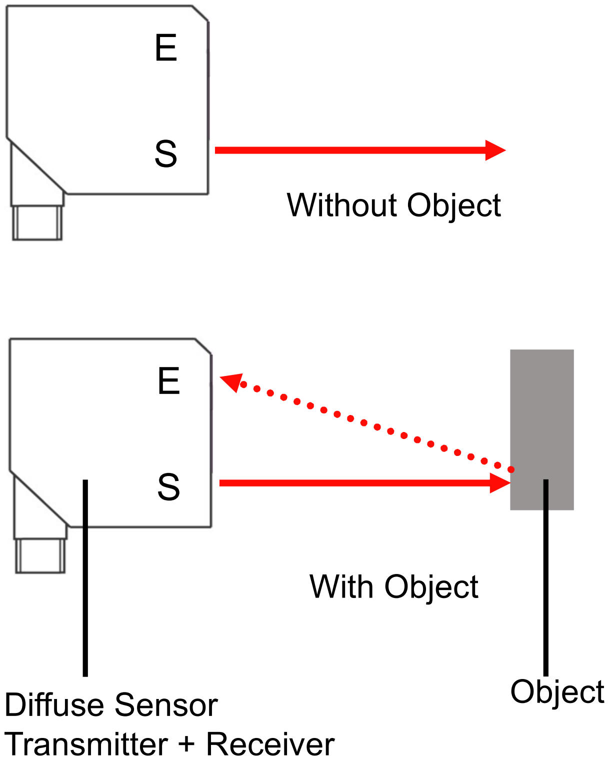


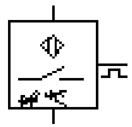
Position control of small objects



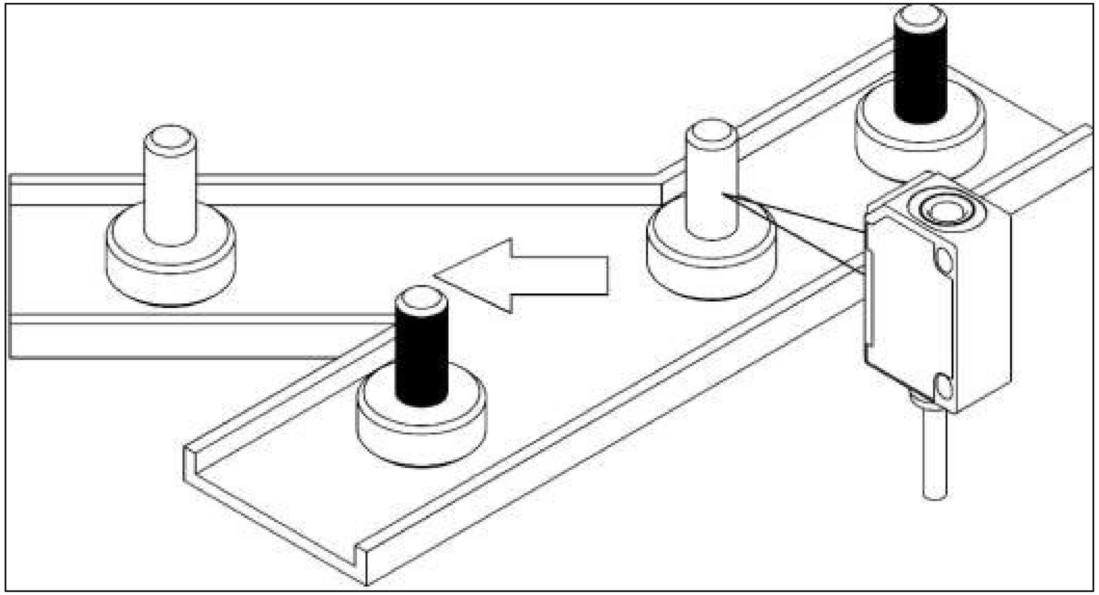
IC-pins control

Optical Proximity Switches – Diffuse Sensor

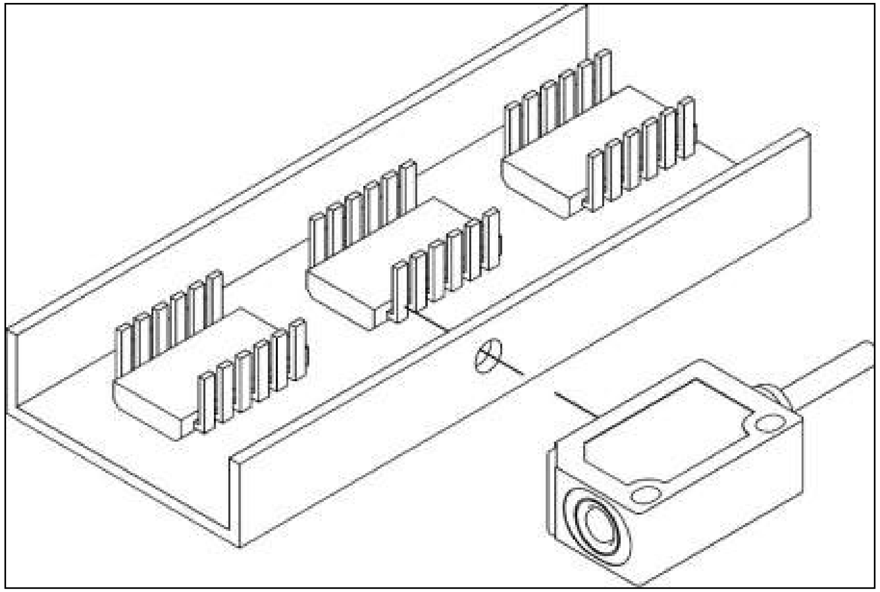




Diffuse Sensor - Applications

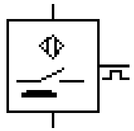


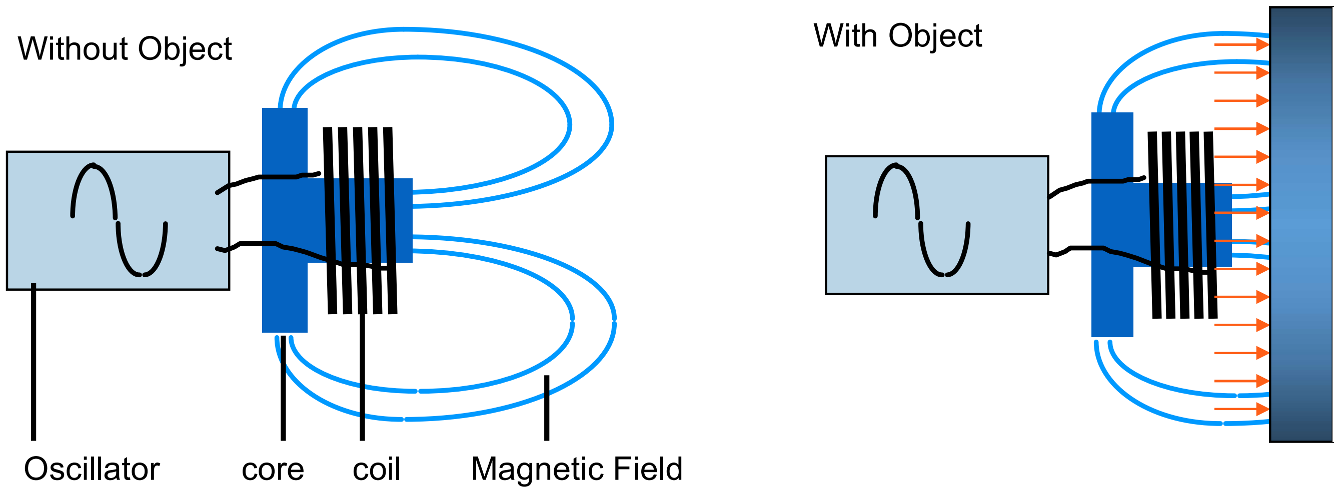
Recognition of objects with different contrasts



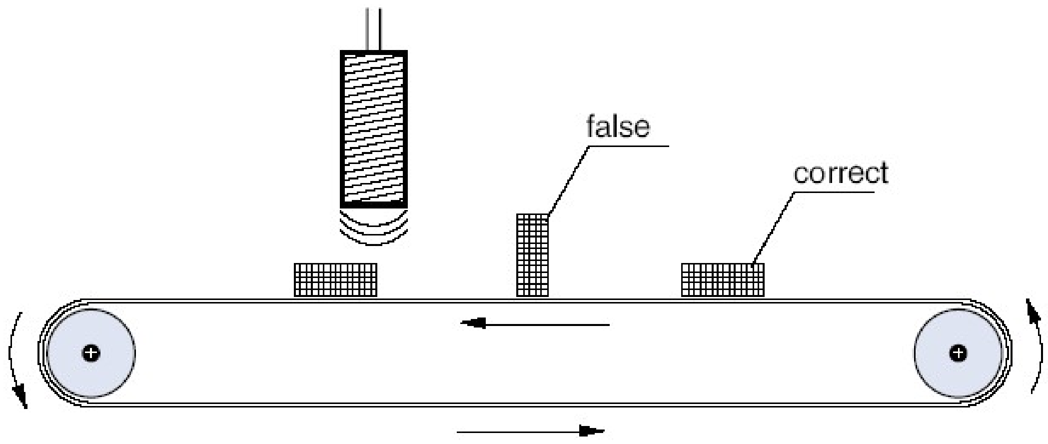
IC-Pin inspection

Inductive Proximity Switches



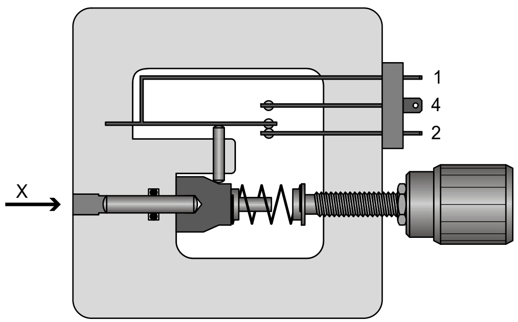


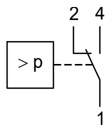
Inductive Sensors - Applications

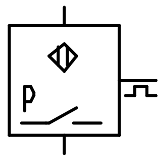


Object Detection

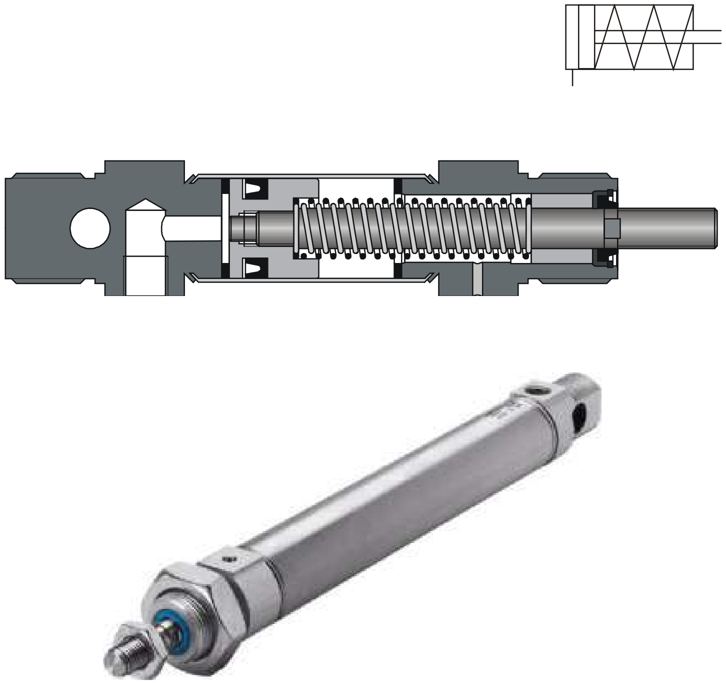
Pressure Switch

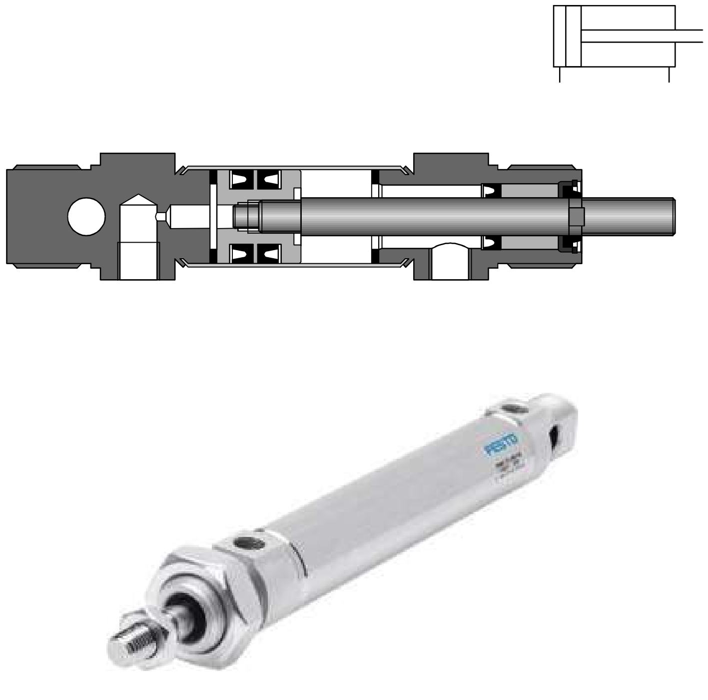




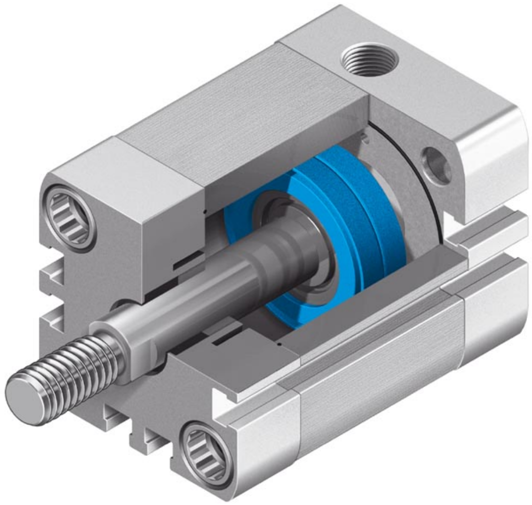


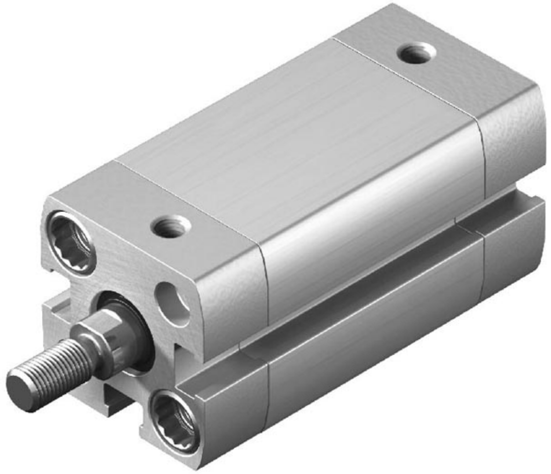
Single and Double Acting Cylinder



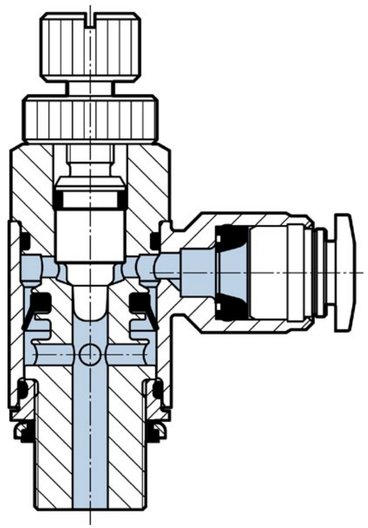


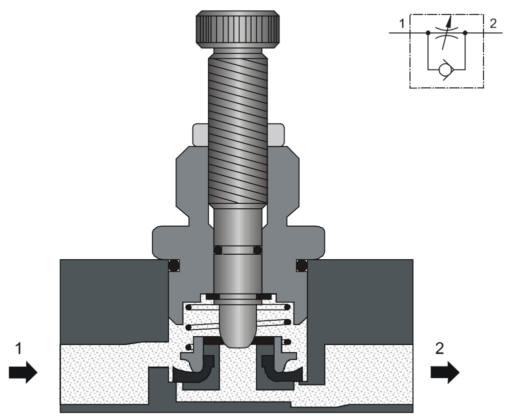
Profile Cylinder



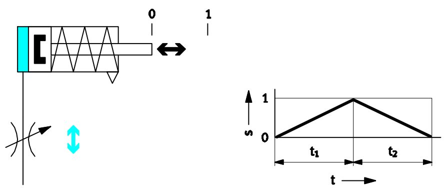


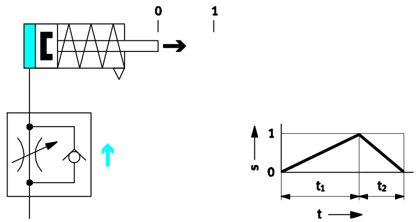
One Way Flow Control Valve

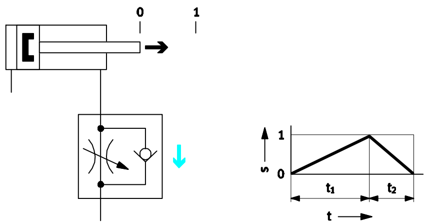


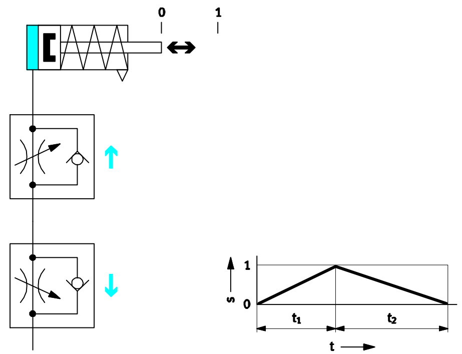


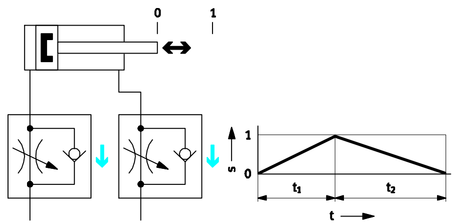
Flow Control









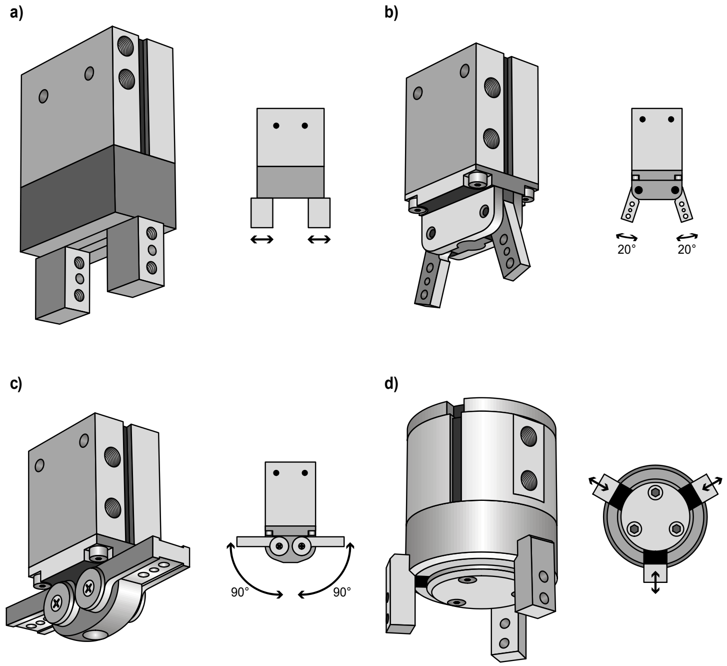


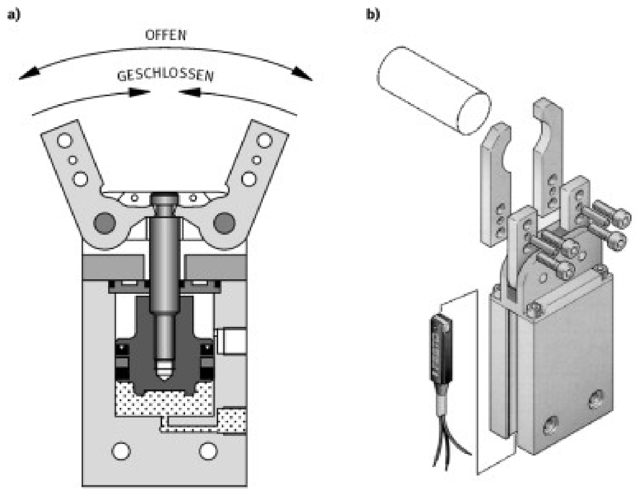
4/2-Way Solenoid Valve

4/2-Way-Valve Monostable

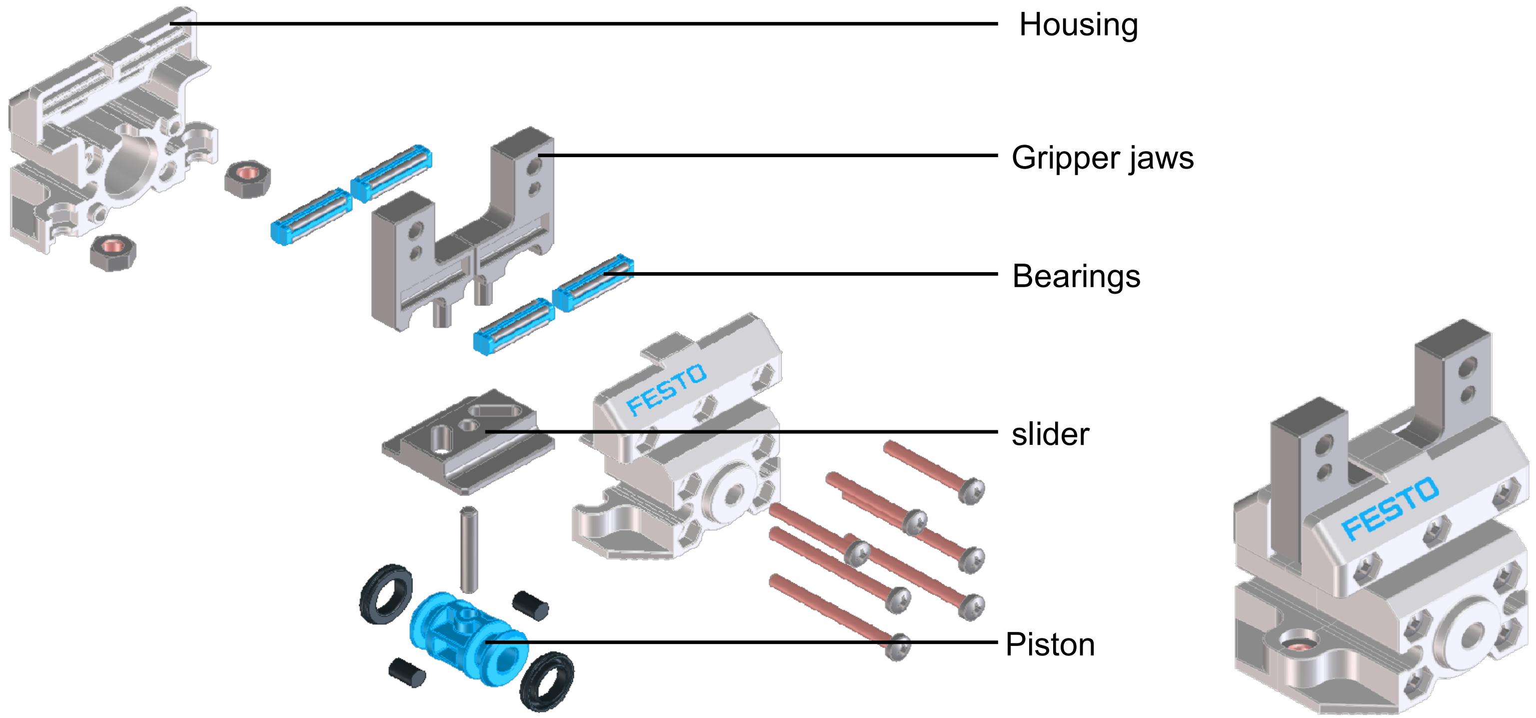
4/2-Way-Valve Bistable

Mechanical Grippers - Types

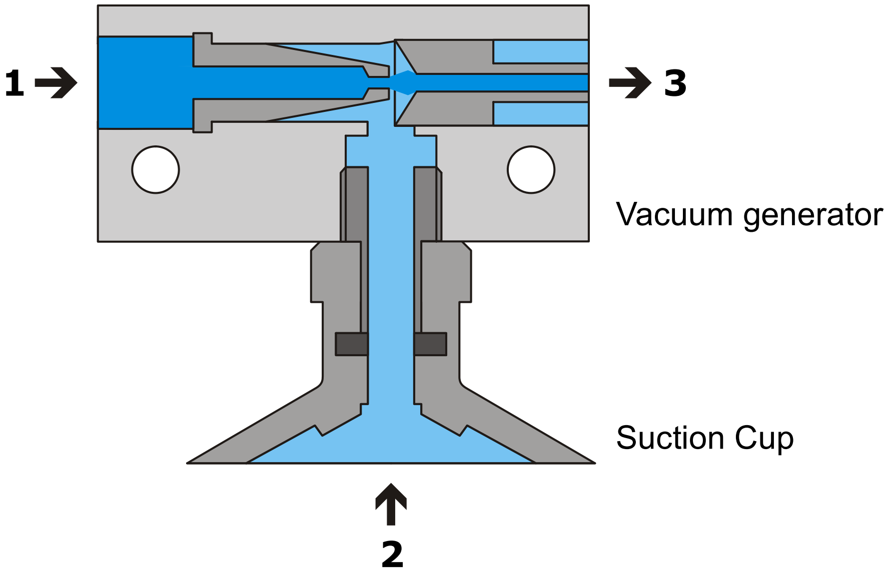


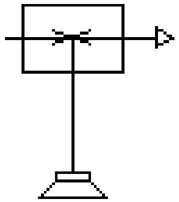


Mechanical Grippers – Working Principle



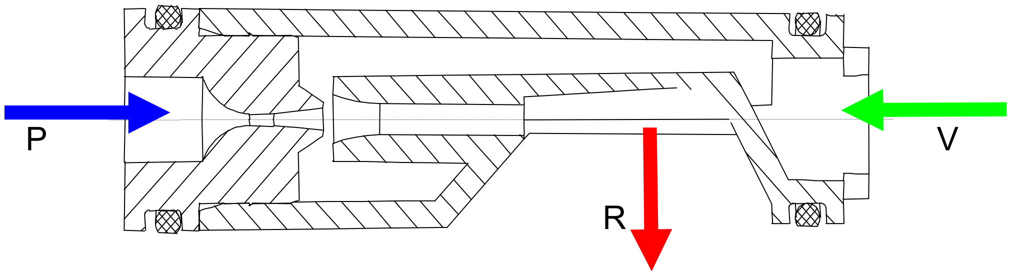
Vacuum Grippers – Working Principle



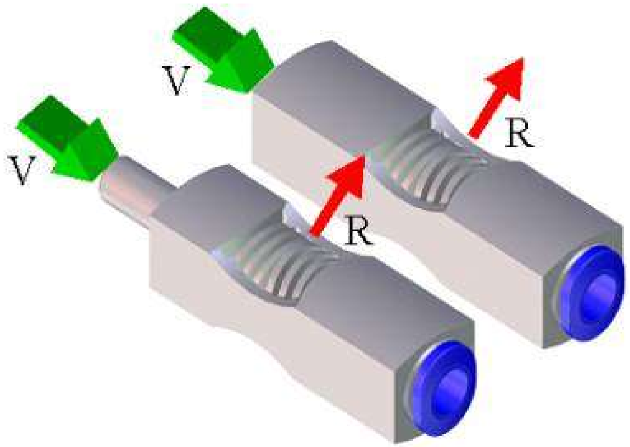


1 Supply port
2 Vacuum port
3 Exhaust port

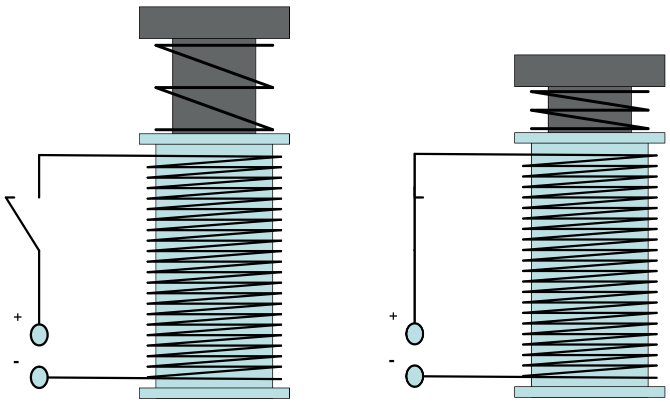
Inline-Type Vacuum Generator

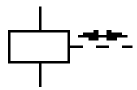


P Supply port
V Vacuum port
R Exhaust port



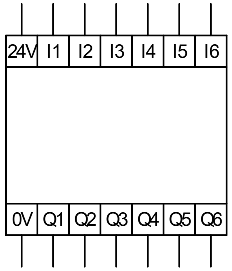
Solenoid



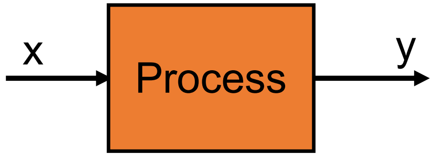


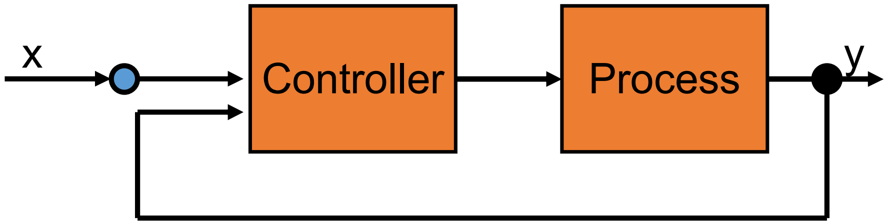
Analyzers - computers

Programmable Logic Controllers

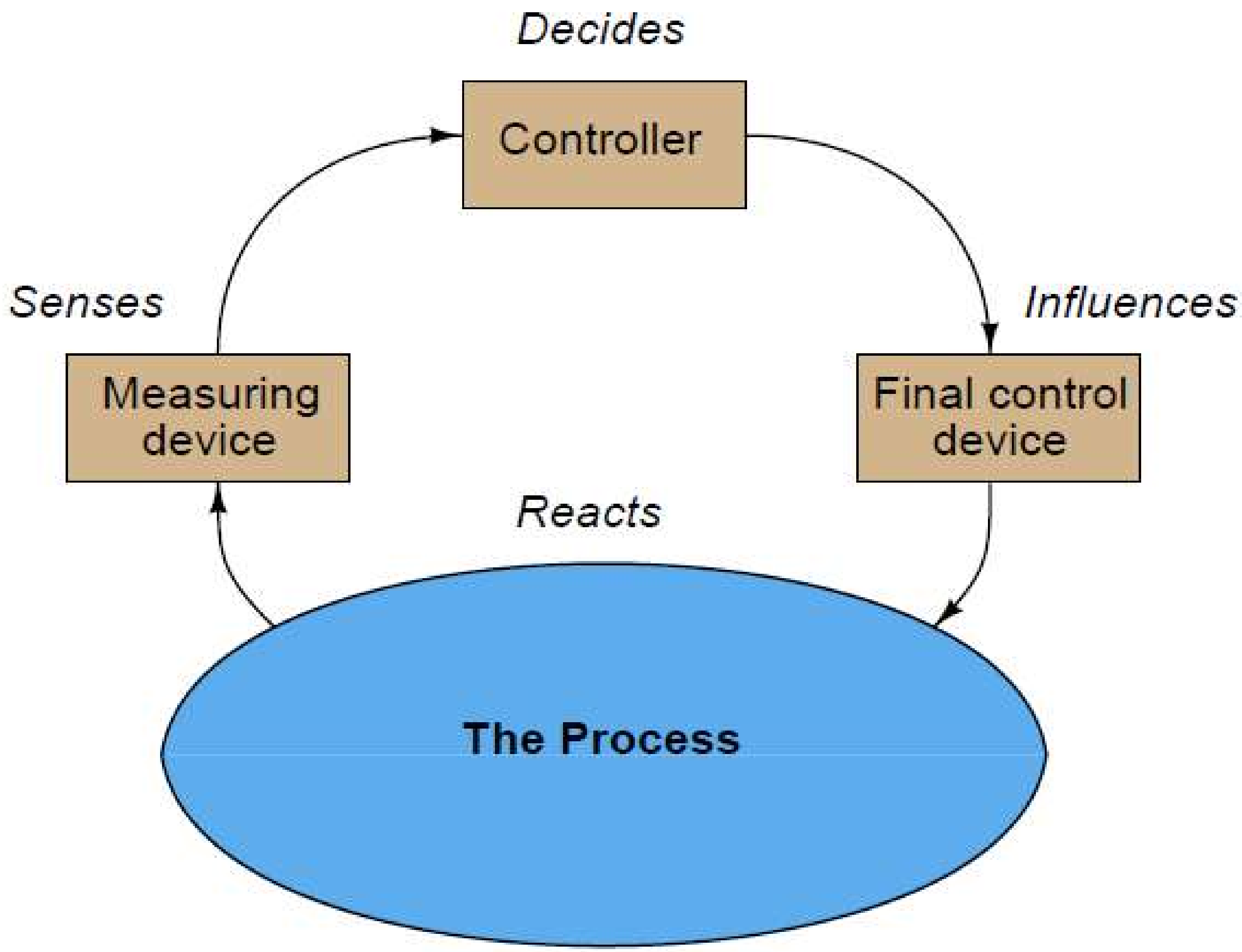


Open and Closed Loop Control

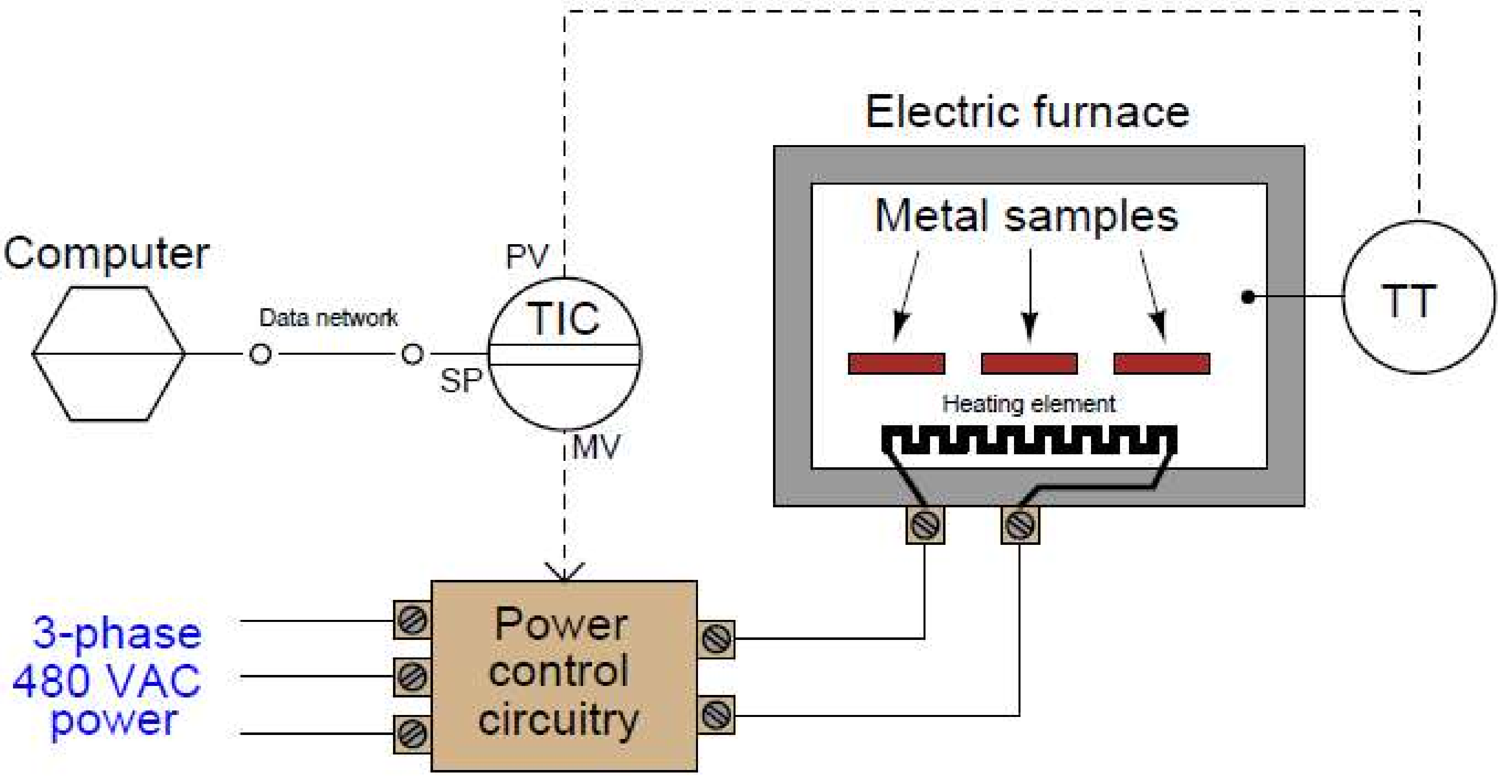




Industrial process



A typical industrial process



What are Programmable logic controllers

- **Programmable logic controllers are essentially nothing more than special-purpose, industrial computers**. As such, they are built far more ruggedly than an ordinary personal computer (PC),and designed to run extremely reliable operating system software.
- **PLCs as a rule do not contain hard disk drives, cooling fans, or any other components with moving parts**. This is an intentional design decision, intended to maximize the reliability of the hardware in harsh industrial environments where the PLC chassis may be subjected to temperature extremes, vibration, humidity, and airborne particulates

- A PLC has many "**input**" terminals, through which it interprets "**high**" and "**low**" logical states from sensors and switches. It also has many output terminals, through which it outputs "**high**" and "**low**" signals to power lights, solenoids, contactors, small motors, and other devices lending themselves to on/off control. In an effort to make PLCs easy to program, their programming language was designed to resemble ladder logic diagrams

PLC Size

<u>SMALL</u> - It covers units with up to 128 I/O's and memories up to 2Kbytes.
- These PLC's are capable of providing simple to advance levels or machine controls.

<u>MEDIUM</u> - have up to 2048 I/O's and memories up to 32 Kbytes.

<u>LARGE</u> - the most sophisticated units of the PLC family. They have up to 8192 I/O's and memories up to 750 Kbytes.
- can control individual production processes or entire plant.

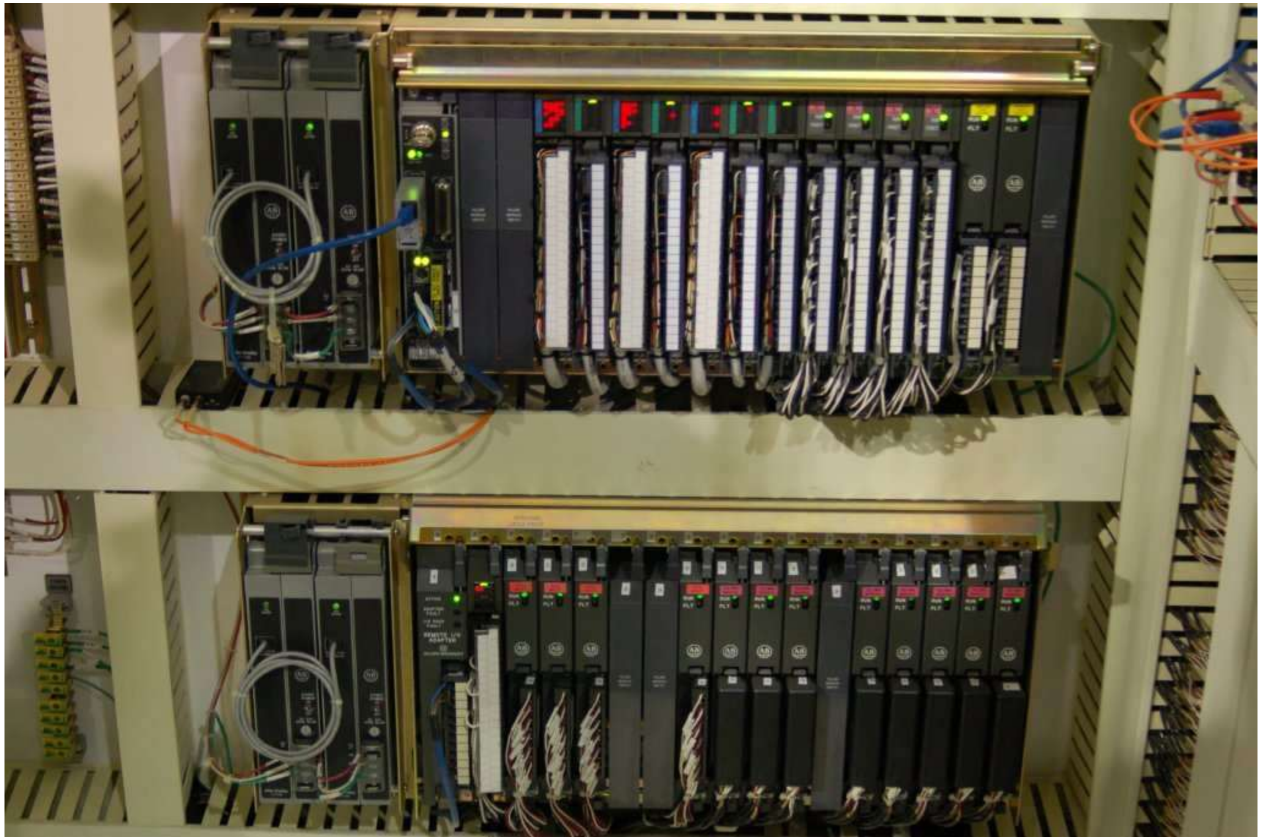
Leading Brands Of PLC

AMERICAN
1. Allen Bradley
2. Gould Modicon
3. Texas Instruments
4. General Electric
5. Westinghouse
6. Cutter Hammer
7. Square D

EUROPEAN
1. Siemens
2. Klockner & Mouller
3. Festo
4. Telemechanique

JAPANESE
1. Toshiba
2. Omron
3. Fanuc
4. Mitsubishi

Allen-Bradley (Rockwell) PLC-5 system, used to monitor and control the operation of a large natural gas compressor

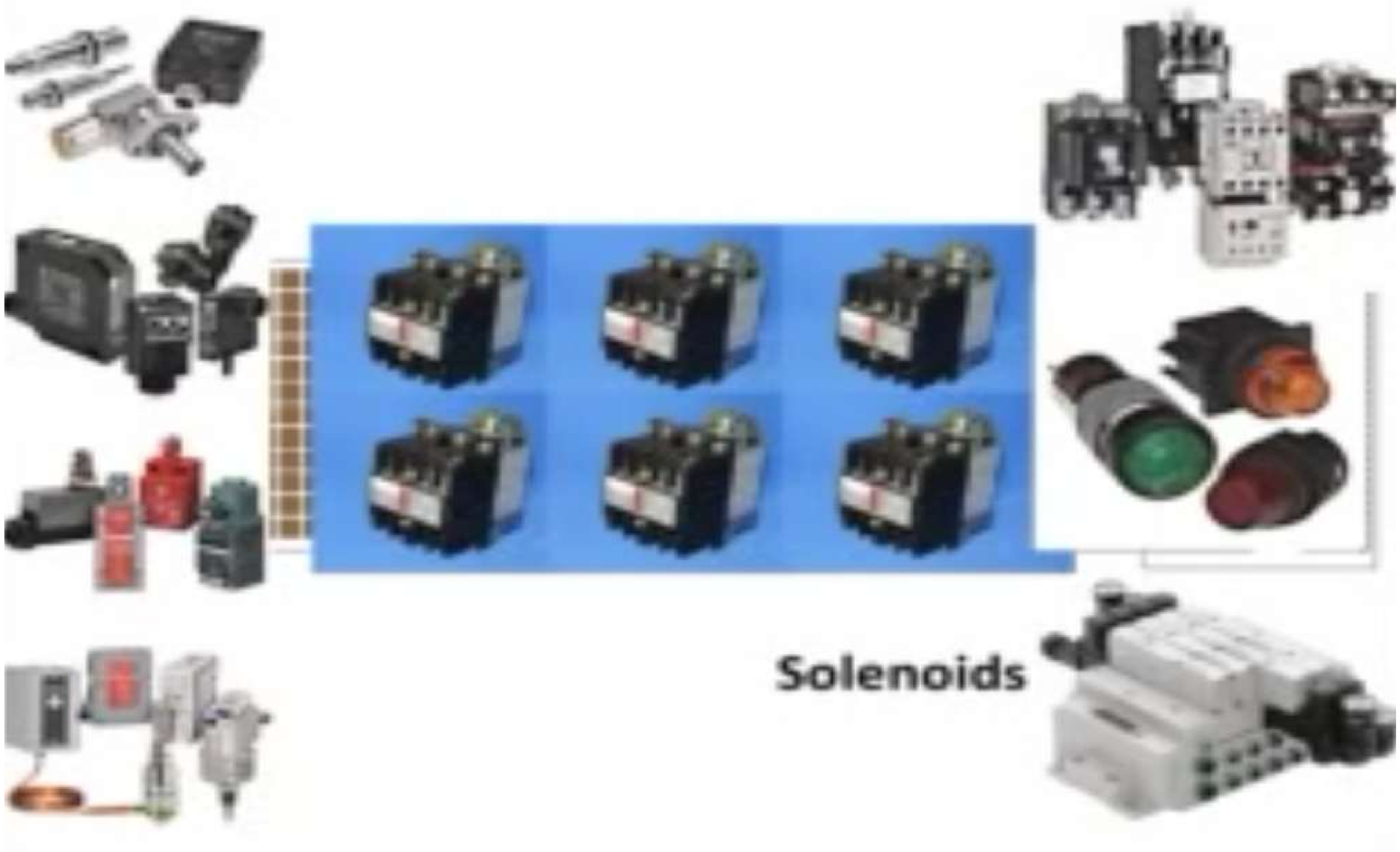


A modern PLC manufactured by Allen-Bradley (Rockwell) ControlLogix 5000 system, used to control a cereal manufacturing process.

Purpose of PLCs

- The purpose of a PLC was to directly replace electromechanical relays as logic elements, substituting instead a solid-state digital computer with a stored program, able to emulate the interconnection of many relays to perform certain logical tasks.

A typical Relay based control panel



Programmable Logic Controller (PLC) development as a low cost computer has brought the most recent revolution,

The advent of the PLC began in the 1970s, and has become the most common choice for manufacturing controls.

PLCs have been gaining popularity on the factory floor and will probably remain predominant for some time to come. Most of this is because of the advantages they offer.

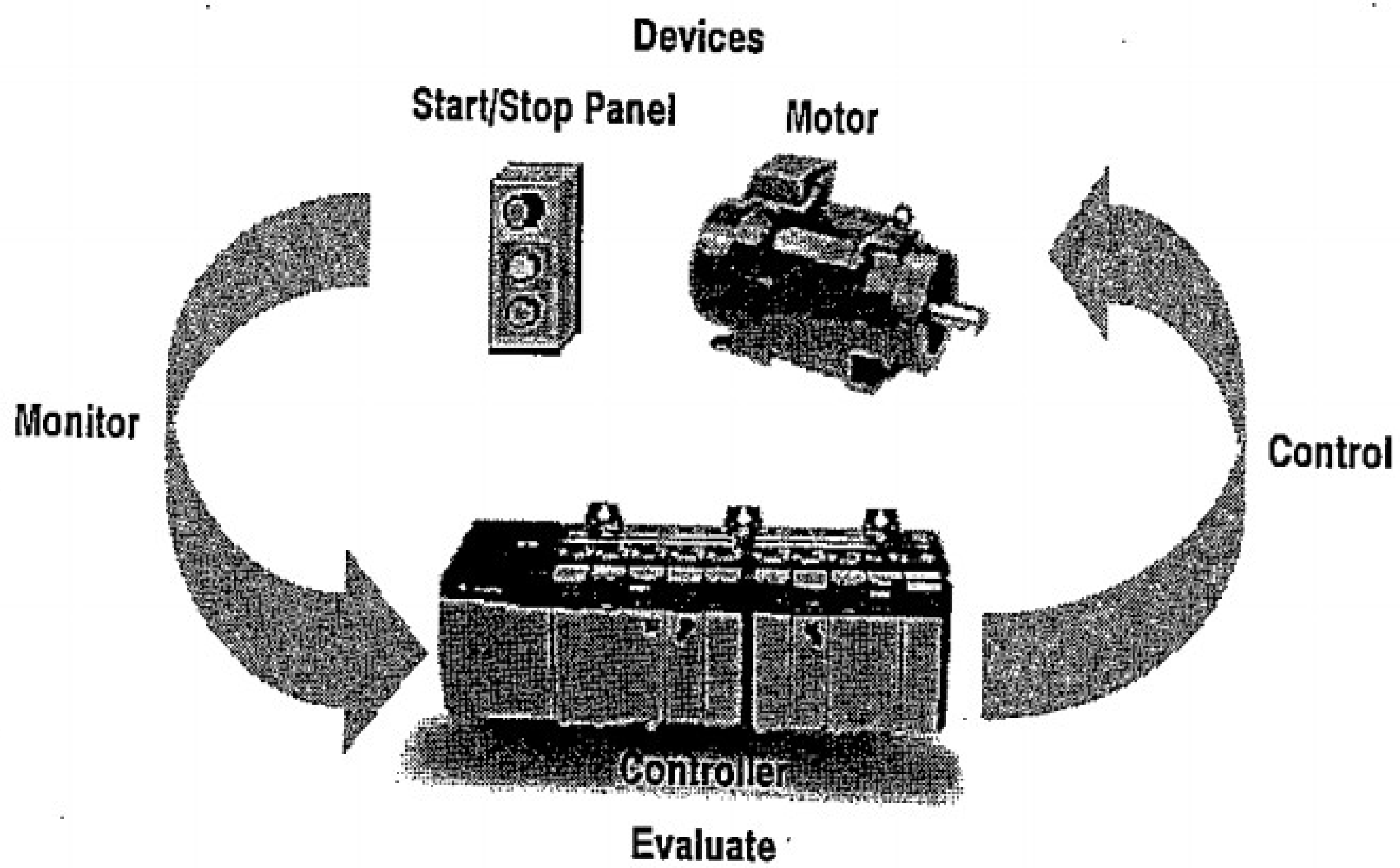
Advantages of PLCs

- Cost effective for controlling complex systems.
- Flexible and can be reapplied to control other systems quickly and easily.
- Computational abilities allow more sophisticated control.
- Trouble shooting aids make programming easier and reduce downtime.
- Reliable components make these likely to operate for years before failure.
- Remote monitoring and control via digital computer networks

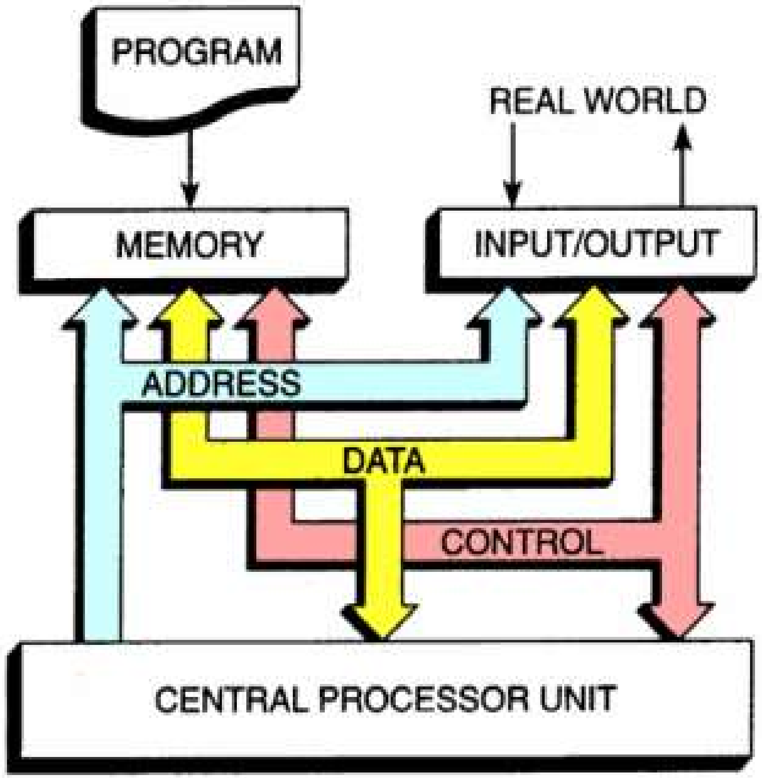
.

This is a semi-modular PLC design with input/output (I/O) channels built into the processor module, but having the capacity to accept multiple I/O modules plugged in to the side (cascade)

PLC COMPONENTS & NETWORK

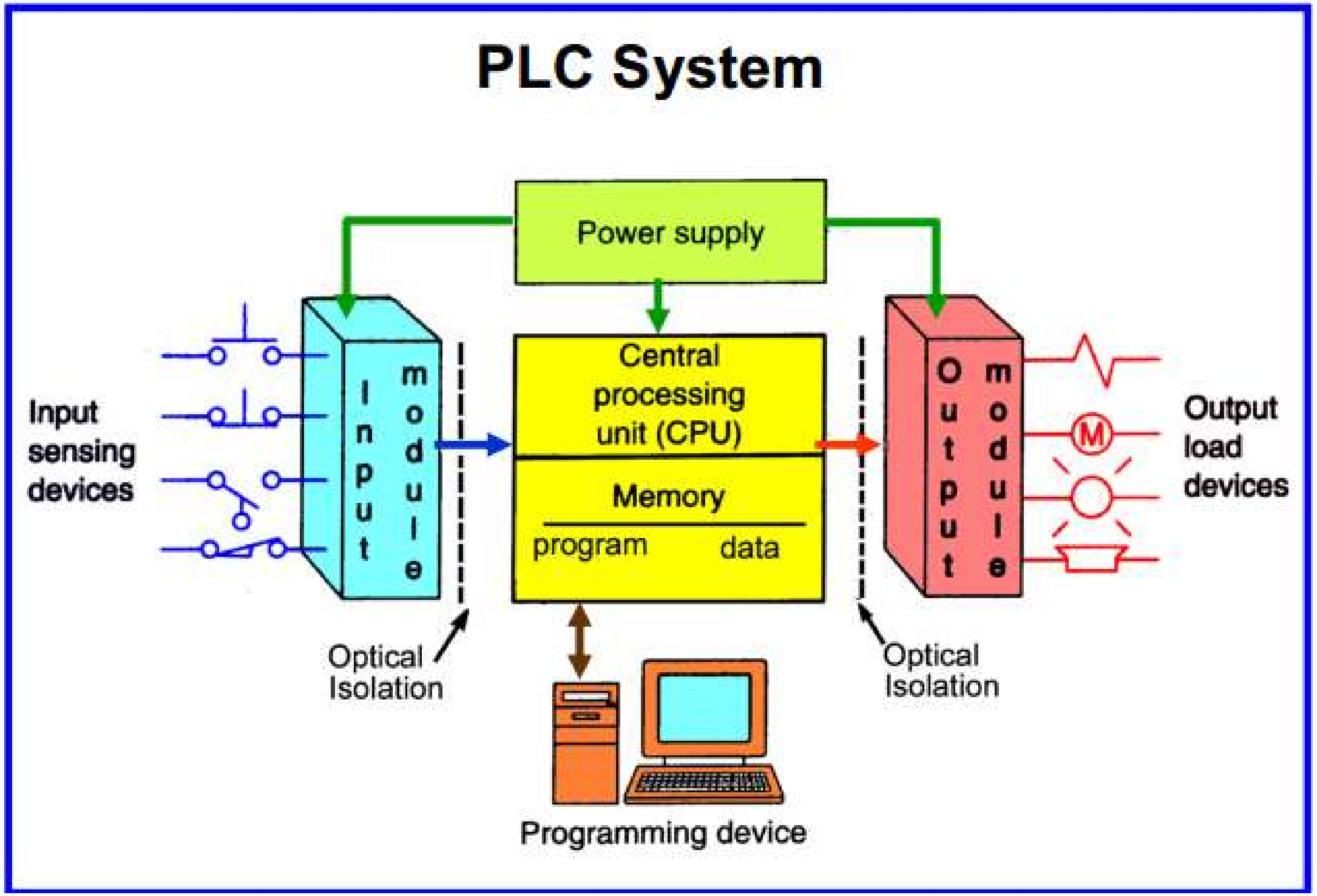


PLC Architecture



The structure of a PLC is based on the same principles as those employed in computer architecture.

Block diagram of PLCs



Plc components (hardware)

- Many PLC configurations are available but, in each of these there are common components and concepts. The most essential components are:
- Power Supply - This can be built into the PLC or be an external unit. Common voltage levels required by the PLC (with and without the power supply) are 24Vdc, 120Vac, 220Vac.

- CPU (Central Processing Unit) - This is a computer where ladder logic is stored and processed.
- I/O (Input/Output) - A number of input/output terminals must be provided so that the PLC can monitor the process and initiate actions.
- Indicator lights - These indicate the status of the PLC including power on, program running, and a fault. These are essential when diagnosing problems.

- Input/output - Every programmable logic controller must have some means of receiving and interpreting signals from real-world sensors such as switches, and encoders, and also be able to effect control over real-world control elements such as solenoids, valves, and motors. This is generally known as input/output or I/O, capability. Monolithic (“brick”) PLCs have a fixed amount of I/O capability built into the unit, while modular (“rack”) PLCs use individual circuit board “cards” to provide customized I/O capability.
- Some PLCs have the ability to connect to processor-less remote racks filled with additional I/O cards or modules, thus providing a way to increase the number of I/O channels beyond the capacity of the base unit. The connection from host PLC to remote I/O racks usually takes the form of a special digital network, which may span a great physical distance

OPERATION SEQUENCE

All PLCs have four basic stages of operations that are repeated many times per second. Initially when turned on the first time it will check it's own hardware and software for faults. If there are no problems it will copy all the input and copy their values into memory, this is called the input scan.

- Using only the memory copy of the inputs the ladder logic program will be solved once, this is called the logic scan. While solving the ladder logic the output values are only changed in temporary memory. When the ladder scan is done the outputs will updated using the temporary values in memory, this is called the output scan.
- The PLC now restarts the process by starting a self check for faults. This process typically repeats 10 to 100 times per second

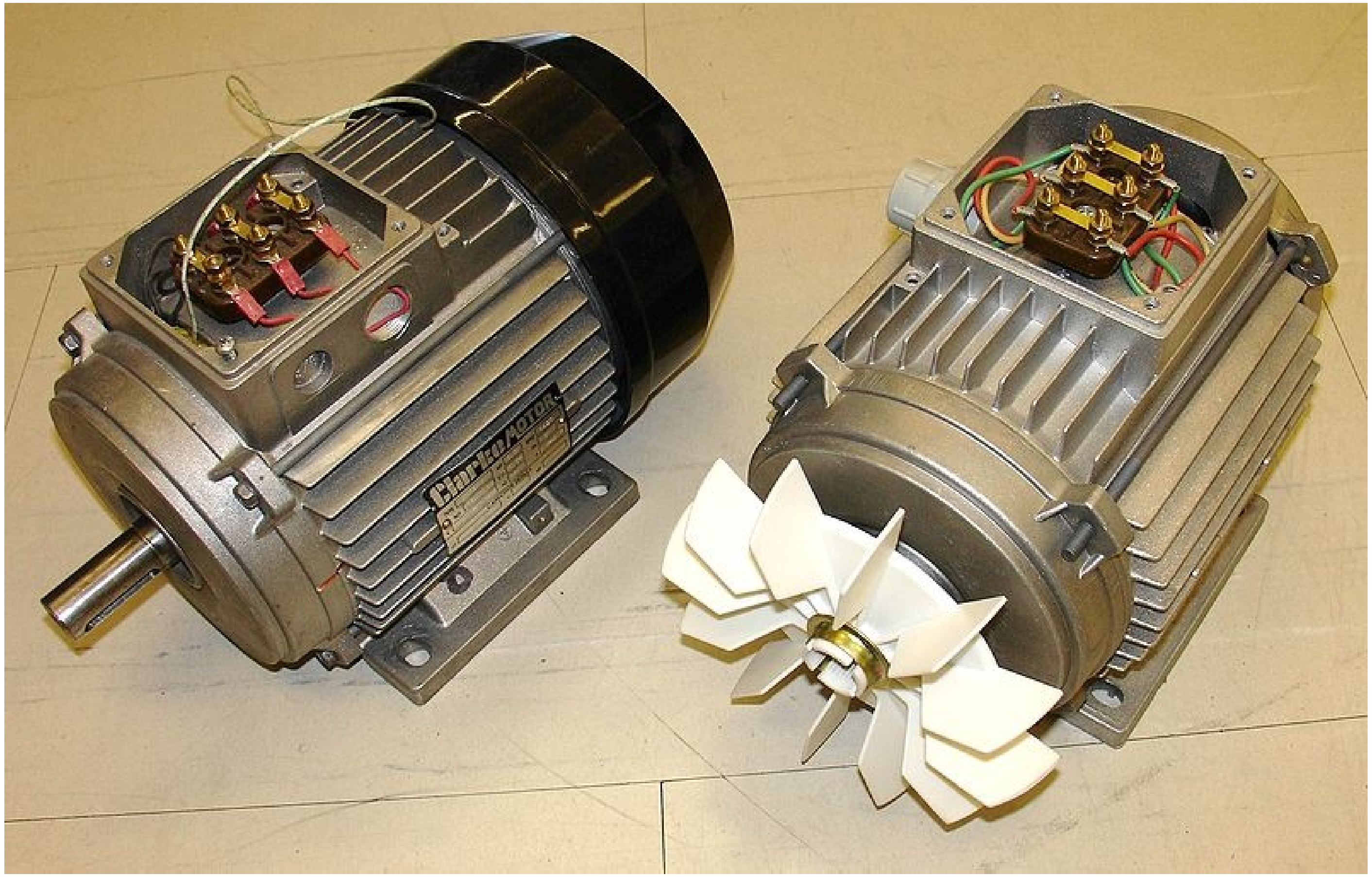
PLCs in rack

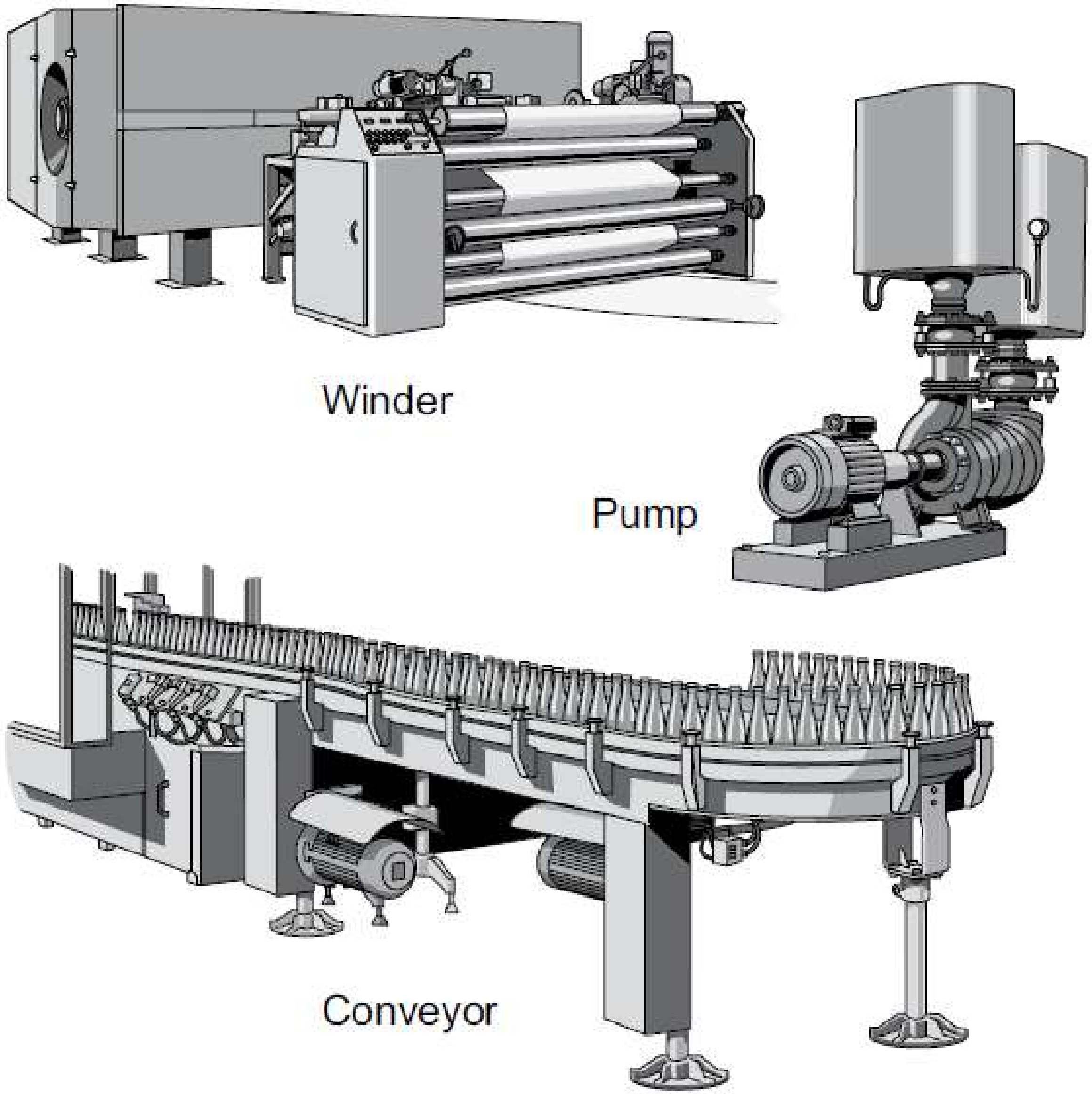
A personal computer displaying a graphic image of a real liquid-level process (a pumping, or "lift," station for a municipal wastewater treatment system) controlled by a PLC. The actual pumping station is located miles away from the personal computer display

Actuators

Drives

Electric Motors



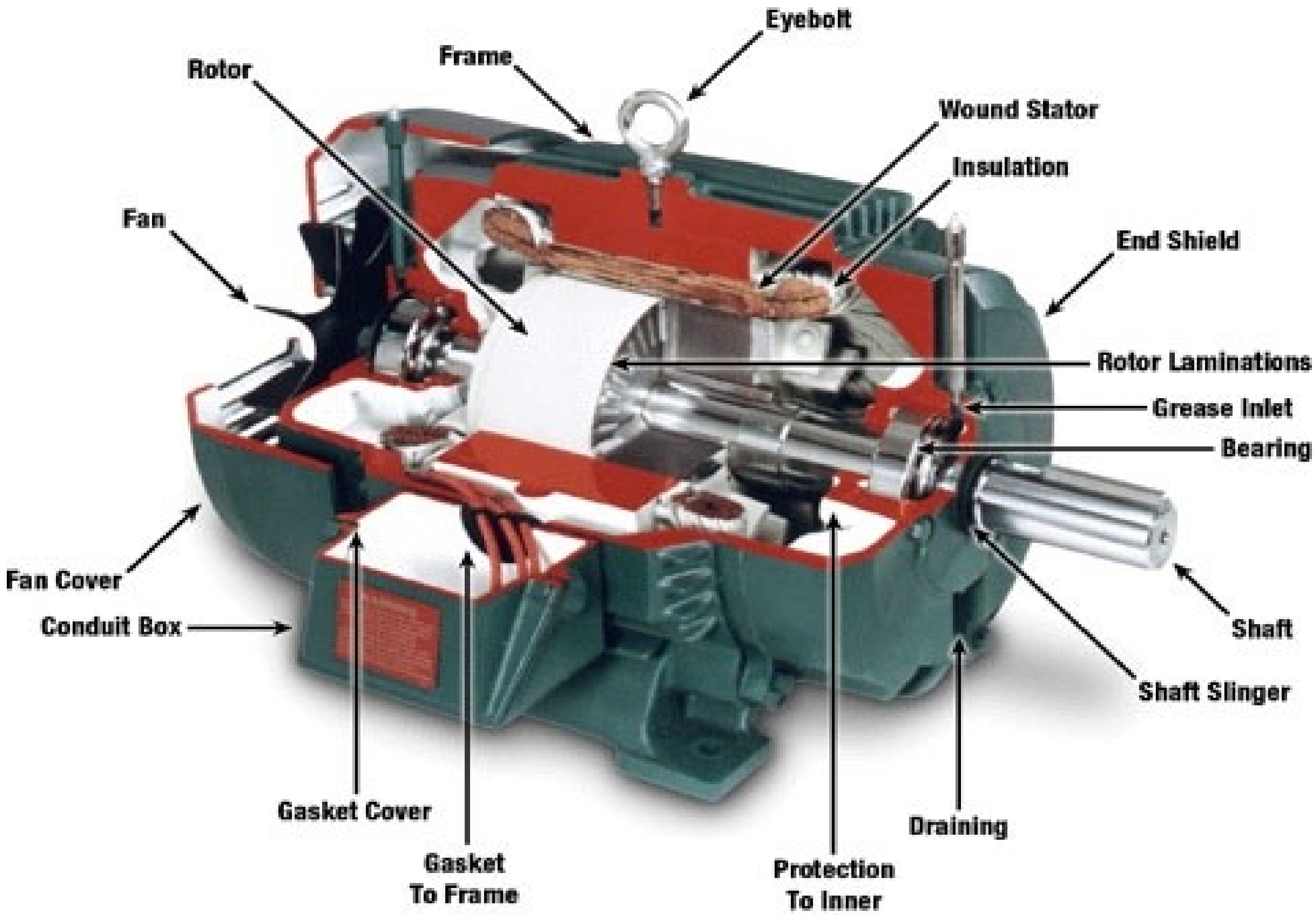

Winder
Pump
Conveyor

Electric motor

An **electric motor** is an electromechanical device that converts electrical energy into mechanical energy.

Most electric motors operate through the interaction of magnetic fields and current-carrying conductors to generate force. The reverse process, producing electrical energy from mechanical energy, is done by generators such as an alternator ; some electric motors can also be used as generators, for example, a traction motor on a vehicle may perform both tasks. Electric motors and generators are commonly referred to as electric machines.

Constructional features



Classification Electric motors

- Electric motors may be classified by the source of electric power, by their internal construction, by their application, or by the type of motion they give.

principles of operation

- In any electric motor, operation is based on simple electromagnetism. A Current-carrying conductor generates a magnetic field; when this is then placed in an external magnetic field, it will experience a force proportional to the Current in the conductor, and to the strength of the external magnetic field.
- The internal configuration of a DC motor is designed to harness the magnetic interaction between a current-carrying conductor and an external magnetic field to generate rotational motion.

Motors comparism

Type	Advantages	Disadvantages	Typical Application	Typical Drive
AC Induction	Long life High power high starting torque Rotation in-sync with freq. long-life (alternator)	Rotation slips from frequency Low starting torque More expensive	Fans Appliances Clocks Audio turntables tape drives	Uni/Poly-phase AC
Stepper DC	Precision positioning High holding torque	Slow speed Requires a controller	Positioning in printers and floppy drives	Multiphase DC
Brushless DC electric motor	Long lifespan low maintenance High efficiency	High initial cost Requires a controller	Hard drives CD/DVD players Electric vehicles	Multiphase DC
Brushed DC electric motor	Low initial cost Simple speed control	High maintenance (brushes) Limited lifespan	Treadmill exercisers Automotive starters Toys and some appliances	Direct (PWM)

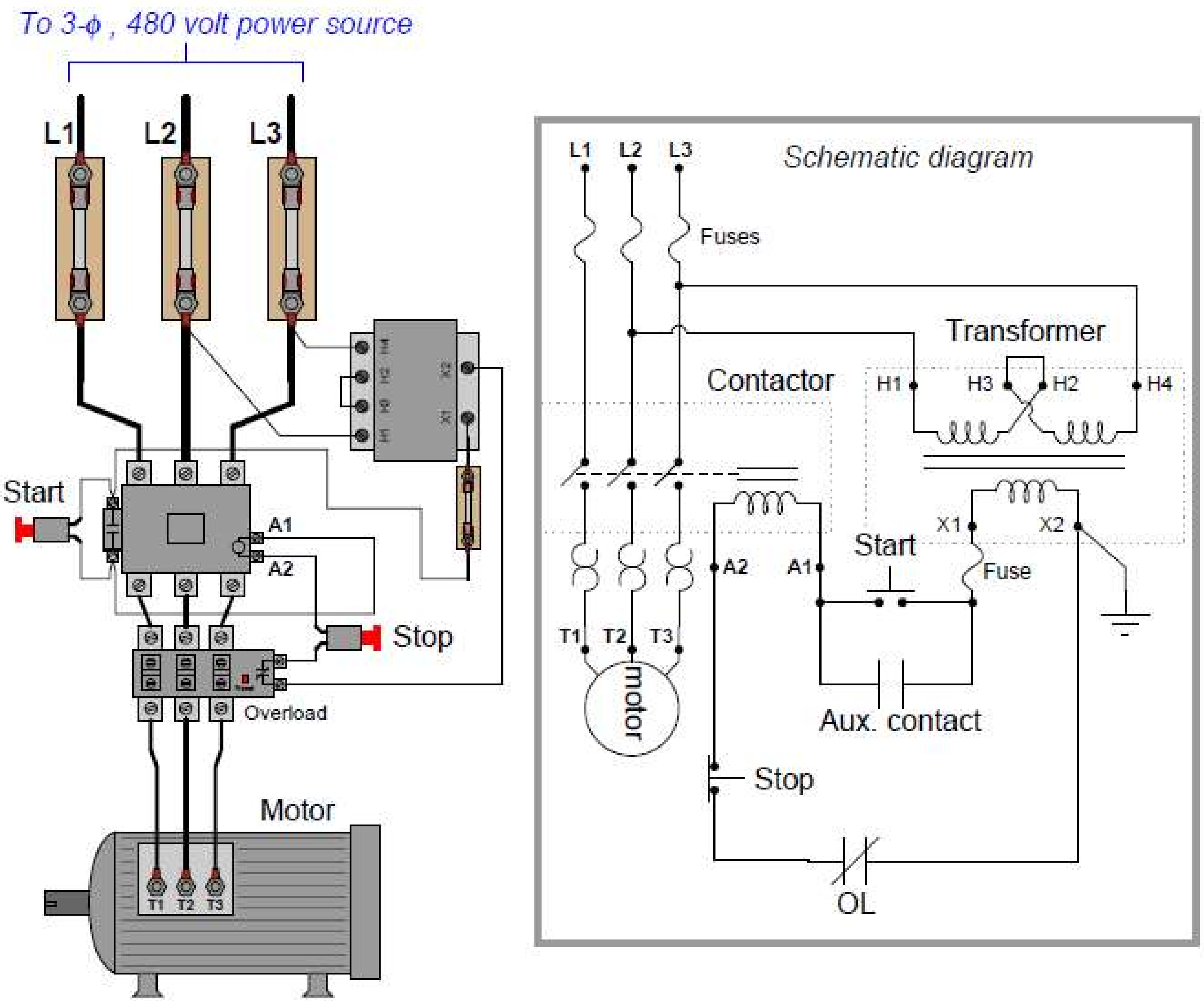
Motor Selection

1. Cost
2. Thermal capacity
3. Efficiency
4. Torque-speed profile
5. Acceleration
6. Power density, volume of the motor
7. Ripple, cogging torque
8. Peak torque capability
9. Suitability for hazardous environment
10. Availability of spare parts

Control issues in motors

The basic issues in motor control are

- How to start
- How to vary the speed
- Which direction to run
- How to stop
- *These will be discussed with reference to DC and AC motors (class discussion)*

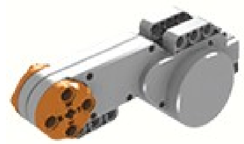


To 3-ϕ , 480 volt power source
L1
L2
L3
Start
A1
A2
Stop
Overload
Motor
T1
T2
T3
Schematic diagram
Fuses
Transformer
Contactor
H1
H3
H2
H4
X1
X2
Fuse
Start
A2
A1
motor
Aux. contact
Stop
OL

Tasks

- State what motors are
- Explain how motors achieve rotation
- Identify the functional parts of a motor and state their importance
- How are motors classified
- State the advantages; disadvantages; limitations and applications of DC-series ; shunt & compound ; AC-synchronous and asynchronous

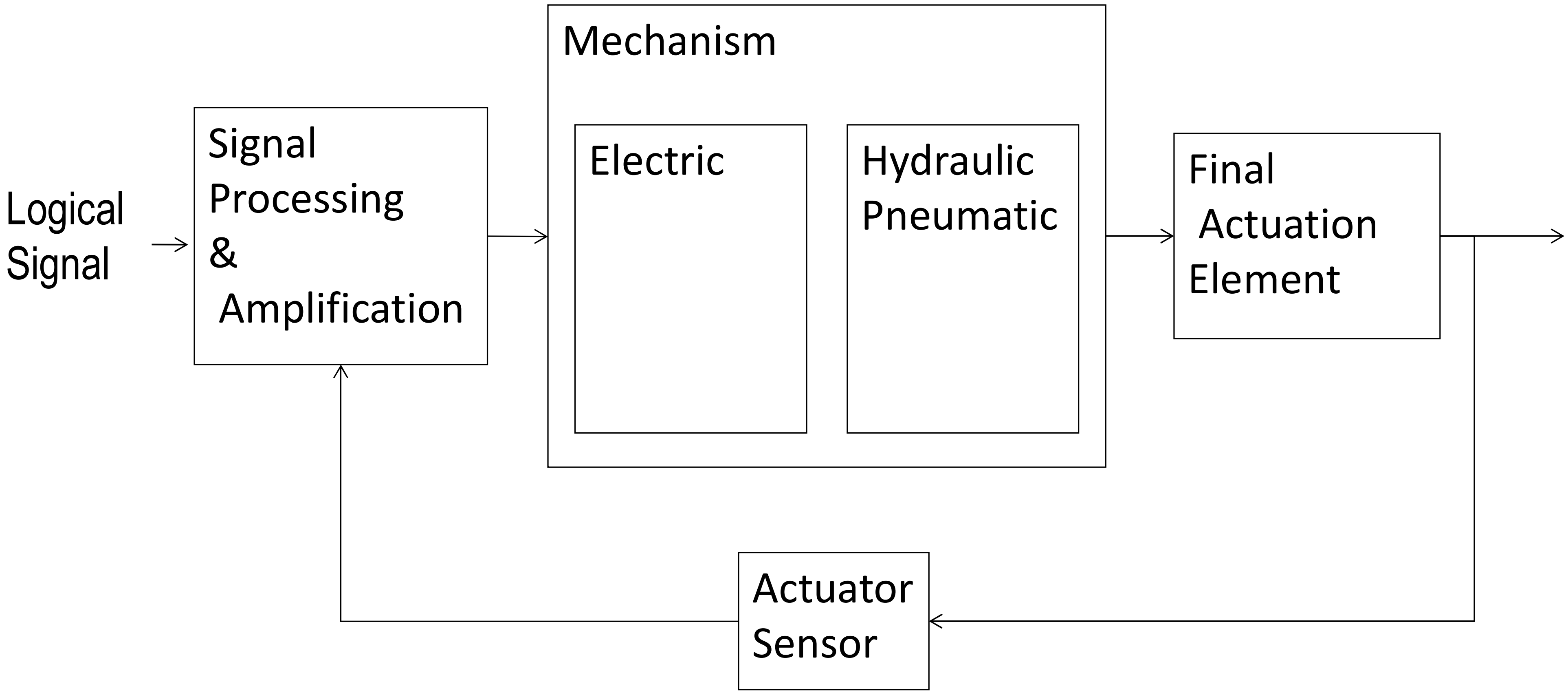
Actuators



Hardware devices that convert a controller command signal into a change in a physical parameter

- The change is usually mechanical (e.g., position or velocity)
- An actuator is also a transducer because it changes one type of physical quantity into some alternative form
- An actuator is usually activated by a low-level command signal, so an amplifier may be required to provide sufficient power to drive the actuator

Actuators


Logical Signal
Signal Processing & Amplification
Mechanism
Electric
Hydraulic Pneumatic
Final Actuation Element
Actuator Sensor

Types of Actuators

1. Electrical actuators
   - Electric motors
     - DC servomotors
     - AC motors
     - Stepper motors
   - Solenoids

2. Hydraulic actuators
   - Use hydraulic fluid to amplify the controller command signal
3. Pneumatic actuators
   - Use compressed air as the driving force

ACTUATORS

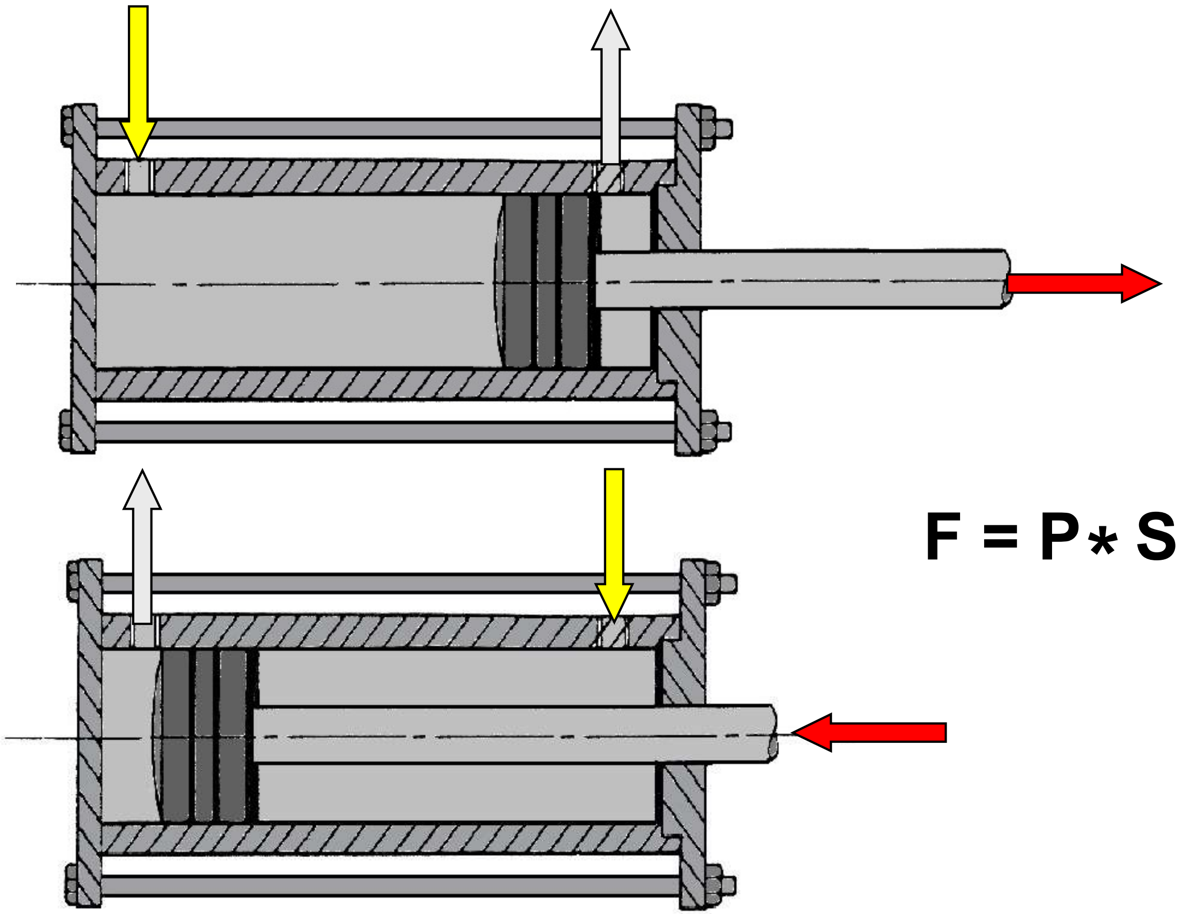
1 – Pneumatic actuators
    Cylinders

2 – Hydraulic actuadors
    Cylinders
    Motors

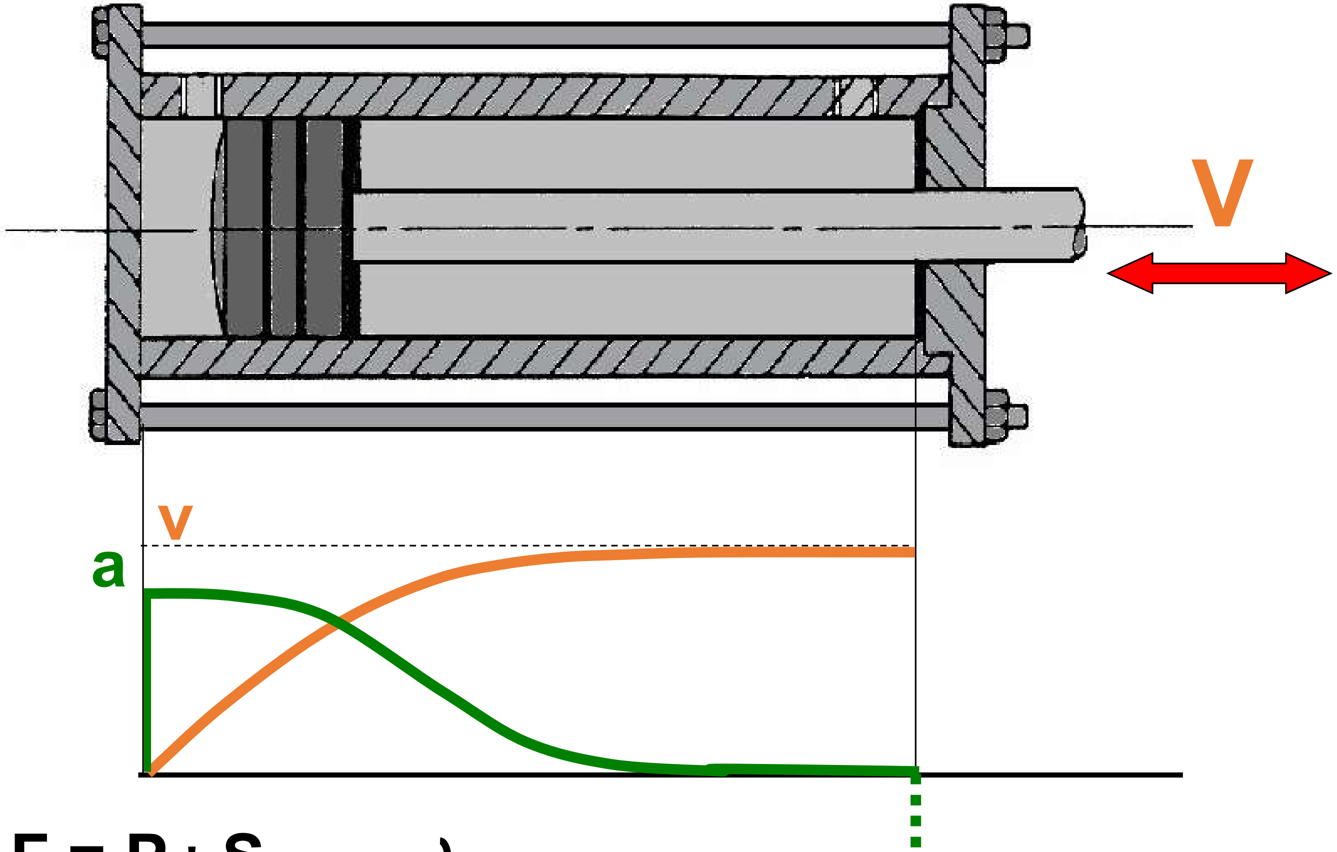
3 – Electrical actuators
    Dc motors.
    Ac motors
    Steeper motors.

1 – Pneumatic actuators (cylinders)

Double effect pneumatic cylinders

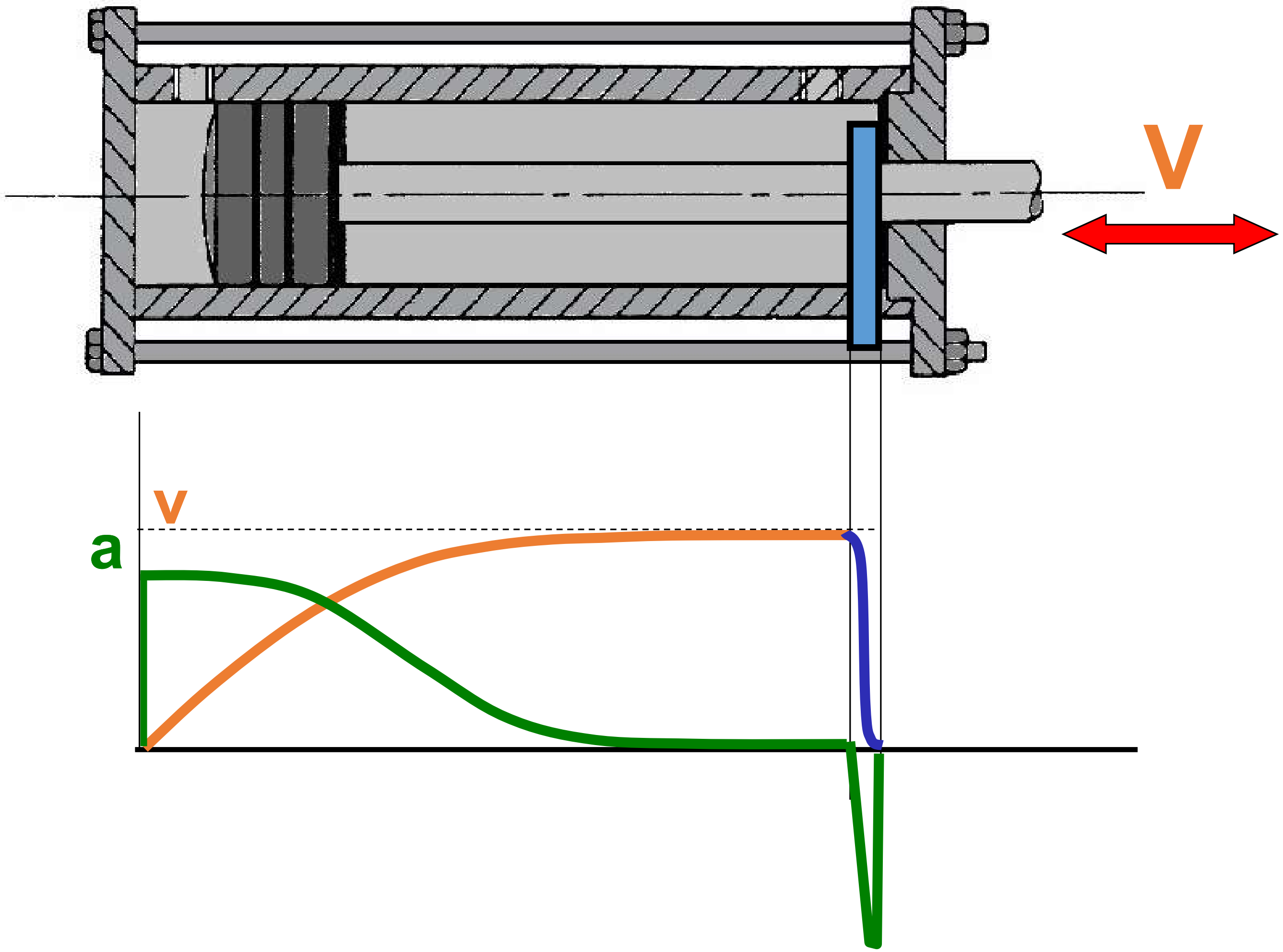


$$F = P * S$$

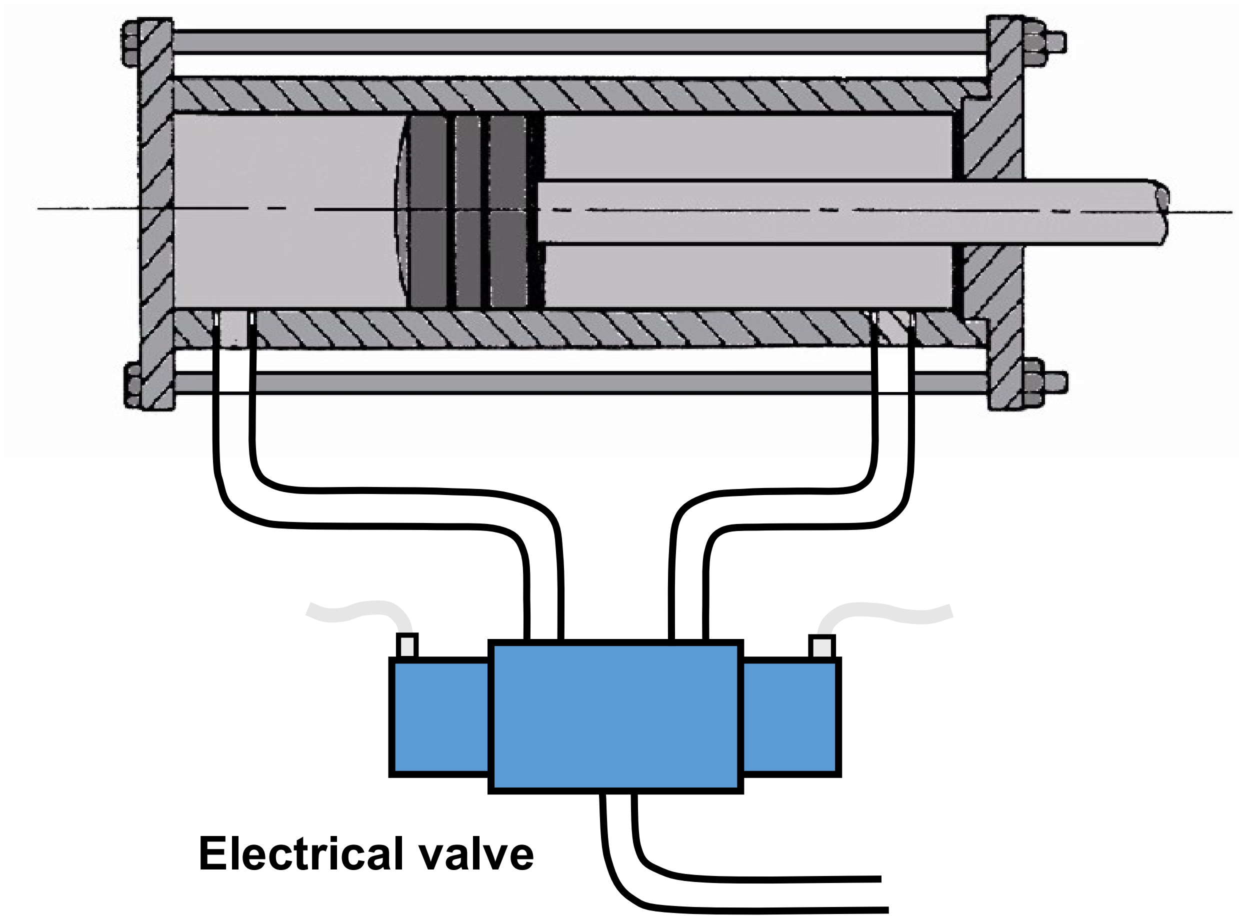


$$\left.\begin{aligned} F &= P * S \\ F - F_f &= M * a \\ F_f &= k * v^2 \end{aligned}\right\} \Rightarrow a = \frac{P * S - k * v^2}{M}$$

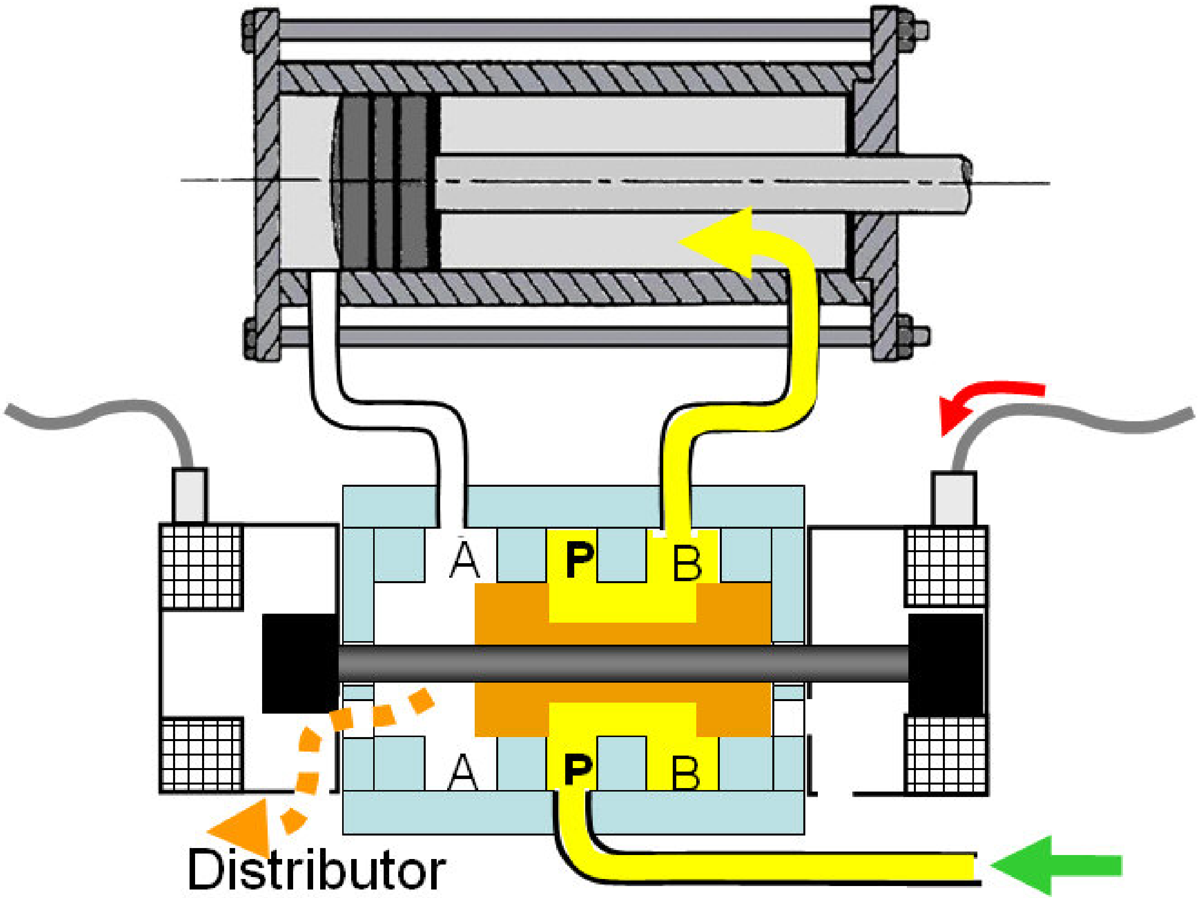
Speed is not controllable. The cylinder maximum speed is achieved when friction forces (kv^2) equal those that produce the advancing movement (F = P.S), and a = 0.



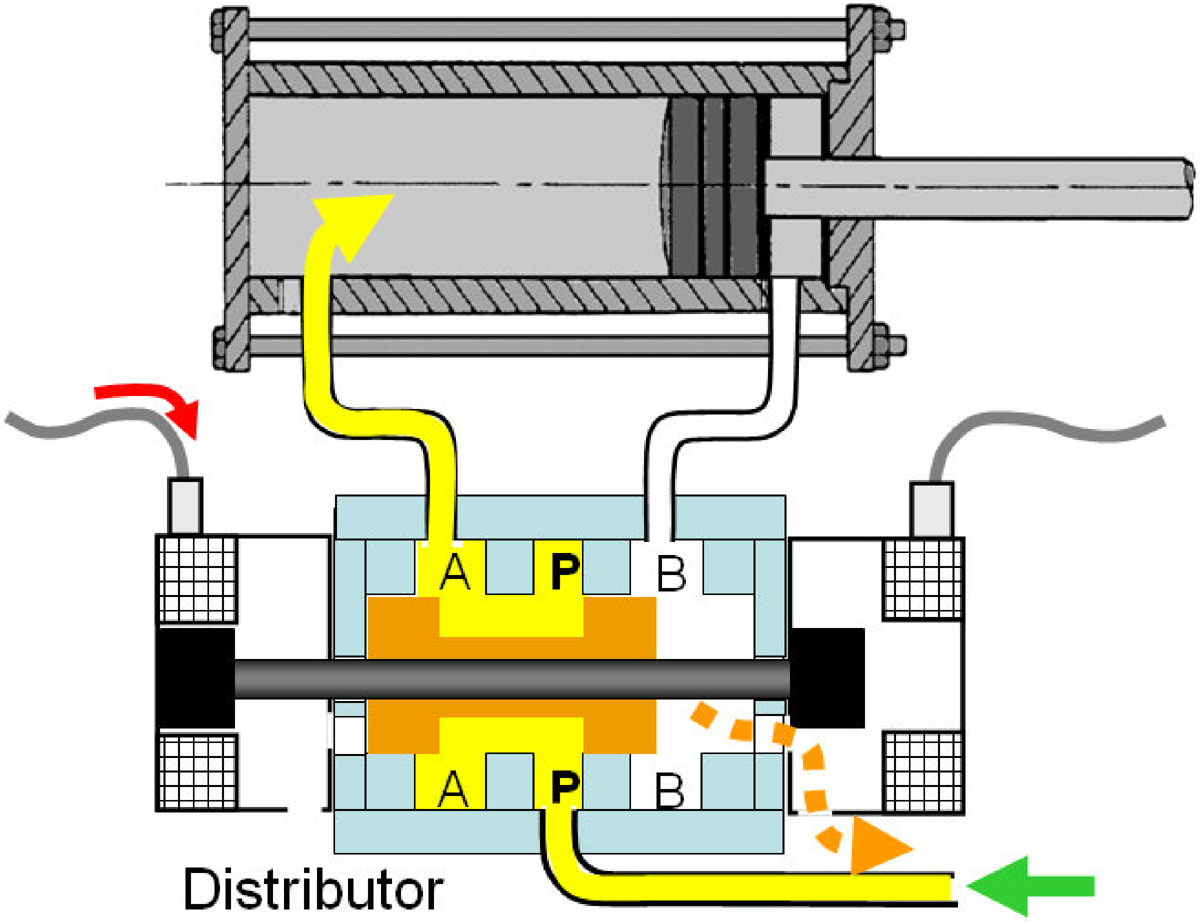
- The impact produced when reaching the end of the run is reduced using a shock absorber.



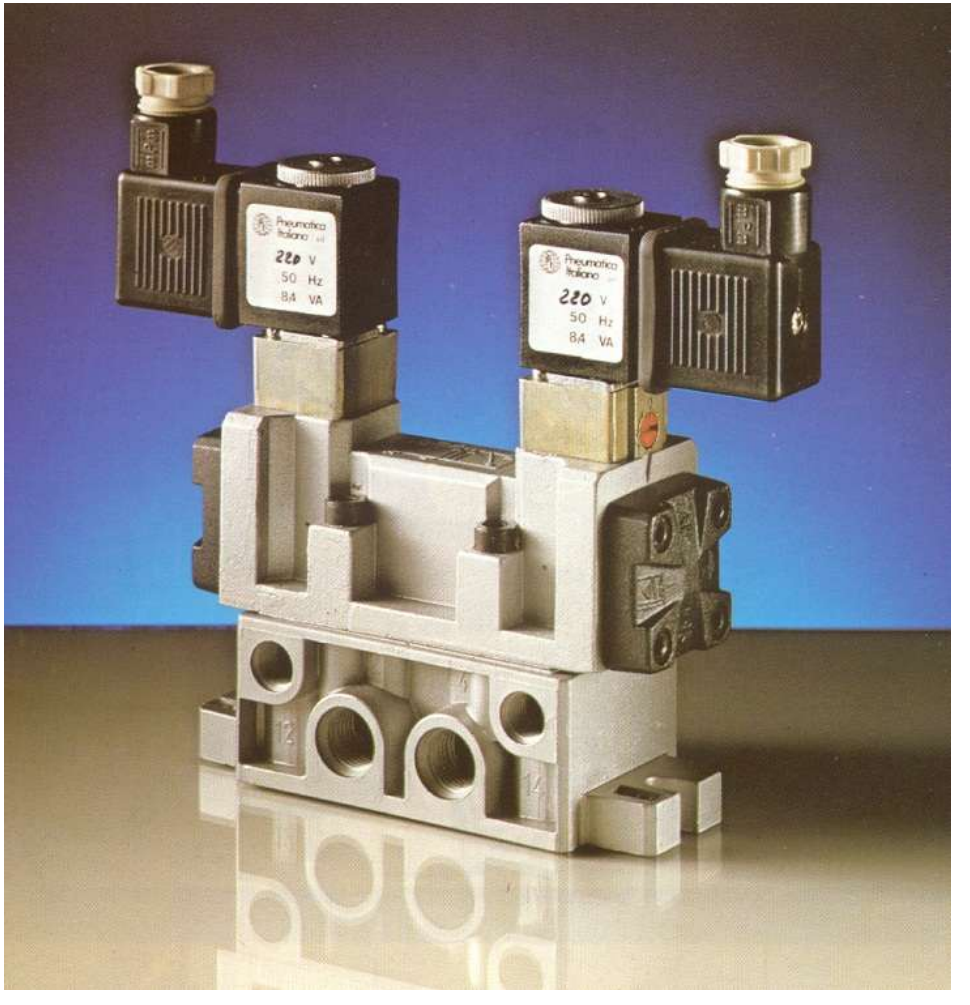
- Electrical valve: the hydraulic-electrical interface



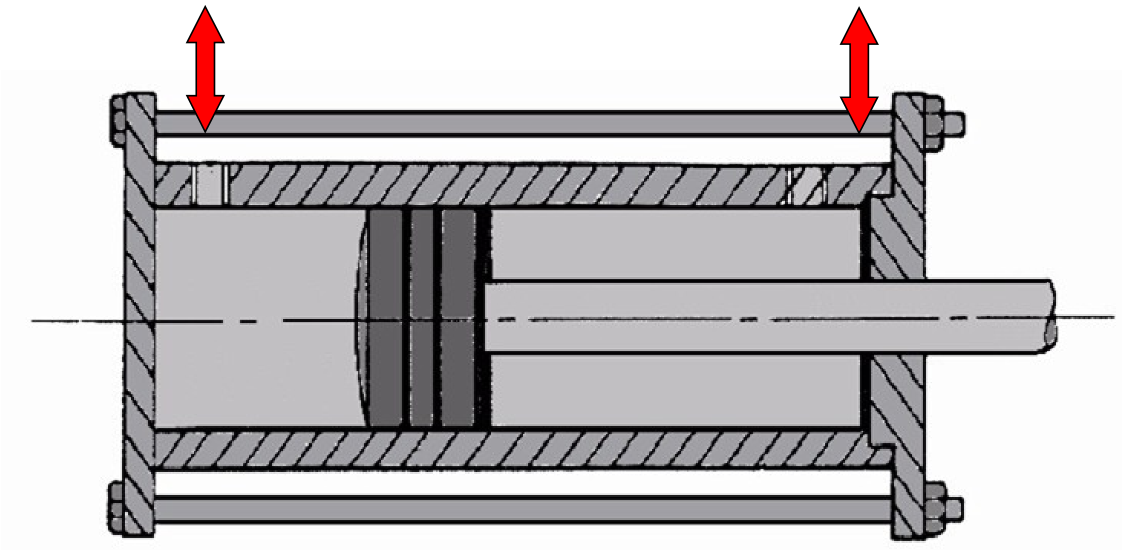
A
P
B
A
P
B
Distributor

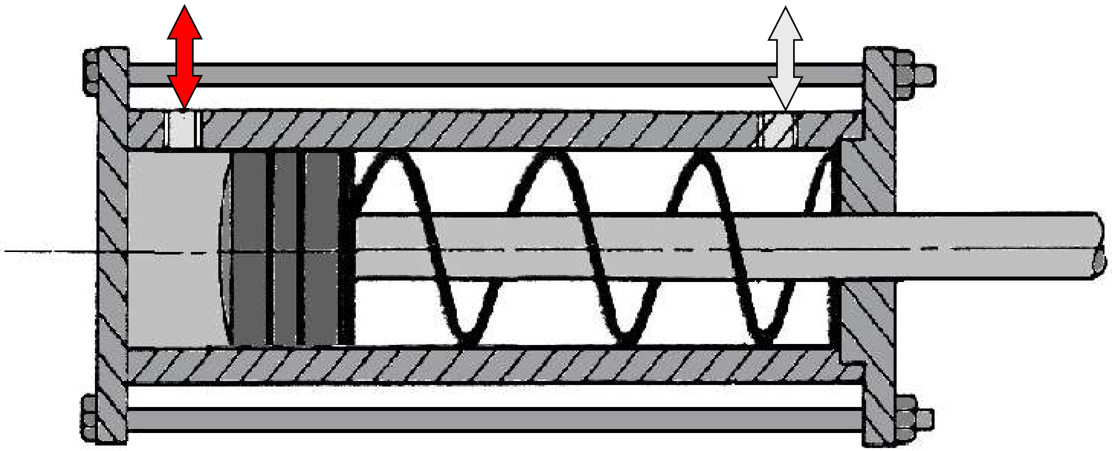


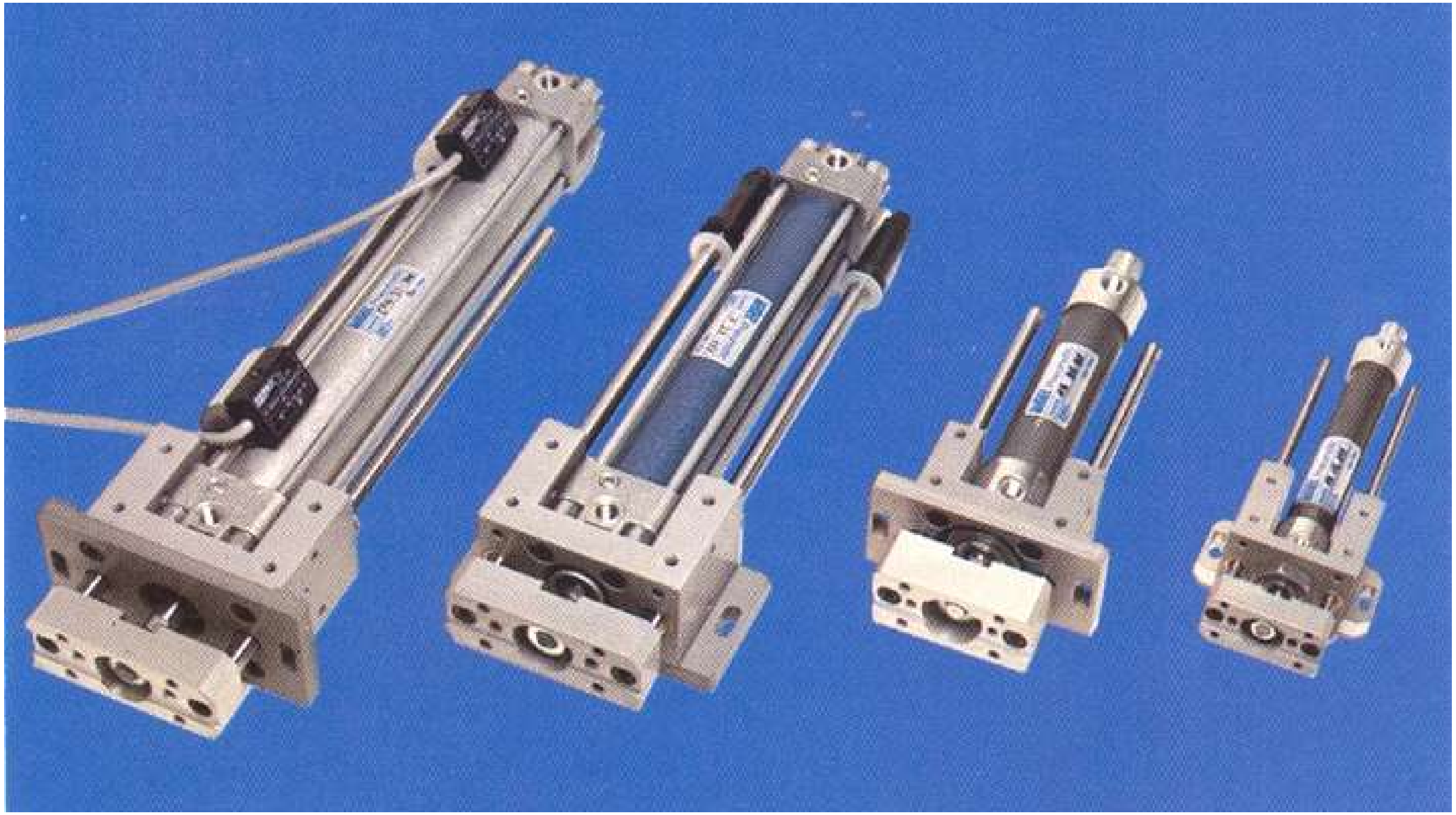
Distributor



Distributor

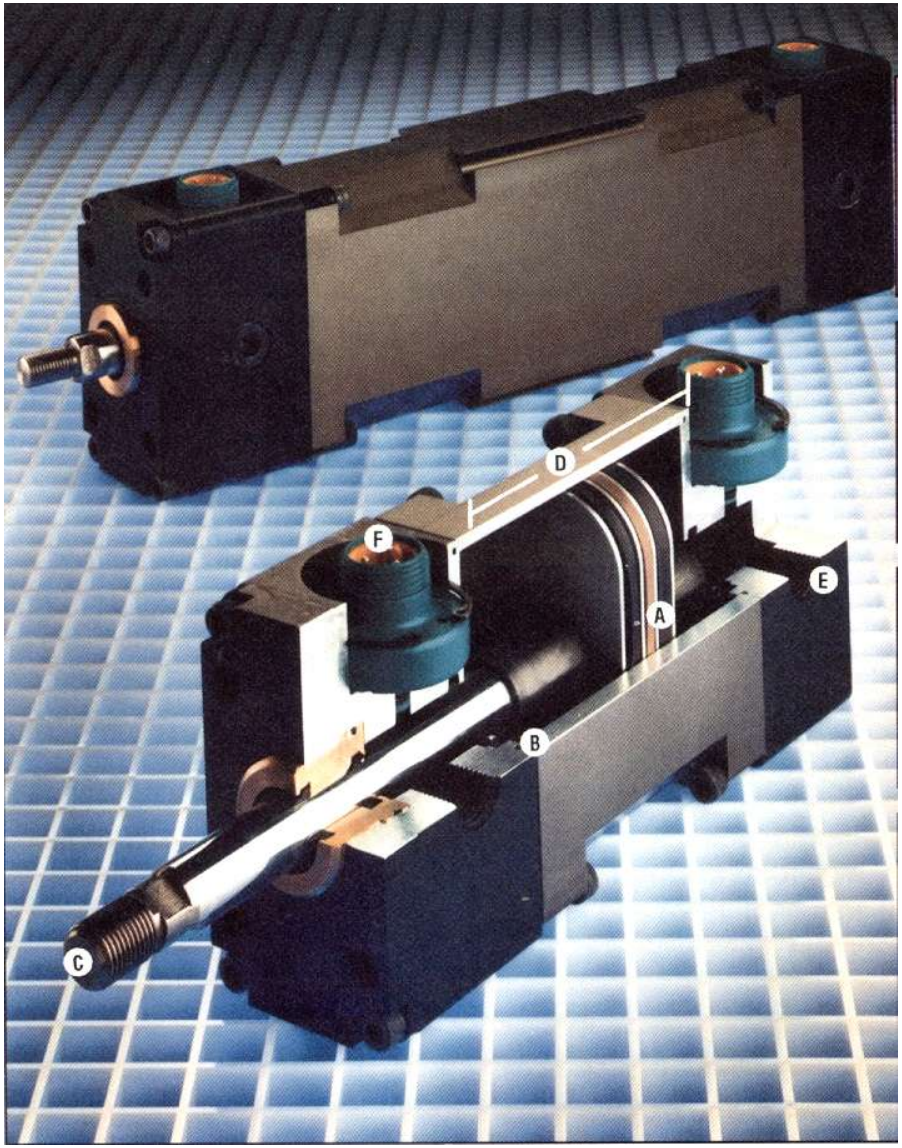

Double effect cylinders


Single effect cylinders

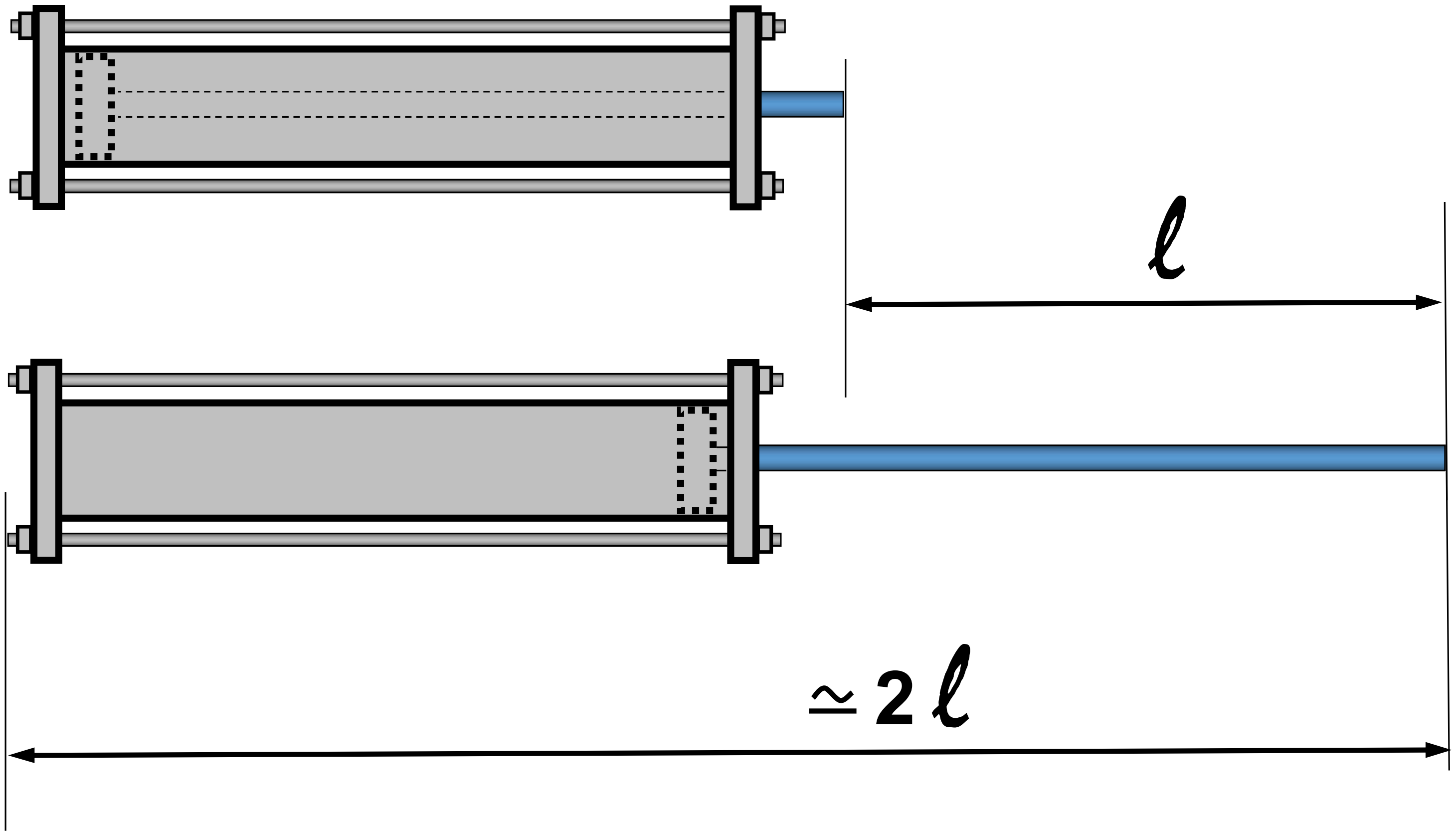


Example of commercial pneumatic cylinders

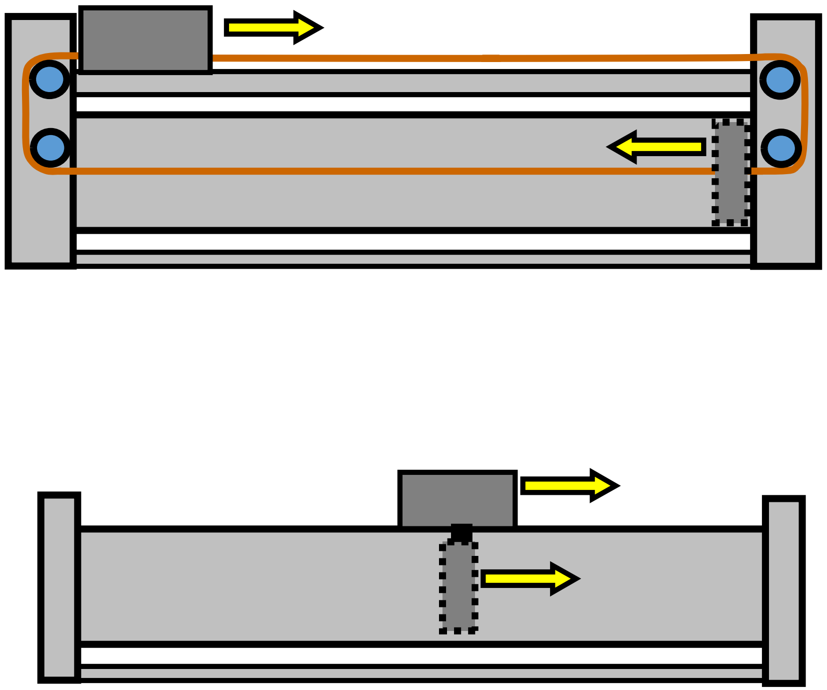
(Lateral guides to prevent axial rotation)



Oval pistons to prevent the rotation of the axis avoiding the need of auxiliary guides

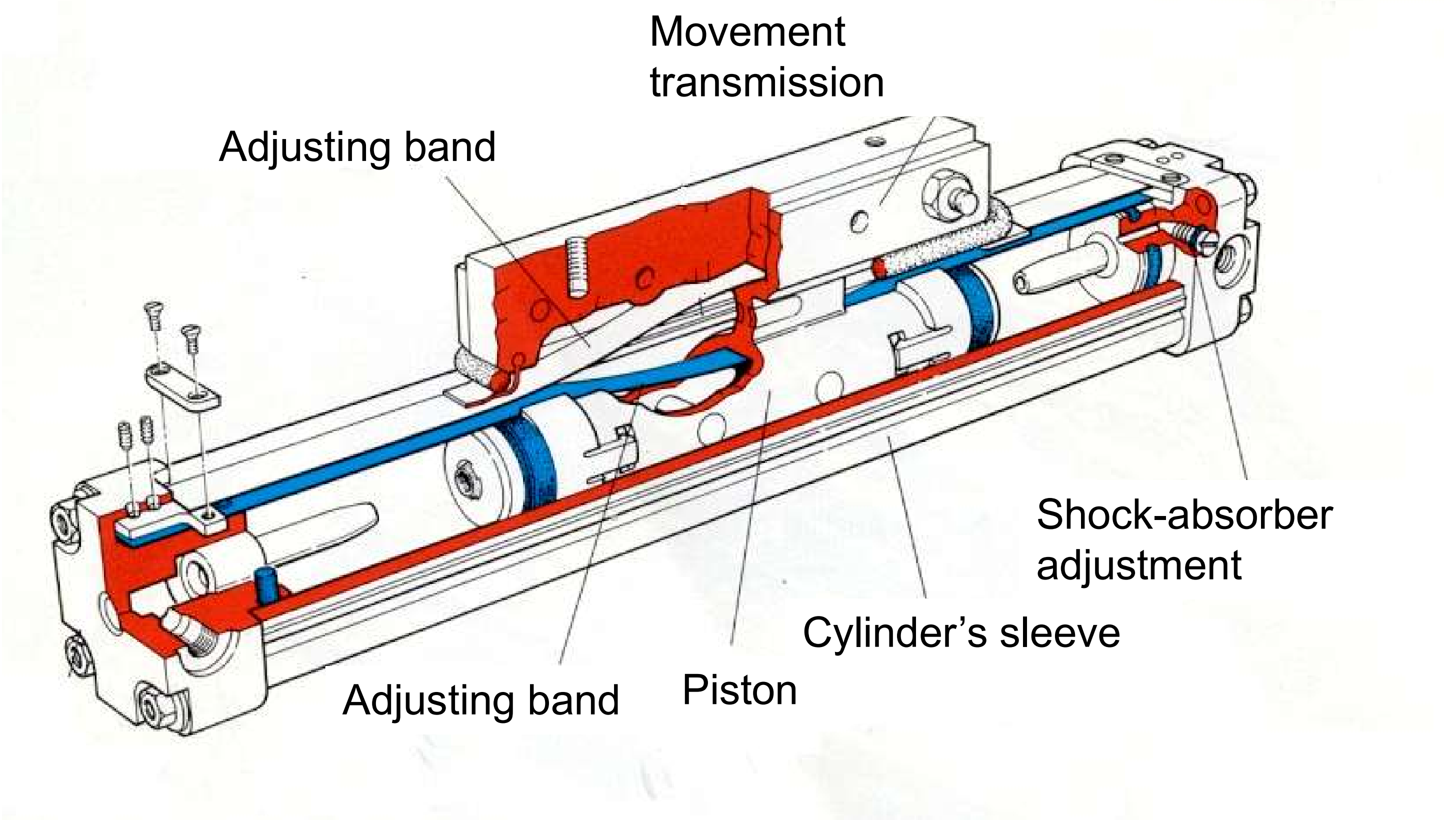


Classical cylinders drawbacks: a displacement of length ℓ requires an additional length ℓ.



$$\Delta \ell \simeq \ell$$

Solutions to reduce the occupied space


Movement transmission
Adjusting band
Shock-absorber adjustment
Cylinder’s sleeve
Adjusting band
Piston


③
④
⑤
⑧
②
* ⑨
⑦
⑥
①
⑪

Pneumatic actuators (cylinders)

- Economic
- Reliable
- High operation speed
- Operation at constant force
- Resistant to overloads

- No speed control
- Poor position speed
- Noisy operation

ACTUATORS

1 – Pneumatic actuators

Cylinders

Motors

2 – Hydraulic actuadors

Cylinders

Motors

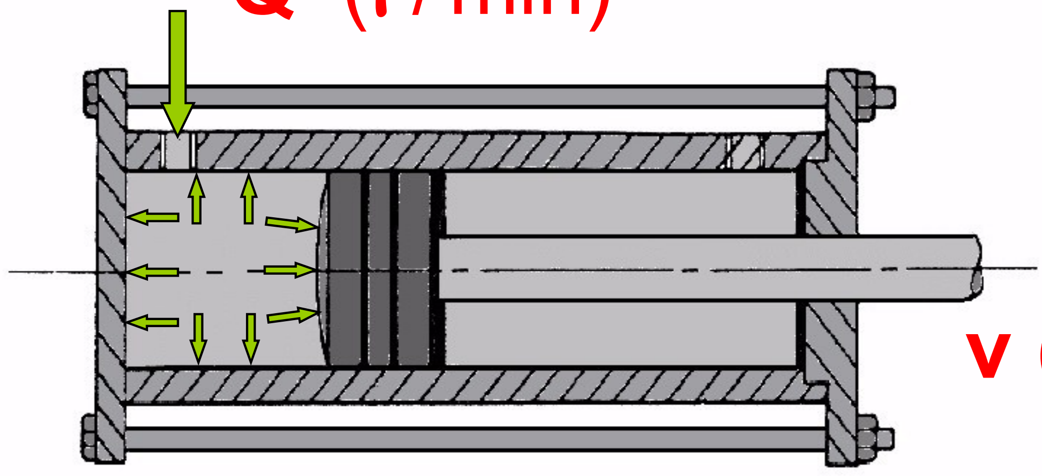
3 – Electrical actuators

Dc motors.

Ac motors

Steeper motors.

2 – Hydraulic actuators (cylinders)



Energy source: oil pressurized between 20 and 300 bars.

$$\mathbf{F = P * S} \qquad \text{If } \mathbf{P} \uparrow\uparrow \;\rightarrow\; \mathbf{F} \rightarrow \infty$$

- Controllable position
- Controllable speed

ACTUATORS

1 – Pneumatic actuators

Cylinders

Motors

2 – Hydraulic actuadors

Cylinders

Motors

3 – Electrical actuators

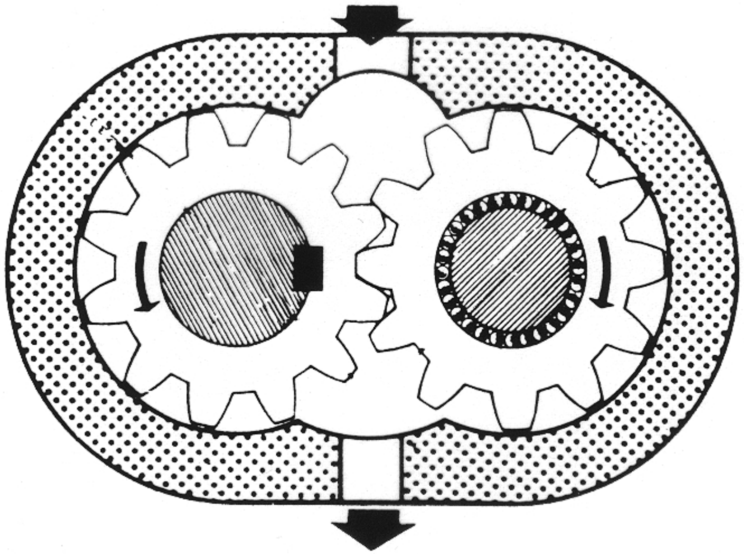
Dc motors.

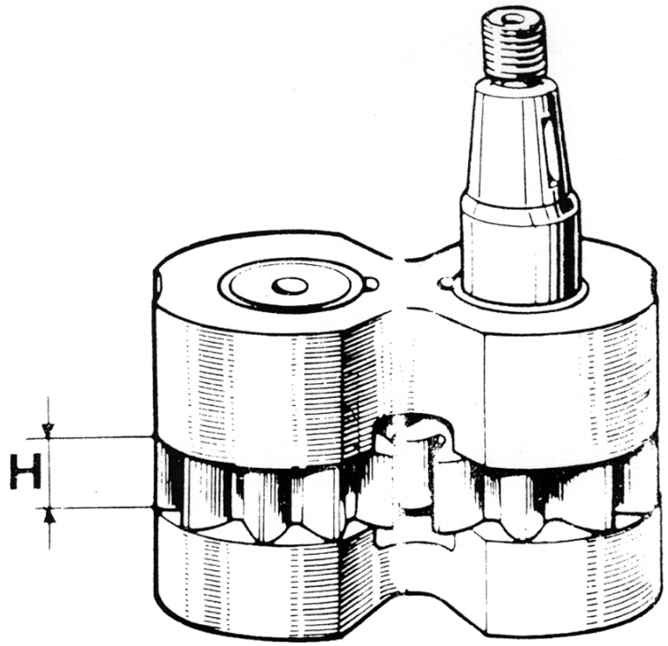
Ac motors

Steeper motors.

Hydraulic pumps and motors

(Kind of gears)

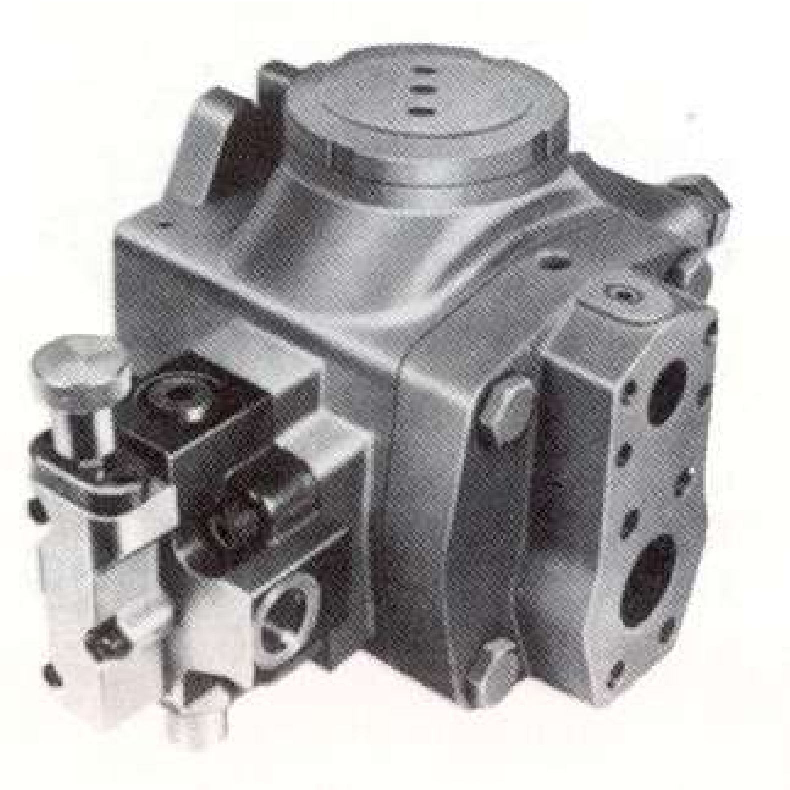




Fix caudal

Hydraulic pumps and motors

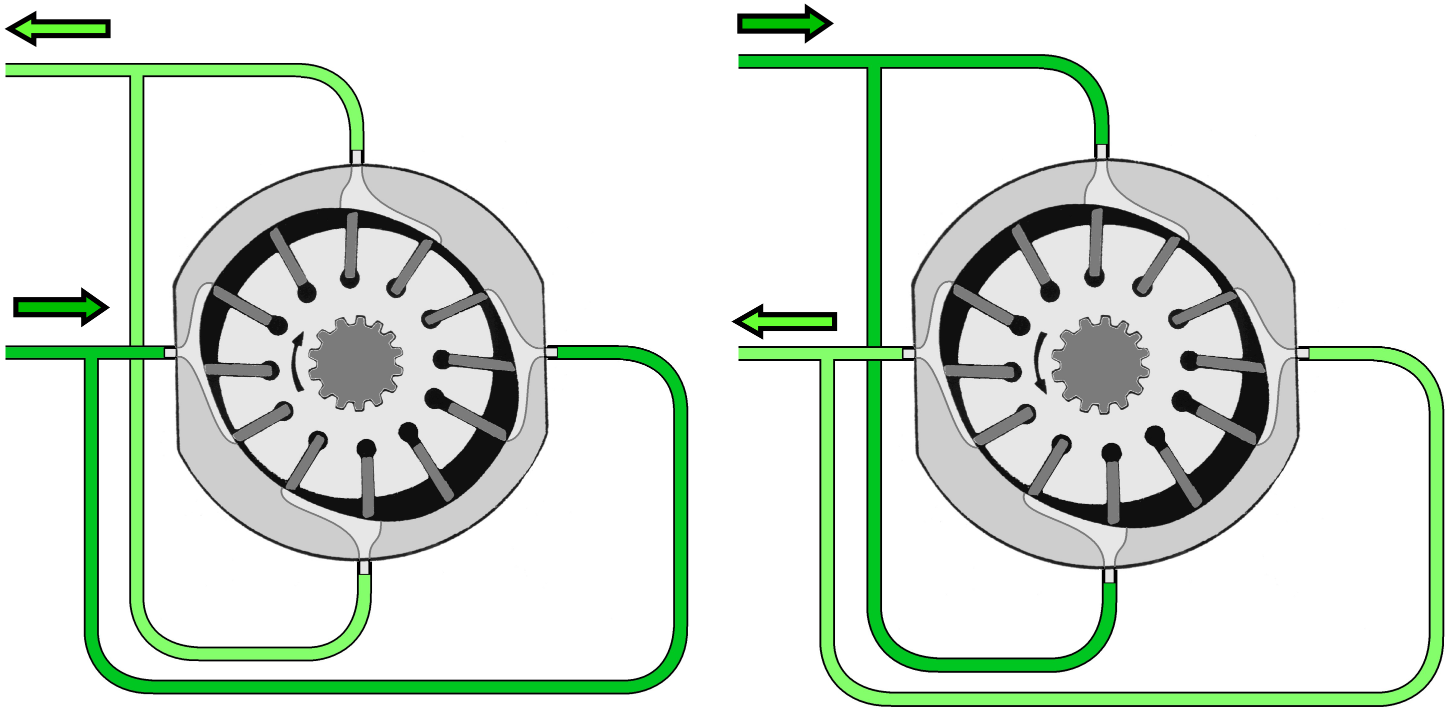
(Kind of gears)



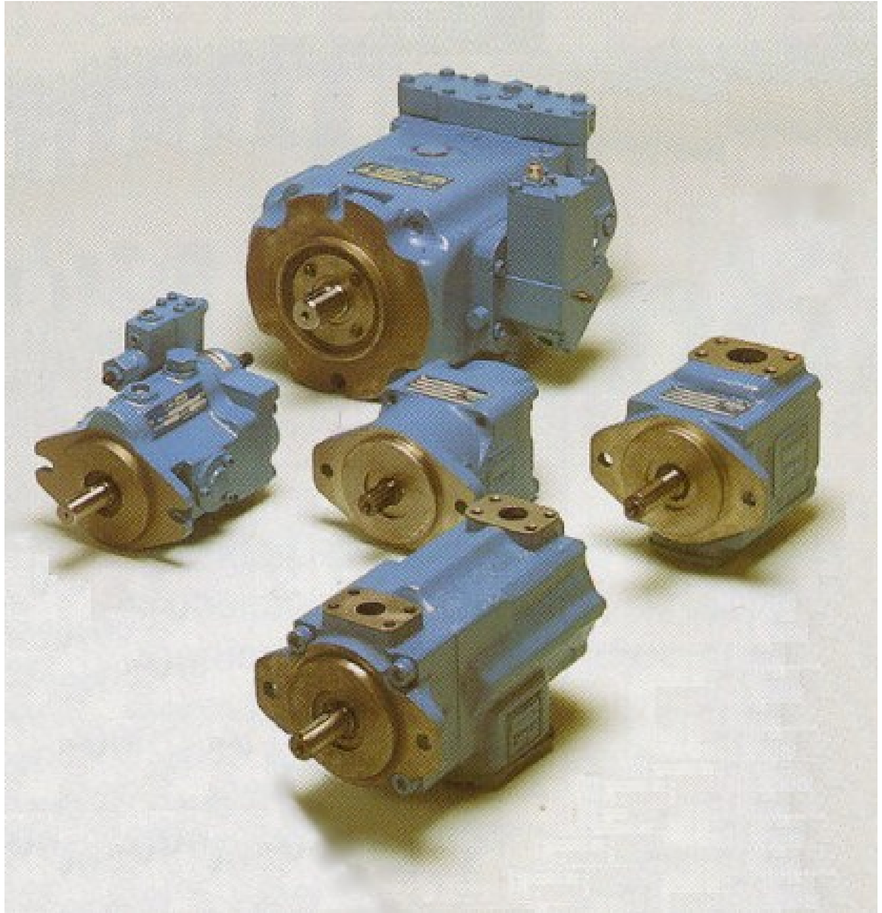
Fix caudal

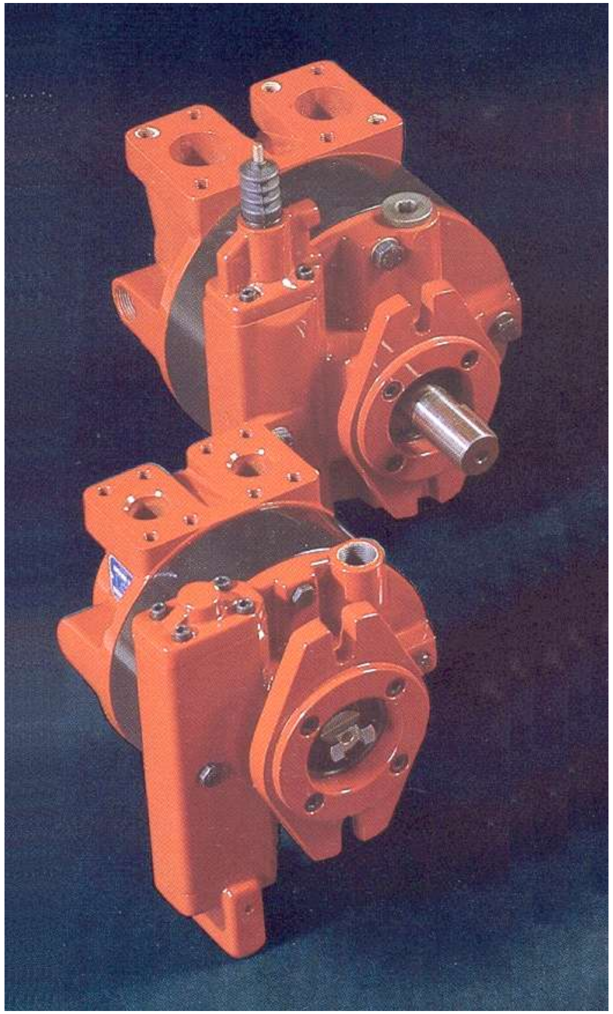
Hydraulic pumps and motors

(Kind of blades)



Hydraulic pumps or motors



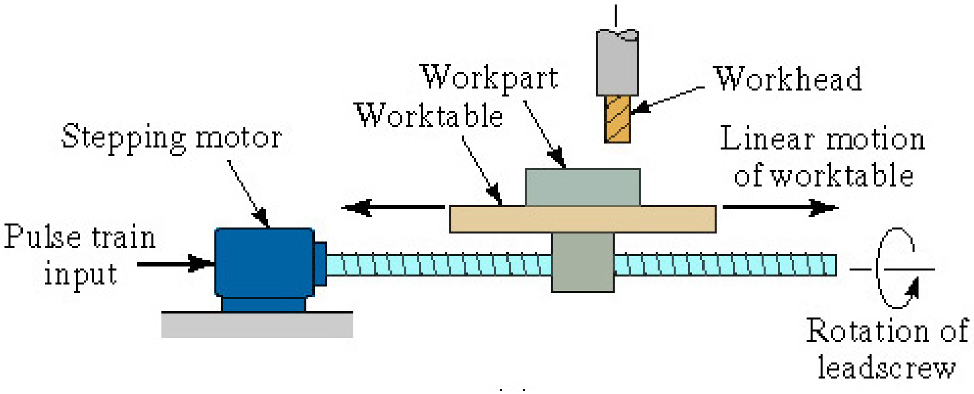


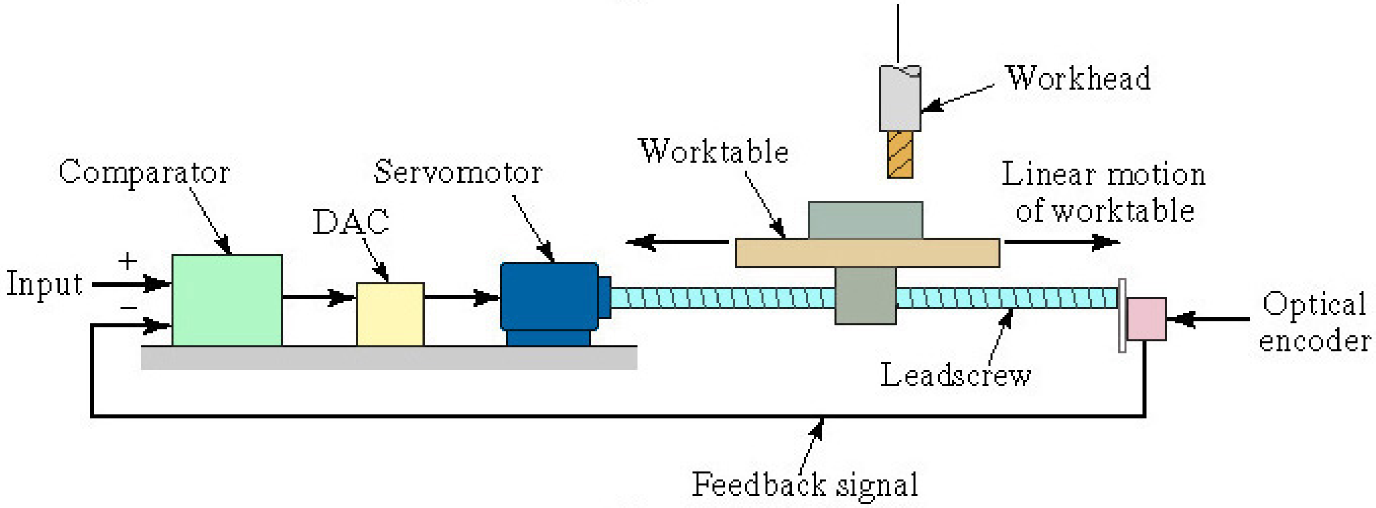
Hydraulic actuators

- Economic
- Reliable
- Able to support heavy loads
- Resistant to overloads

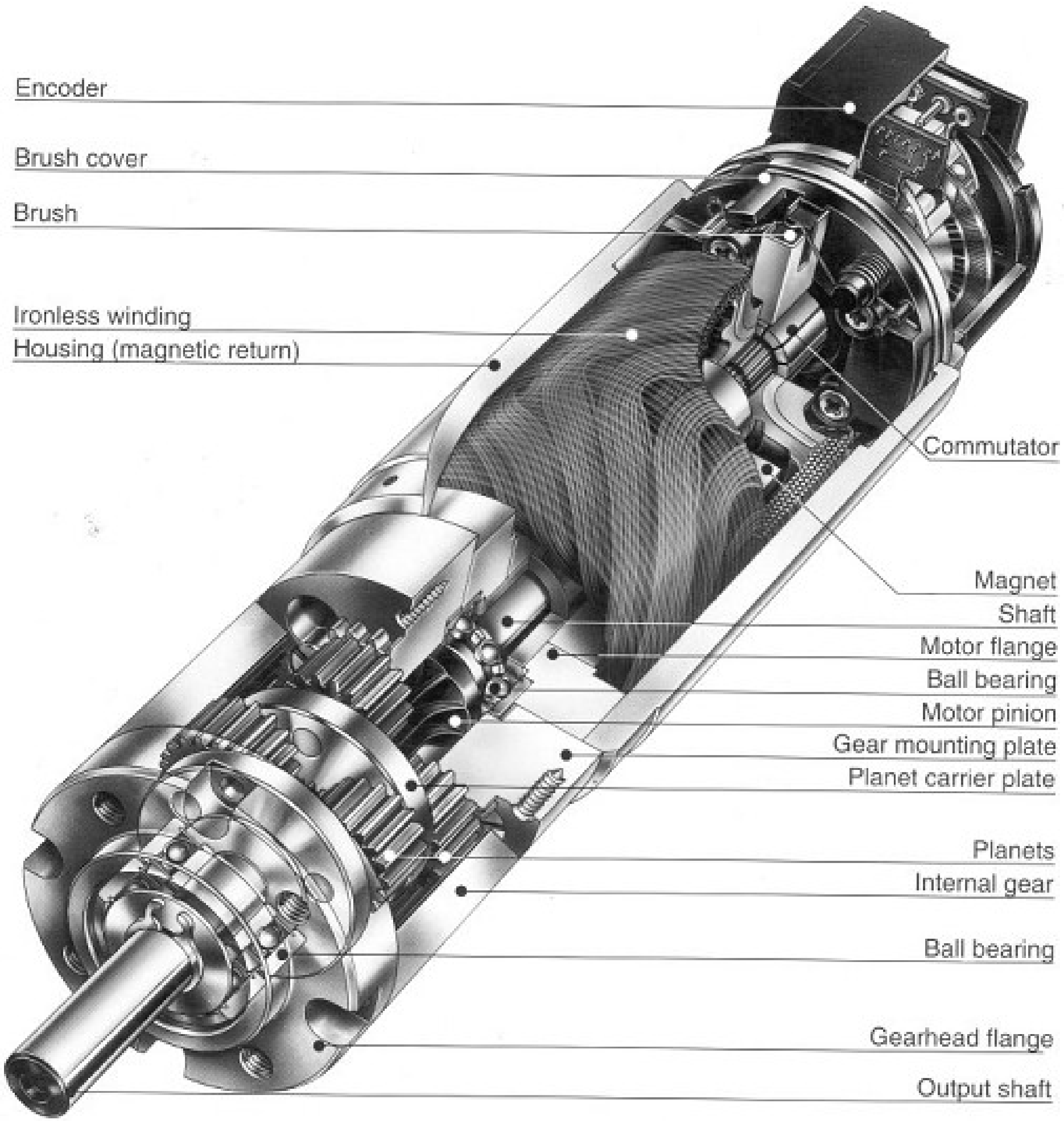
- Low working speed
- Hydraulic group noisy in operation
- Possible oil leakage

Stepper motor and Servomotor



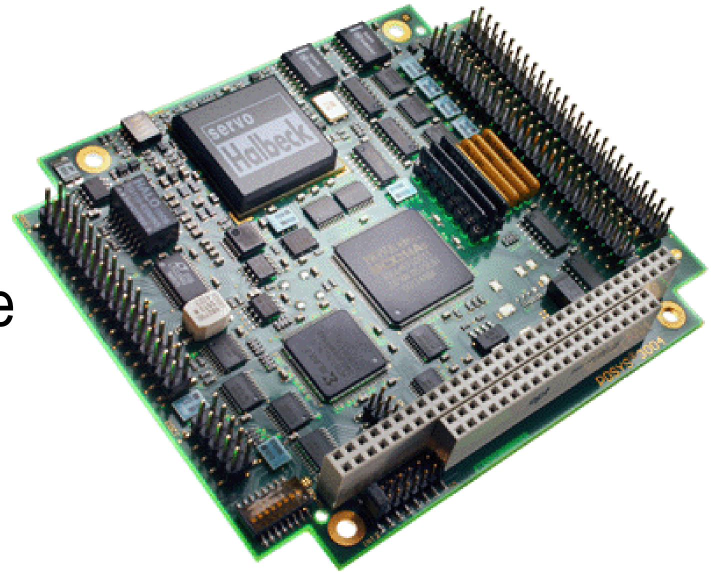


Servo Motor

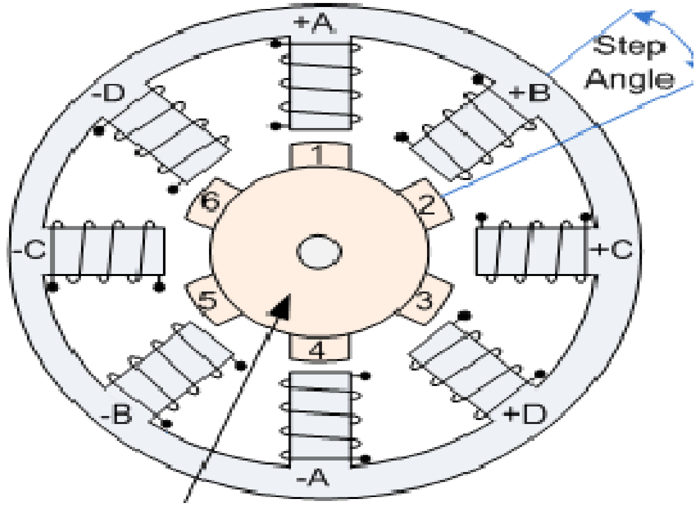

Encoder
Brush cover
Brush
Ironless winding
Housing (magnetic return)
Commutator
Magnet
Shaft
Motor flange
Ball bearing
Motor pinion
Gear mounting plate
Planet carrier plate
Planets
Internal gear
Ball bearing
Gearhead flange
Output shaft

Motor Controllers

The POSYS® 3004 (Designed & Made in Germany) is a PC/104 form factor board dedicated to high performance motion control applications with extensive interpolation functionality. The POSYS® 3004 is designed to control up to 4 axes of servo and stepper motors and provides **hardware linear, circular, Bit Pattern and continuous interpolation** which allow to perform the **most complex motion profiles**. Update rates per axis do not exist as each axis runs in absolute real-time mode simultaneously which makes these boards to one of the best performing motion controllers for up to 4 axes in the market.



Stepper Motors



Step angle is given by: : $$\alpha = \frac{360}{n_s}$$

where n_s is the number of steps for the stepper motor (integer)

Total angle through which the motor rotates (A_m) is given by: $$A_m = n_p \alpha$$

where n_p = number of pulses received by the motor.

Angular velocity is given by: $$\omega = \frac{2\pi f_p}{n_s}$$ where f_p = pulse frequency

Speed of rotation is given by: $$N = \frac{60 f_p}{n_s}$$

Example

- A stepper motor has a step angle = 3.6°. (1) How many pulses are required for the motor to rotate through ten complete revolutions? (2) What pulse frequency is required for the motor to rotate at a speed of 100 rev/min?

Solution

$$\alpha = \frac{360}{n_s}$$

(1) 3.6° = 360 / n_s;   3.6° (n_s) = 360; n_s = 360 / 3.6 = 100 step angles

$$A_m = n_p \alpha$$

(2) Ten complete revolutions: 10(360°) = 3600° = A_m

Therefore n_p = 3600 / 3.6 = 1000 pulses

$$N = \frac{60 f_p}{n_s}$$

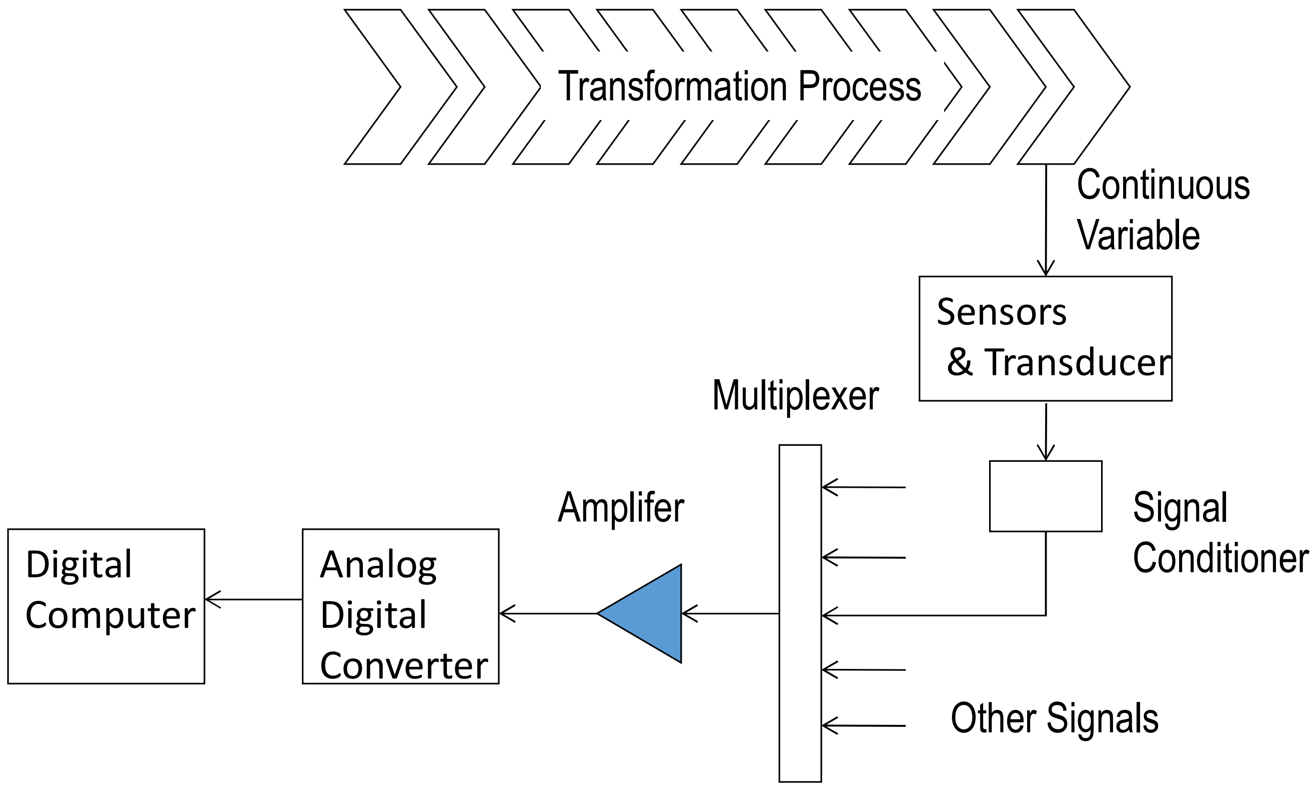
Where N = 100 rev/min:

100 = 60 f_p / 100

10,000 = 60 f_p

f_p = 10,000 / 60 = 166.667 = 167 Hz

Hardware Devices in Analog-to-Digital Conversion



Features of an ADC

- Sampling rate – rate at which continuous analog signal is polled e.g. 1000 samples/sec
- Quantization – divide analog signal into discrete levels
- Resolution – depends on number of quantization levels
- Conversion time – how long it takes to convert the sampled signal to digital code
- Conversion method – means by which analog signal is encoded into digital equivalent
    - Example – Successive approximation method

Digital-to-Analog Conversion

- Convert digital values into continuous analogue signal
  - Decoding digital value to an analogue value at discrete moments in time based on value within register

$$E_0 = E_{ref}\left\{0.5B_1 + 0.25B_2 + \cdots + \left(2^n\right)^{-1}B_n\right\}$$

Where E_0 is output voltage; E_{ref} is reference voltage; B_n is status of successive bits in the binary register

  - Data Holding that changes series of discrete analogue signals into one continuous signal

Example

- A DAC has a reference voltage of 100 V and has 6-bit precision. Three successive sampling instances 0.5 sec apart have the following data in the data register:

Instant	Binary Data
1	101000
2	101010
3	101101

- Output Values:

$E_{01} = 100\{0.5(1)+0.25(0)+0.125(1)+0.0625(0)+0.03125(0)+0.015625(0)\}$
$E_{01} = 62.50V$
$E_{02} = 100\{0.5(1)+0.25(0)+0.125(1)+0.0625(0)+0.03125(0)+0.015625(0)\}$
$E_{02} = 65.63V$
$E_{03} = 100\{0.5(1)+0.25(0)+0.125(1)+0.0625(0)+0.03125(0)+0.015625(0)\}$
$E_{03} = 70.31V$

Input/Output Devices

Binary data:

- Contact input interface – input data to computer
- Contact output interface – output data from computer

Discrete data other than binary:

- Contact input interface – input data to computer
- Contact output interface – output data from computer

Pulse data:

- Pulse counters - input data to computer
- Pulse generators - output data from computer

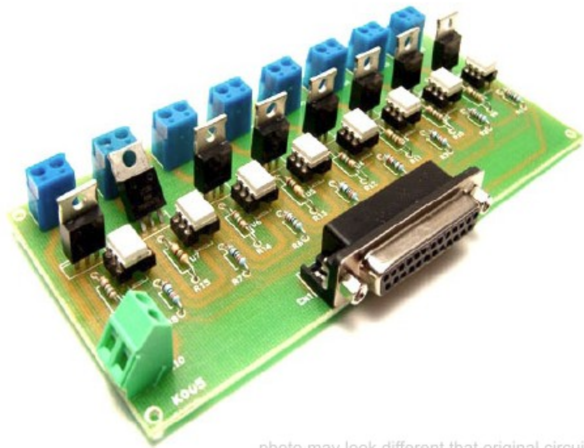


photo may look different that original circuit

CEN520

Part 2

automation

Computer-Integrated- Manufacturing

(CIM)

[redacted]

Introduction

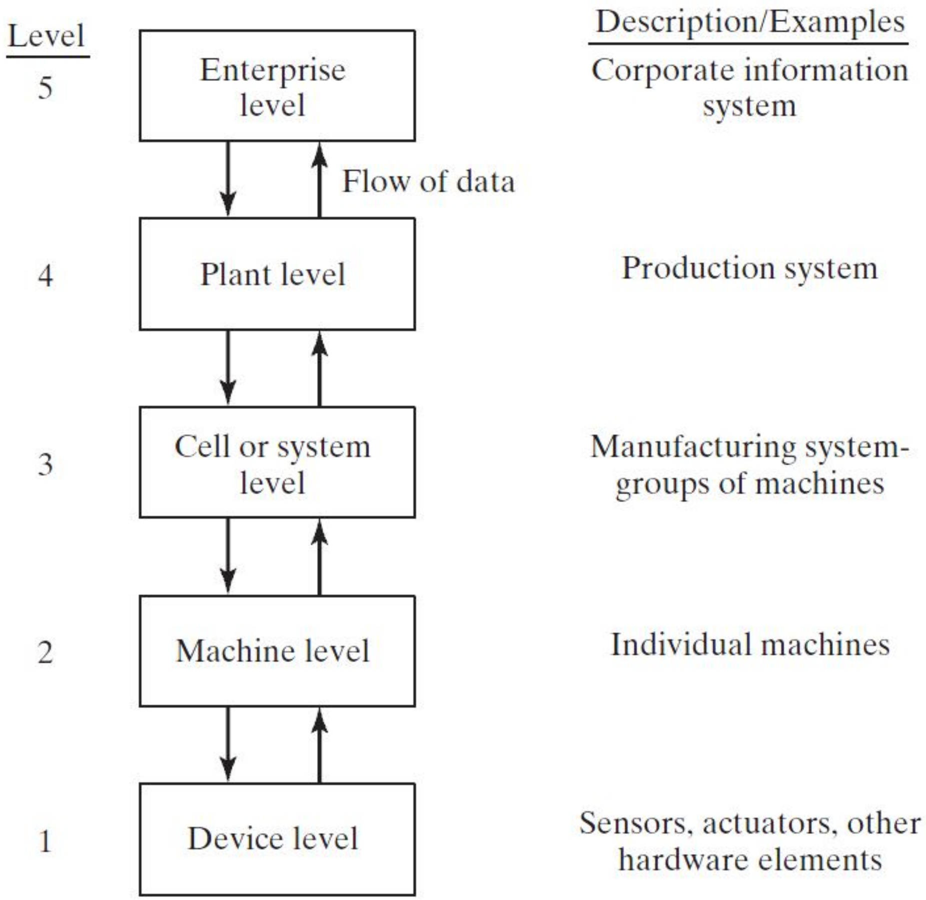
- Manufacturing is a collection of interrelated activities that includes product design and documentation, material selection, planning, production, quality assurance, management, and marketing of goods.

- The fundamental goal of manufacturing is to use these activities to convert raw materials into finished goods on a profitable basis.

Levels of Automation

1. Device level – actuators, sensors, and other hardware components to form individual control loops for the next level.
2. Machine level – CNC machine tools and similar production equipment, industrial robots, material handling equipment.
3. Cell or system level – manufacturing cell or system.
4. Plant level – factory or production systems level.
5. Enterprise level – corporate information system.

Levels of Automation



CIM has Different Definitions for Different Users

i. Shop communications

ii. Recurring processes

iii. Non-recurring processes

iv. Engineering/manufacturing communication

v. Other users

vi. Improving communication through CIM

Computer Integrated Manufacturing

Refers to the technology, tool or method used to improve entirely the design and manufacturing process and increase productivity, to help people and machines to communicate. It includes:

CAD (Computer-Aided Design),

CAM (Computer- Aided Manufacturing),

CAPP (Computer-Aided Process Planning,

CNC (Computer Numerical Control Machine tools),

DNC (Direct Numerical Control Machine tools),

FMS (Flexible Machining Systems),

ASRS (Automated Storage and Retrieval Systems),

AGV (Automated Guided Vehicles),

use of robotics and automated conveyance,

computerized scheduling and production control,

business system integrated by a common database. (Houston Cole Library)

Computer Integrated Manufacturing

Is the process of automating various functions in a manufacturing company (business, engineering, and production) by integrating the work through computer networks and common databases. CIM is a critical element in the competitive strategy of global manufacturing firms because it lowers costs, improves delivery times and improves quality. (Amatrol)

Computer-integrated Manufacturing Defined:

- CIM is the integration of the total manufacturing enterprise through the use of integrated systems and data communications coupled with new managerial philosophies that improve organizational and personal efficiency.

What is CIM?

- C + I + M
- C = Computer

  i. Enabling tool

  ii. Information flow

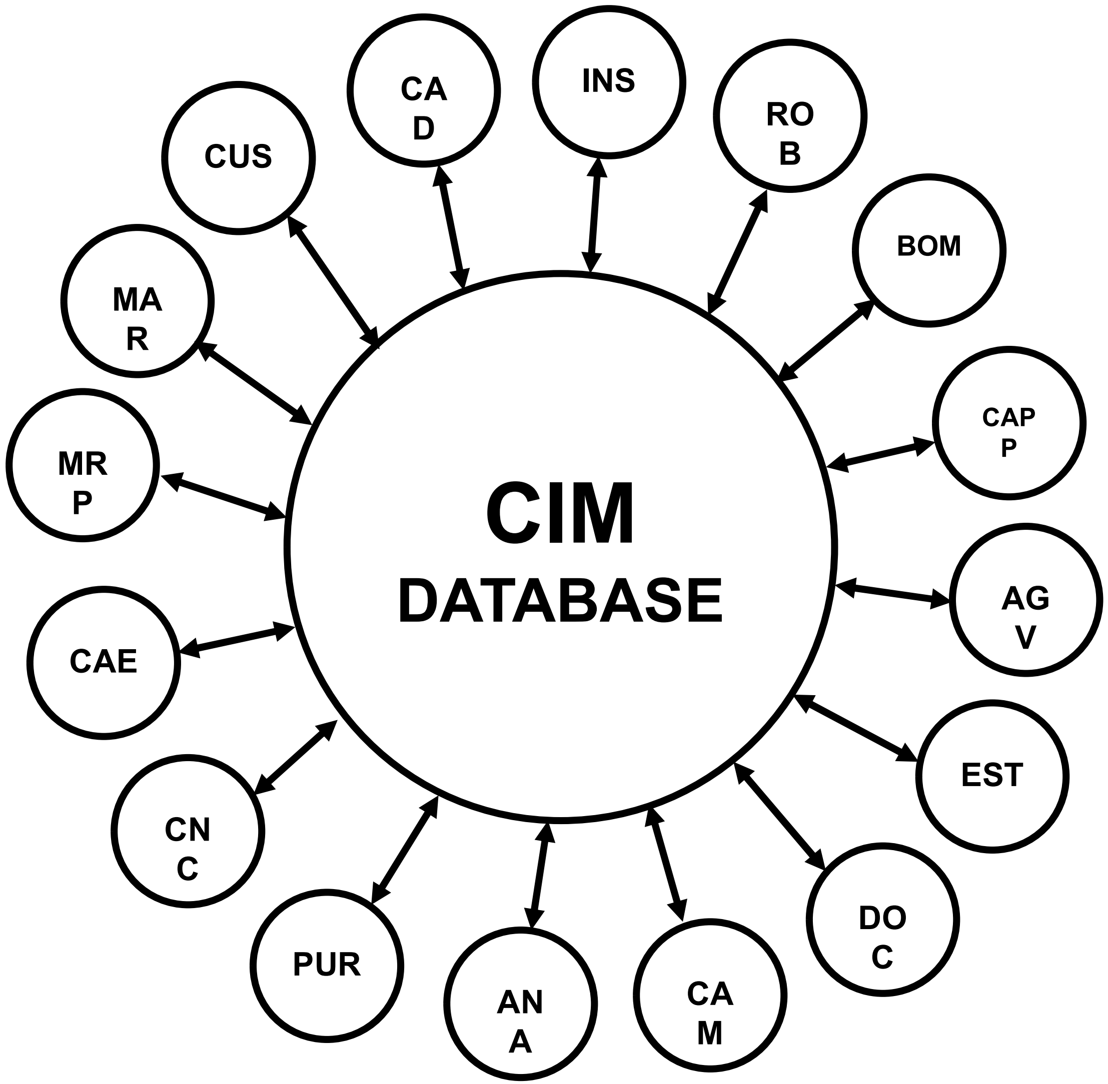
  iii. Information management

What is CIM?

- I = Integrated
    - i. Integration vs. interfacing
    - ii. Shared information
    - iii. Shared functionality
- M = Manufacturing
    - i. Production control
    - ii. Production scheduling
    - iii. Process design
    - iv. Product design
    - v. Manufacturing enterprise

CIM Components or Subsystems Include:

- CAD
- CAM
- CAPP
- CAE
- ERP
- PLC
- CAE
- Computers
- Automated conveyors
- CNC
- DNC
- Robotics
- Controllers
- FMS
- ASRS
- AGV
- Monitoring equipment
- Others


CIM
DATABASE
CUS
CA
D
INS
RO
B
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P
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AN
A
PUR
CN
C
CAE
MR
P
MA
R

Computerized Manufacturing Support Systems

Objectives of automating the manufacturing support systems:

- To reduce the manual and clerical effort in product design, manufacturing planning and control, and the business functions.
- Integrates computer-aided design (CAD) and computer-aided manufacturing (CAM) in CAD/CAM.
- CIM includes CAD/CAM and the business functions of the firm.

Reasons for Automating

1. Increase labor productivity
2. Reduce labor cost
3. Mitigate the effects of labor shortages
4. Reduce or remove routine manual and clerical tasks
5. Improve worker safety
6. Improve product quality
7. Reduce manufacturing lead time
8. Accomplish what cannot be done manually
9. Avoid the high cost of not automating

Manual Labor in Production Systems

Is there a place for manual labor in the modern production system?

- Answer: YES

- Two aspects:
  1. Manual labor in factory operations
  2. Labor in manufacturing support systems

Manual Labor in Factory Operations

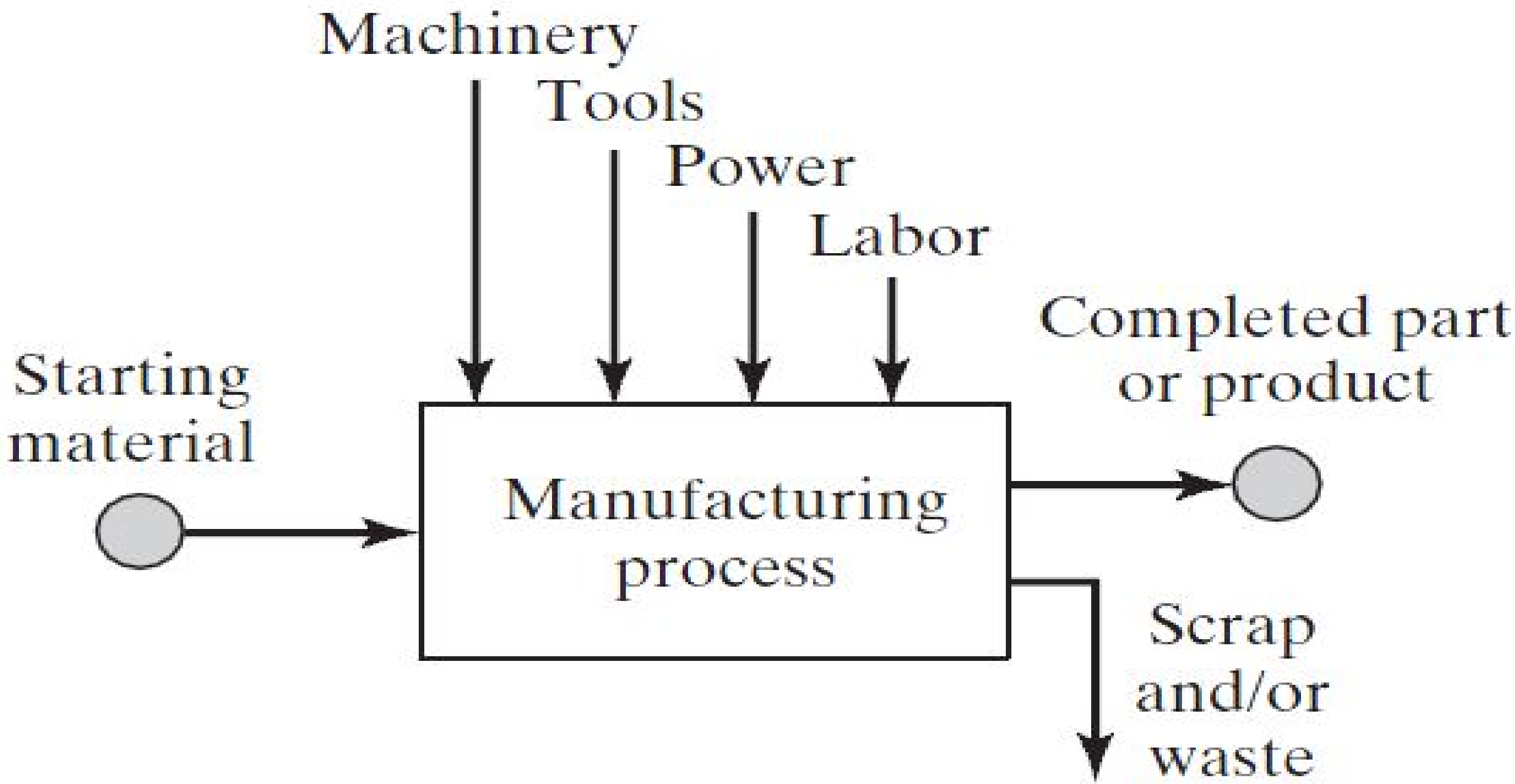
The long term trend is toward greater use of automated systems to substitute for manual labor.

- When is manual labor justified?
  - Some countries have very low labor rates and automation cannot be justified.
  - Task is technologically too difficult to automate
  - Short product life cycle.
  - Customized product requires human flexibility.
  - To cope with ups and downs in demand.
  - To reduce risk of new product failure.

Labor in Manufacturing Support Systems

- Product designers who bring creativity to the design task.
- Manufacturing engineers who:
  - Design the production equipment and tooling.
  - And plan the production methods and routings.
- Equipment maintenance.
- Programming and computer operation.
- Engineering project work.
- Plant management.

Manufacturing: Technological Definition

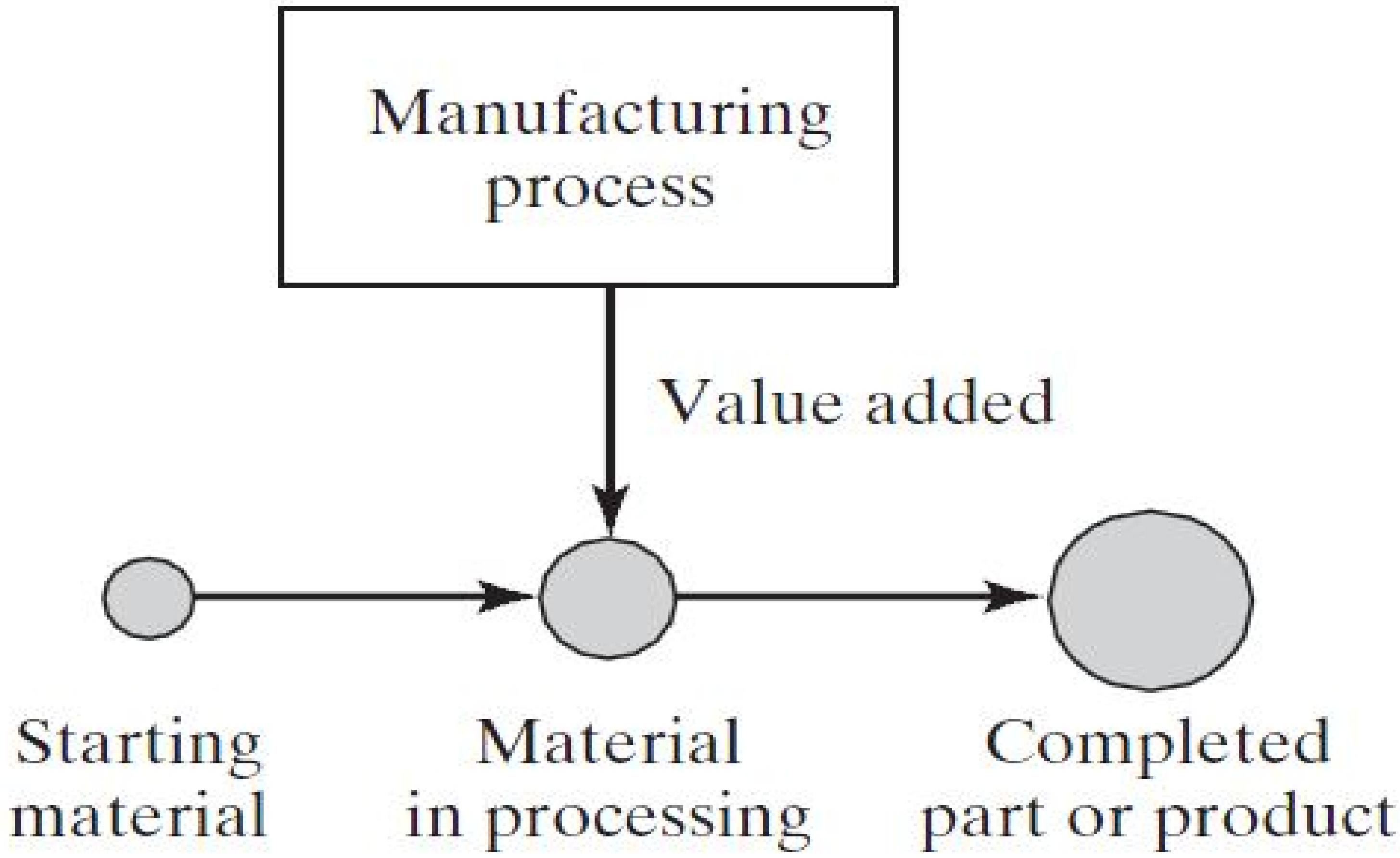


Manufacturing: Economic Definition

Transformation of materials into items of greater value by means of one or more processing and/or assembly operations.

- Manufacturing *adds value* to the material
- Examples:
  - Converting iron ore to steel adds value
  - Transforming sand into glass adds value
  - Refining petroleum into plastic adds value

Manufacturing: Economic Definition



Classification of Industries

1. Primary industries – cultivate and exploit natural resources
   - Examples: agriculture, mining
2. Secondary industries – convert output of primary industries into products
   - Examples: manufacturing, power generation, construction
3. Tertiary industries – service sector
   - Examples: banking, education, government, legal services, retail trade, transportation

Manufacturing Operations

- There are certain basic activities that must be carried out in a factory to convert raw materials into finished products
- For discrete products:
  1. Processing and assembly operations
  2. Material handling
  3. Inspection and testing
  4. Coordination and control

Processing Operations

- Shaping operations
  1. Solidification processes
  2. Particulate processing
  3. Deformation processes
  4. Material removal processes
  5. Additive manufacturing (a.k.a. rapid prototyping)
- Property-enhancing operations (heat treatments)
- Surface processing operations
  - Cleaning and surface treatments
  - Coating and thin-film deposition

Assembly Operations

- Joining processes
  - Welding
  - Brazing and soldering
  - Adhesive bonding
- Mechanical assembly
  - Threaded fasteners (e.g., bolts and nuts, screws)
  - Rivets
  - Interference fits (e.g., press fitting, shrink fits)
  - Other

Other Factory Operations

- Material handling and storage
- Inspection and testing
- Coordination and control

Material Handling and Storage

- Material transport:
  - Vehicles, e.g., forklift trucks, AGVs, monorails
  - Conveyors
  - Hoists and cranes
- Storage systems
- Automatic identification and data capture: (AIDC)
  - Bar codes
  - RFID
  - Other AIDC

Inspection and Testing

Inspection – examination of the product and its components to determine whether they conform to design specifications:

- Inspection for variables – measuring
- Inspection for attributes – gaging

Testing – observing the product (or part, material, subassembly) during actual operation or under conditions that might occur during operation.

Coordination and Control

- Regulation of the individual processing and assembly operations:
  - Process control
  - Quality control
- Management of plant level activities:
  - Production planning and control
  - Quality control

Automation Defined

Automation is the technology by which a process or procedure is accomplished without human assistance.

- Basic elements of an automated system:
  1. Power - to accomplish the process and operate the automated system
  2. Program of instructions – to direct the process
  3. Control system – to actuate the instructions

Power to Accomplish the Automated Process

- Power for the process
  - To drive the process itself
  - To load and unload the work unit
  - Transport between operations
- Power for automation
  - Controller unit
  - Power to actuate the control signals
  - Data acquisition and information processing

Program of Instructions

Set of commands that specify the sequence of steps in the work cycle and the details of each step.

- Example: NC part program.
- During each step, there are one or more activities involving changes in one or more process parameters.
  - Examples:
    - Temperature setting of a furnace
    - Axis position in a positioning system
    - Motor on or off

Decision-Making in a Programmed Work Cycle

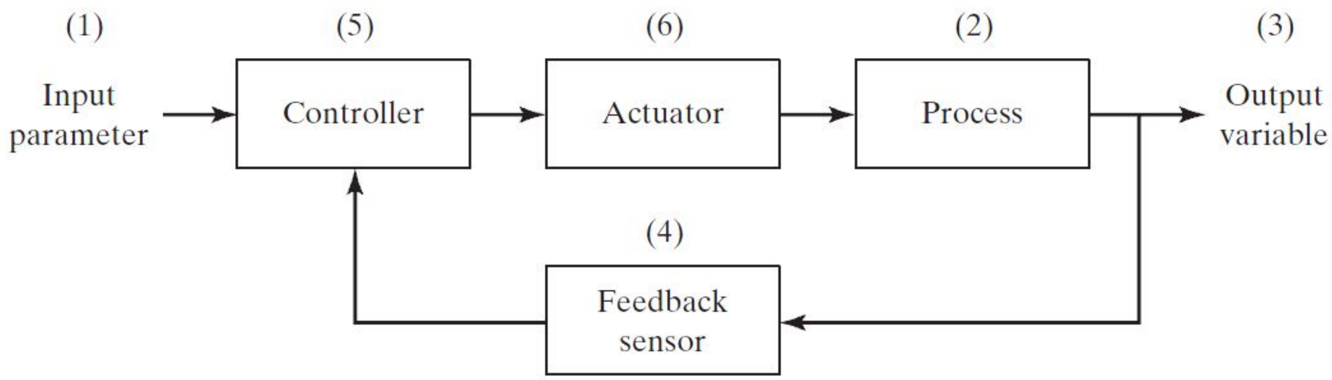
- Following are examples of automated work cycles in which decision making is required:
  - Operator interaction:
    - Automated teller machine
  - Different part or product styles processed by the system:
    - Robot welding cycle for two-door vs. four door car models
  - Variations in the starting work units:
    - Additional machining pass for oversized sand casting

Control System – Two Types

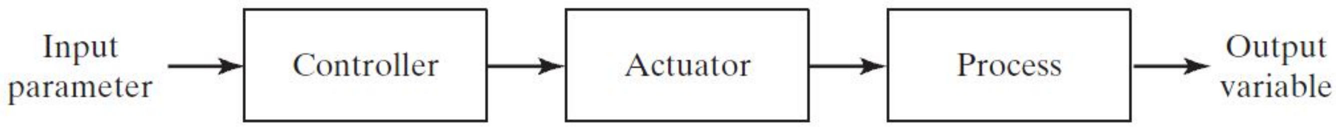
1. Closed-loop (feedback) control system – a system in which the output variable is compared with an input parameter, and any difference between the two is used to drive the output into agreement with the input
2. Open-loop control system – operates without the feedback loop
    - Simpler and less expensive
    - Risk that the actuator will not have the intended effect

(a) Feedback Control System and (b) Open-Loop Control System

(a)

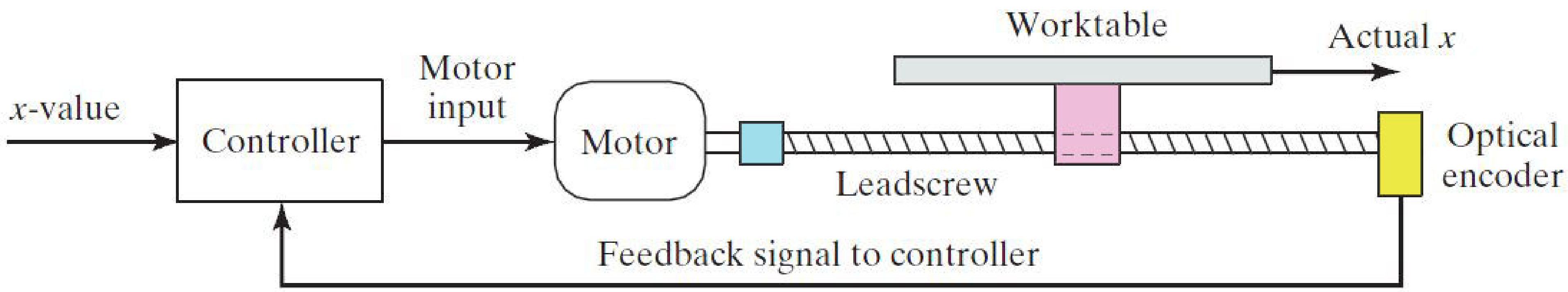


(b)



Positioning System Using Feedback Control

A one-axis position control system consisting of a leadscrew driven by a dc servomotor and using an optical encoder as the feedback sensor



When to Use an Open-Loop Control System

- Actions performed by the control system are simple.
- Actuating function is very reliable.
- Any reaction forces opposing the actuation are small enough as to have no effect on the actuation.
- If these conditions do not apply, then a closed-loop control system should be used.

Advanced Automation Functions

1. Safety monitoring
2. Maintenance and repair diagnostics
3. Error detection and recovery

Safety Monitoring

Use of sensors to track the system's operation and identify conditions that are unsafe or potentially unsafe

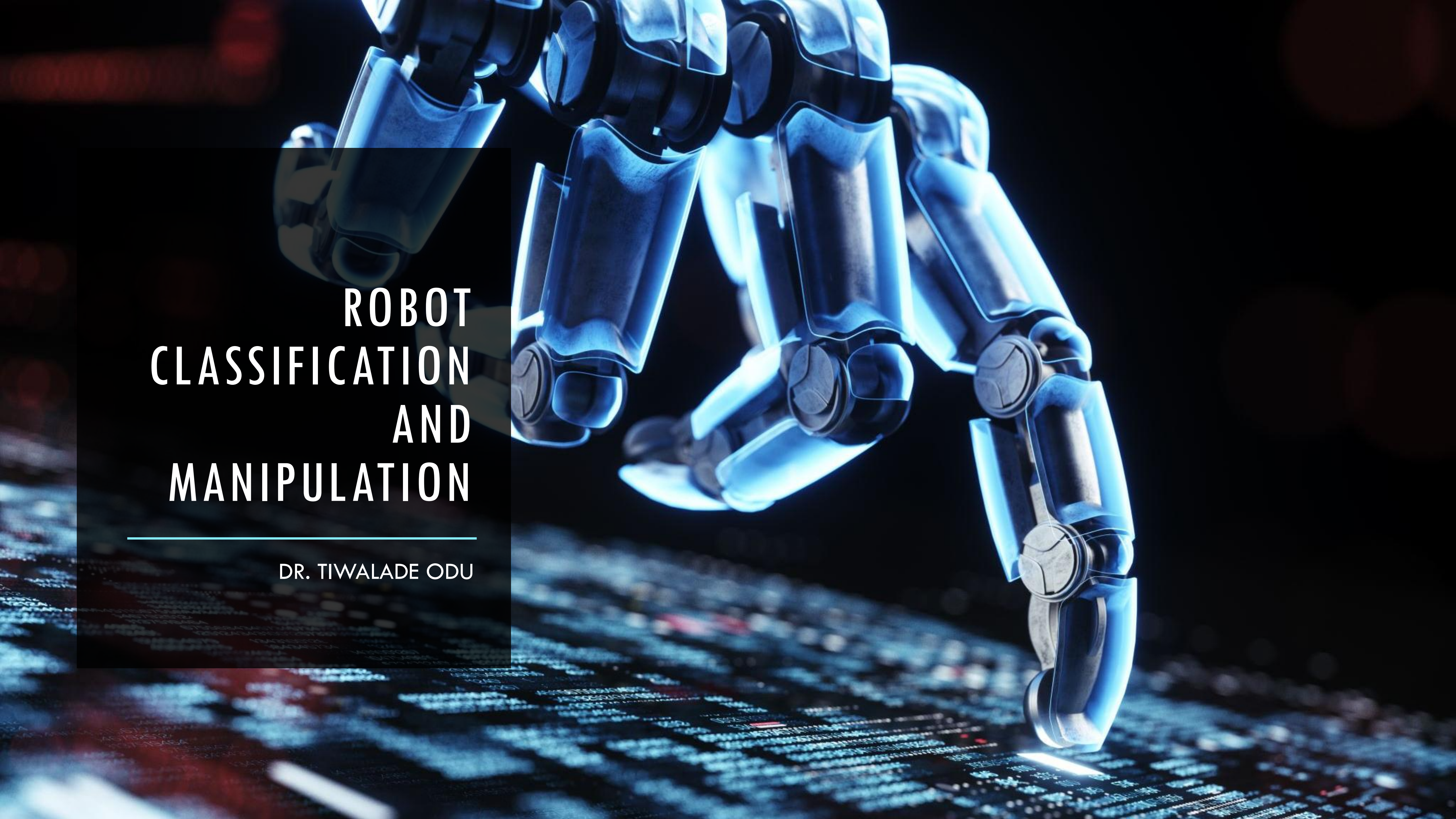
- Reasons for safety monitoring
  - To protect workers and equipment
- Possible responses to hazards:
  - Complete stoppage of the system
  - Sound an alarm
  - Reduce operating speed of process
  - Take corrective action to recover from the safety violation

Maintenance and Repair Diagnostics

- Status monitoring:
  - Monitors and records status of key sensors and parameters during system operation
- Failure diagnostics:
  - Invoked when a malfunction occurs
  - Purpose: analyze recorded values so the cause of the malfunction can be identified
- Recommendation of repair procedure:
  - Provides recommended procedure for the repair crew to effect repairs

Error Detection and Recovery

1. Error detection – functions:
    - Use the system's available sensors to determine when a deviation or malfunction has occurred
    - Correctly interpret the sensor signal
    - Classify the error
2. Error recovery – possible strategies:
    - Make adjustments at end of work cycle
    - Make adjustments during current work cycle
    - Stop the process to invoke corrective action
    - Stop the process and call for help



ROBOT CLASSIFICATION AND MANIPULATION

DR. TIWALADE ODU

INTRODUCTION

- An industrial robot is an automatically controlled, programmable, multipurpose manipulator that is programmable in 3 or more axes, which can either be fixed or mobile for use in industrial automation applications.'
- Robots vary in their autonomy: **unsupervised** and **supervised**.

INTRODUCTION (CONTD.)

- Service Robots: they perform tasks for humans or equipment excluding industrial automation applications.
  - They require a degree of autonomy and generally include systems based on some degree of human-robot interaction to perform tasks.
- Examples include:
  - BellaBot – A restaurant assistant that delivers food to tables and interacts with customers.
  - Pepper – A humanoid robot used in retail and healthcare for customer engagement and assistance.
  - KettyBot – A logistics robot that helps with inventory management and goods transportation in warehouses.
  - Household Robots – Automated vacuum cleaners, lawn-mowers, and cooking bots that make daily chores easier.
  - Medical Robots – Robots used in surgeries, patient care, and rehabilitation to improve healthcare efficiency.

INTRODUCTION (CONTD.)

- **Cobots**: they are collaborative robots, designed for direct physical interaction with a human operator, within a shared workspace.
  - They are produced to be safe for use in environments where humans are present, and therefore have certain constraints built into their design.
- Examples include:
  - Automotive Manufacturing – Companies like BMW use cobots for welding, bolting, and assembling parts, reducing worker fatigue and improving precision.
  - Healthcare – Surgical robots, such as the da Vinci surgical system, assist surgeons in performing complex procedures with high precision.
  - Agriculture – Singapore-based Singrow developed an AI-powered cobot for harvesting and pollination, increasing efficiency and yield.
  - Hospitality – Cobots are used to serve food and drinks, clean high-touch areas, and even scoop ice cream in restaurants and hotels.
  - Material Handling – Cobots help with packing, palletizing, labeling, and inspecting product quality in warehouses and factories.

INTRODUCTION (CONTD.)

- **Autonomous Mobile Robots (AMRs):** they can make decisions based on their surrounding using machine learning and artificial intelligence.
- Examples include:
  - Warehouse Robots: Companies like Amazon use AMRs to transport goods efficiently within fulfillment centers.
  - Delivery Robots: Self-driving robots deliver food and packages in urban areas.
  - Healthcare Robots: Mobile robots assist in hospitals by transporting medical supplies and even disinfecting rooms.
  - Agricultural Robots: Autonomous machines help with planting, harvesting, and monitoring crops, improving efficiency in farming.
  - Security and Surveillance Robots: Some robots patrol areas, detect intruders, and provide real-time monitoring.

INTRODUCTION (CONTD.)

- **Robotic Process Automation (RPA):** This software technology is easy for anyone to use to automate digital tasks.
  - As a software tool, it can partially or fully automate repetitive or rule-based human activities such as data entry or simple customer service queries.
- Examples include:
  - Banking & Finance: Automating fraud detection, invoice processing, and compliance monitoring.
  - Healthcare: Managing patient records, automating insurance claims, and tracking health metrics.
  - Retail & E-commerce: Handling inventory optimization, order processing, and customer service automation.
  - Human Resources: Automating payroll, onboarding, and benefits management.
  - Supply Chain & Logistics: Managing procurement, distribution, and tracking shipments.

INTRODUCTION (CONTD.)

- **Robotics as a Service (RaaS):** the robotics infrastructure is leased.
- Examples include:
  - Healthcare: Robots are used in hospitals for tasks such as delivering medications, transporting supplies, and assisting in surgeries. e.g., TUG robots by Aethon are used to transport medications and supplies withing hospitals.
  - Hospitality: In hotels and restaurants, robots can handle room deliveries, greet guests, and even cook meals. e.g., the Relay robot by Savioke is used for room service deliveries in hotels.
  - Logistics and Warehousing: Companies like Fetch Robotics provide robots that help with picking, packing, and transporting goods within warehouses. These robots can be leased on a subscription basis, making it easier for companies to scale their operations without significant upfront costs.
  - Retail: Robots like Simbe Robotics' Tally are used for inventory management in retail stores. They can scan shelves to check stock levels and ensure products are correctly placed.
  - Agriculture: Autonomous robots are used for tasks such as planting, weeding, and harvesting. e.g. Naio Technologies offers robots that can perform weeding and hoeing in fields, helping farmers reduce labour costs and increase efficiency.

HISTORY OF THE DEVELOPMENT OF ROBOTS

Ancient Greece Zeus gives Europa a 'metal man', Talos, as a gift
Organs and clocks driven by water power were created

AD 10–70 Hero of Alexandria wrote a book called *Automata* and invented a wind-powered organ, animated statues, and the first steam engine

1495 Leonardo da Vinci designed the first humanoid robot

1645 The first calculator, the Pascaline, was invented

1913 Henry Ford installed first conveyor belt assembly line

1941–42 Isaac Asimov wrote the Three Laws of Robotics

1942 Paint-spraying robot designed by the DeVilbiss Company

1950 Alan Turing proposed the Turing Test to determine if a machine has the power to think for itself

1963 First robotic arm

1969 Man landed on the Moon

1970 WABOT-1 created in Japan, able to walk, grip, transport objects, sense objects and communicate – the first android robot

1970 KUKA built the first industrial robot with six electromechanical axes

1977 The first *Star Wars* movie was released, depicting a galaxy shared with robots

1981 First quadruped robot created by Shigeo Hirose

1990 iRobot Corporation founded

1996 Honda revealed its first humanoid robot, P2

1997 NASA's Pathfinder robot landed on Mars

1999 Sony revealed AIBO, a robotic dog

2000 Honda showed its most advanced humanoid, ASIMO, which could run, walk, communicate and interact with its environment

2001 PackBot robots sent in after the World Trade Centre attack to search through debris

2002 iRobot released the first Roomba – a vacuuming robot for the home

2004 The Mars rovers Spirit and Opportunity landed on Mars

2004 Epsom released the smallest-known robot of the time

2012 Mars rover Curiosity landed on the Red Planet

2013 iRobot revealed its first robot doctor

2017 A robot called Sophia was granted Saudi Arabian citizenship

Advantages of Robots

- They can be used to boost productivity, improve safety, efficiency, quality, and consistency of products.
- They can work unaided in hazardous and dangerous situations, i.e., nuclear stations, undersea, space, extremely hot or cold environments, etc.
- They work with high accuracy and consistent precision.
- They can be used continuously without adverse effects on their accuracy and precision.
- They can be used to perform extremely arduous tasks.
- They can handle multiple tasks simultaneously.

Disadvantages of Robots

- Their initial investment cost is high.
- They can pose economic hardship as they tend to replace human workers.
- They can not work in all terrains due to their limitations in terms of sensing, computation, actuation, dexterity and maneuvering, degrees of freedom, etc.

Terminologies

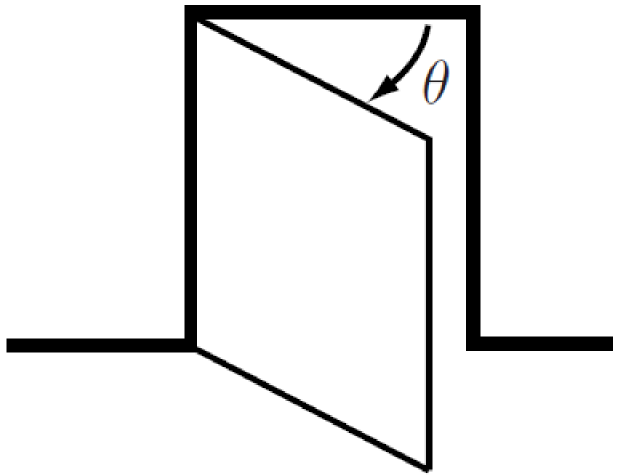
A **mechanism** is constructed by connecting rigid bodies, called **links**, together using **joints** to make relative motion between adjacent links possible.

Actuation of the joints, typically by electric motors, causes the robot to move and exert forces in desired ways.

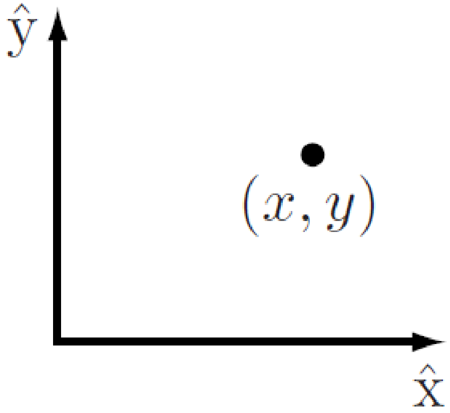
A robot is mechanically constructed by connecting a set of bodies, called **links**, using various **joints**.

Actuators, such as electric motors, deliver forces or torques that cause the robot's links to move. Usually an **end-effector**, such as a gripper or hand for grasping and manipulating objects, is attached to a specific link.

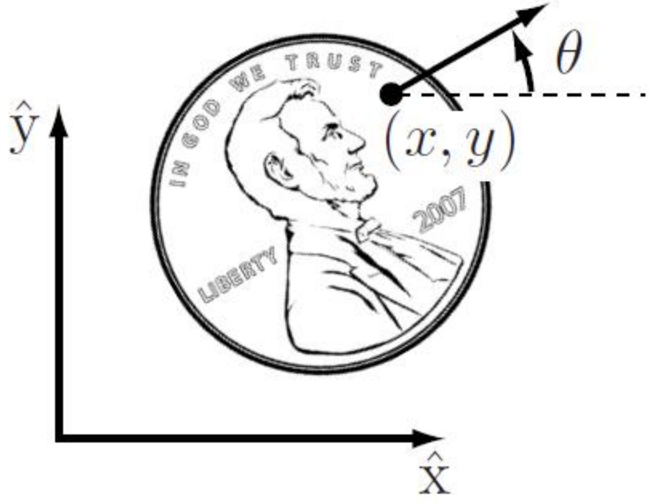
A robot’s **configuration** provides information on its location; it specifies the positions of all points of the robot.



The configuration of a door can be represented by a single number, the angle θ about its hinge.



The configuration of a point in a plane can be described by coordinates (x, y).



The configuration of a coin lying heads up on a table can be described by three coordinates: two coordinates (x, y) that specify the location of a particular point on the coin, and one coordinate (θ) that specifies the coin's orientation.

Terminologies (Contd.)

The number of **degrees of freedom (DOF)** of a robot is the smallest number of real-valued coordinates needed to represent its configuration.

The n-dimensional space containing all possible robot configurations is called the **configuration space (C-space)**.

The configuration of a robot is represented by a point in its C-space.

The C-space of a robot's end-effector is its **task space**.

Configuration Space

A robot is mechanically constructed by connecting a set of bodies, called **links**, to each other using various types of **joints**.

Actuators, such as electric motors, deliver forces or torques that cause the robot's links to move.

An **end-effector**, such as a gripper or hand for grasping and manipulating objects, is attached to a specific link.

The location , in space, of a robot is derived from its **configuration**.

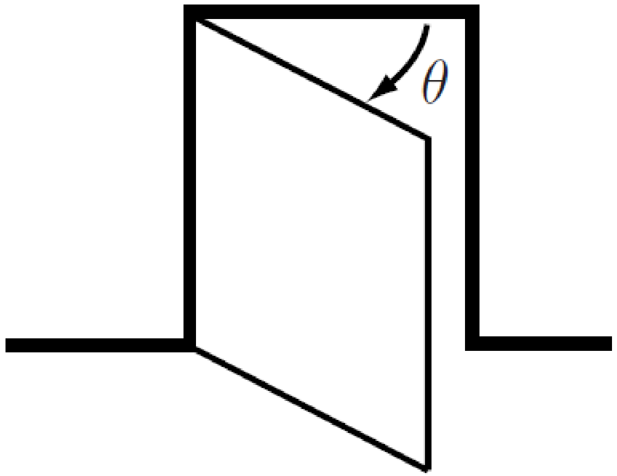
The configuration of a robot is the specification of the positions of all points of the robot.

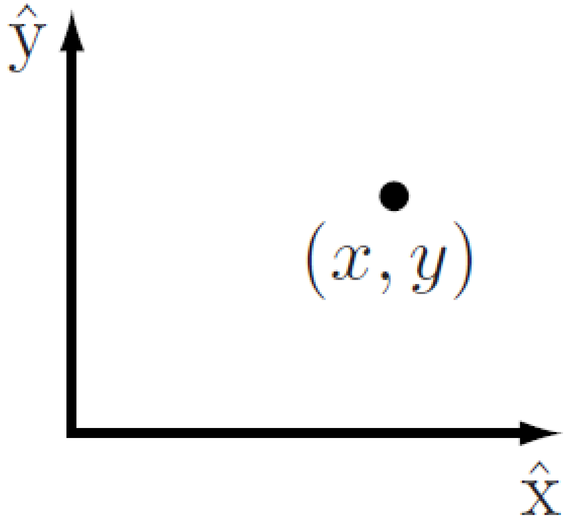
The robot's links are rigid and of a known shape,therefore only a few numbers are needed to represent its configuration.

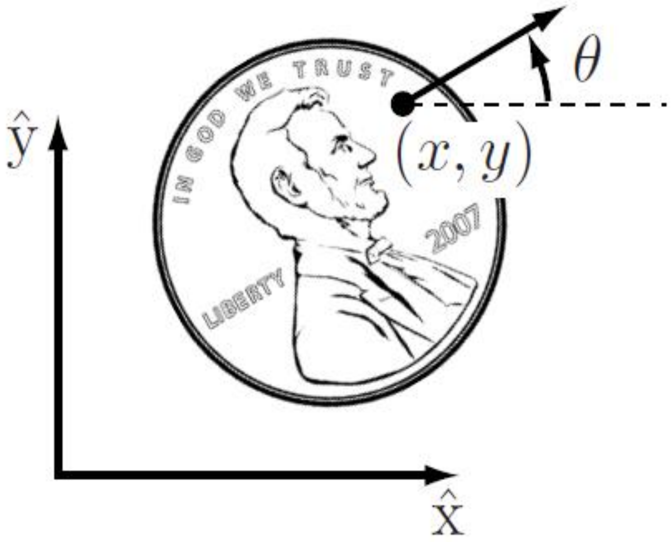
- Examples:
  - The configuration of a door can be represented by a single number, the angle θ about its hinge.
  - The configuration of a point on a plane can be described by two coordinates, (x, y).
  - The configuration of a coin lying heads up on a table can be described by three coordinates: two coordinates (x, y) that specify the location of a particular point on the coin, and one coordinate (θ) that specifies the coin's orientation.

The values of these coordinates are real, continuous numbers.

- The number of **degrees of freedom (dof)** of a robot is the smallest number of real-valued coordinates needed to represent its configuration.





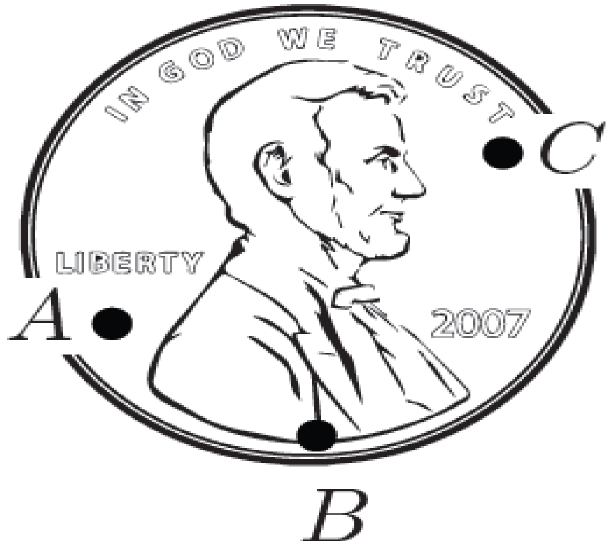


- The door has one degree of freedom
- The coin lying heads up, on a table has three degrees of freedom.

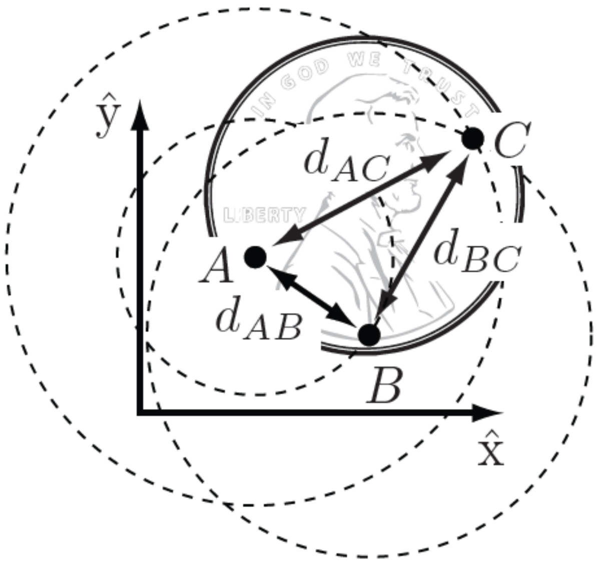
- The **configuration** of a robot is a complete specification of the position of every point of the robot.
- The minimum number, **n** of real-valued coordinates needed to represent the configuration is the number of **degrees of freedom (dof)** of the robot.
- The n-dimensional space containing all possible configurations of the robot is called the **configuration space (C-space).**
- The configuration of a robot is represented by a point in its C-space.

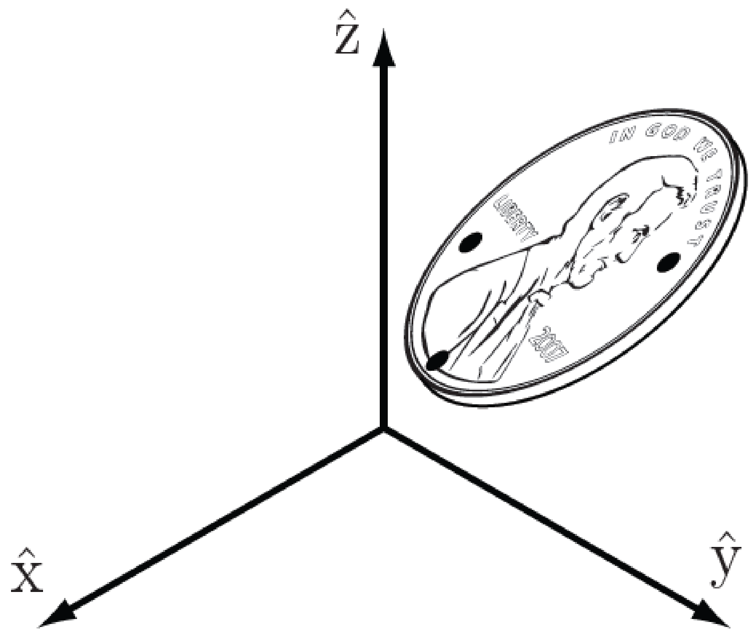
Degrees of Freedom of a Rigid Body

- Choose three fixed points A, B, and C on a coin lying on a table



- Once the location of A is chosen, B must lie on a circle of radius, $\mathbf{d_{AB}}$ centered at A. Once the location of B is chosen, C must lie at the intersection of circles centered at A and B. Only one of these two intersections corresponds to the "heads up" configuration.





- The configuration of a coin in three-dimensional space is given by the three coordinates of A, two angles to the point B on the sphere of radius, $\mathbf{d_{AB}}$ centered at A, and one angle to the point C on the circle defined by the intersection of the sphere centered at A and a sphere centered at B.

- Once a coordinate frame $\hat{x} - \hat{y}$ is attached to the plane, the positions of these points A, B, and C in the plane are written as $(x_A, y_A), (x_B, y_B), and\ (x_C, y_C)$
- If the points could be placed independently anywhere in the plane, the coin would have six degrees of freedom - two for each of the three points.
- But, according to the definition of a rigid body, the distance between point A and point B, denoted d(A,B), is always constant regardless of where the coin is.

- Similarly, the distances d(B,C) and d(A,C) must be constant.
- The following equality constraints on the coordinates $(x_A, y_A), (x_B, y_B), and\ (x_C, y_C)$ must therefore always be satisfied:

$$d(A,B) = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2} = d_{AB},$$

$$d(B,C) = \sqrt{(x_B - x_C)^2 + (y_B - y_C)^2} = d_{BC},$$

$$d(A,C) = \sqrt{(x_A - x_C)^2 + (y_A - y_C)^2} = d_{AC}.$$

- To determine the number of degrees of freedom of the coin on the table, first choose the position of point A in the plane.

- We may choose it to be anything we want, so we have two degrees of freedom to specify, namely (x_A, y_A).

- Once (x_A, y_A) is specified, the constraint d(A,B) = $\mathbf{d_{AB}}$ restricts the choice of (x_B, y_B) to those points on the circle of radius $\mathbf{d_{AB}}$ centered at A.

- A point on this circle can be specified by a single parameter, e.g., the angle specifying the location of B on the circle centered at A.

- Let's call this angle $\emptyset_{AB}$ and define it to be the angle that the vector $\overrightarrow{AB}$ makes with the $\hat{x}$-axis.
- Once we have chosen the location of point B, there are only two possible locations for C: at the intersections of the circle of radius $\mathbf{d_{AC}}$ centered at A and the circle of radius $\mathbf{d_{BC}}$ centered at B.
- These two solutions correspond to heads or tails.
- This implies once we have placed A and B and chosen heads or tails, the two constraints d(A,C) = $\mathbf{d_{AC}}$ and d(B,C) = $\mathbf{d_{BC}}$ eliminate the two apparent freedoms provided by (x_C, y_C), and the location of C is fixed.
- The coin has exactly three degrees of freedom in the plane, which can be specified by $(x_A, y_A, \emptyset_{AB})$.

- If we choose to specify the position of an additional point D on the coin.
- This introduces three additional constraints: d(A,D) = $\mathbf{d}_{\mathbf{AD}}$, d(B,D) = $\mathbf{d}_{\mathbf{BD}}$, and d(C,D) = $\mathbf{d}_{\mathbf{CD}}$.
- One of these constraints is redundant, i.e., it provides no new information; only two of the three constraints are independent.
- The two freedoms apparently introduced by the coordinates (x_D, y_D) are then immediately eliminated by these two independent constraints.
- The same would hold for any other newly chosen point on the coin, so that there is no need to consider additional points.

- The general rule for determining the number of degrees of freedom of a system is:

$$\boldsymbol{degrees\ of\ freedom\ =\ (sum\ of\ freedoms\ of\ the\ points)\ -\ (number\ of\ independent\ constraints)}$$

- Expressing this rule in terms of the number of variables and independent equation that describe the system:

$$\boldsymbol{degrees\ of\ freedom\ =\ (number\ of\ variables) -\ (number\ of\ independent\ equations)}$$

- A rigid body moving in three-dimensional space is called a **spatial rigid body**, and it has **six degrees of freedom**.

- A rigid body moving in a two-dimensional plane is called a **planar rigid body**, and it has **three degrees of freedom**.

- For robots with rigid bodies,

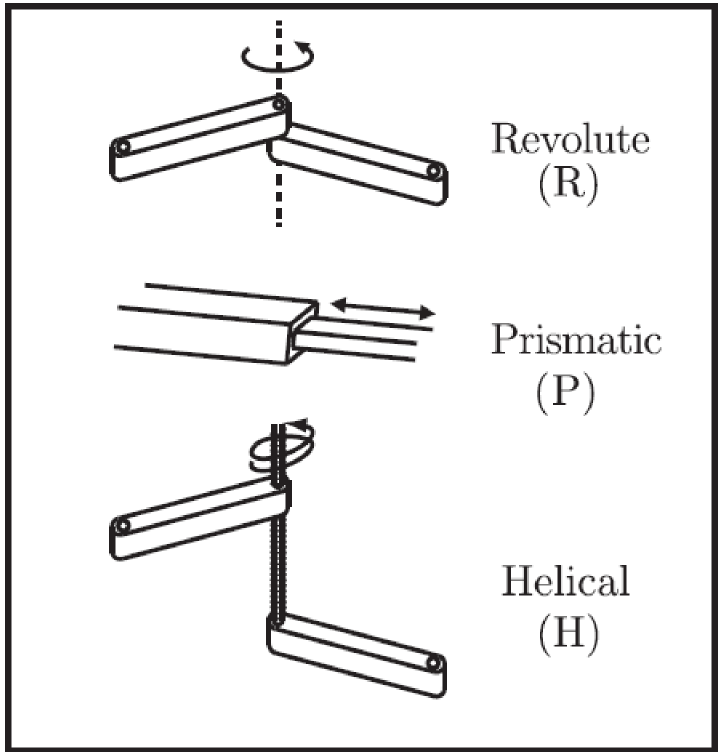
$$\boldsymbol{degrees\ of\ freedom\ =\ (sum\ of\ freedoms\ of\ the\ bodies)\ -\ (number\ of\ independent\ constraints)}$$

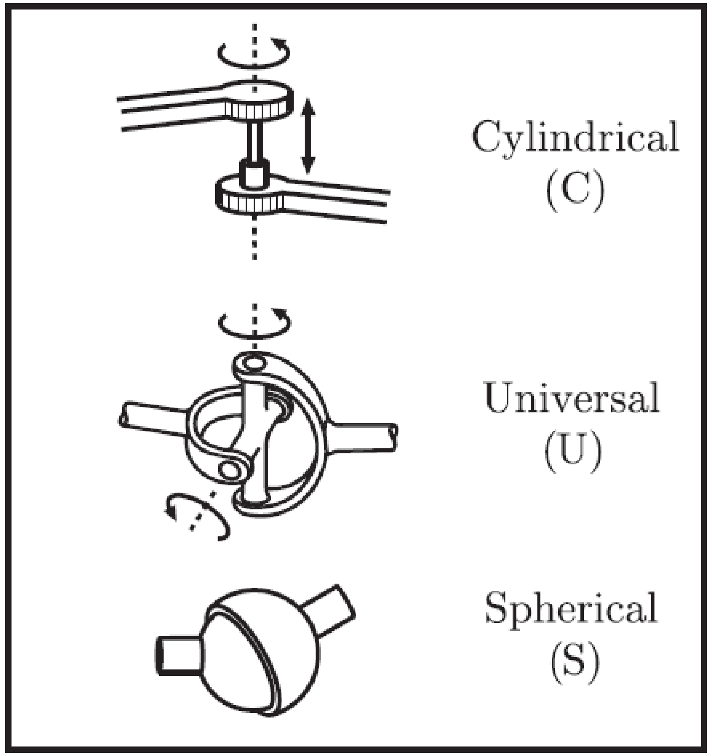
Degrees of Freedom of a Robot

- The previous example of a door, which is a single rigid body connected to a wall by a hinge joint; has only one degree of freedom, represented by the hinge joint angle, θ.
  - Without the hinge joint, the door would be free to move in three-dimensional space and would have six degrees of freedom.
  - By connecting the door to the wall via the hinge joint, five independent constraints are imposed on the motion of the door, leaving only one independent coordinate (θ).
  - Alternatively, the door can be viewed from above and regarded as a planar body, which has three degrees of freedom.
  - The hinge joint then imposes two independent constraints, again leaving only one independent coordinate (θ).

- The door's C-space is represented by some range in the interval $[0, 2\pi)$ over which θ is allowed to vary.
- In both cases the joints constrain the motion of the rigid body, thus reducing the overall degrees of freedom.
- This suggests a formula for determining the number of degrees of freedom of a robot, simply by counting the number of rigid bodies and joints.
- The Grubler's formula is used to determine the number of degrees of freedom of planar and spatial robots.

Robot Joints





Typical Robot Joints

- Every joint connects exactly two links; joints that simultaneously connect three or more links are not allowed.
- The **revolute joint (R**), also called a **hinge joint**, allows rotational motion about the joint axis.
- The **prismatic joint (P**), also called a **sliding or linear joint**, allows translational (or rectilinear) motion along the direction of the joint axis.
- The **helical joint (H**), also called a **screw joint**, allows simultaneous rotation and translation about a screw axis.
- Revolute, prismatic, and helical joints all have one degree of freedom.

- Joints can also have multiple degrees of freedom.
- The **cylindrical joint (C**) has two degrees of freedom and allows independent translations and rotations about a single fixed joint axis.
- The **universal joint (U)** is another two-degree-of-freedom joint that consists of a pair of revolute joints arranged so that their joint axes are orthogonal.
- **The spherical joint (S**), also called a **ball-and-socket joint**, has three degrees of freedom and functions much like our shoulder joint.

- A joint can be viewed as providing freedoms to allow one rigid body to move relative to another.
- It can also be viewed as providing constraints on the possible motions of the two rigid bodies it connects.
- For example, a revolute joint can be viewed as allowing one freedom of motion between two rigid bodies in space, or it can be viewed as providing five constraints on the motion of one rigid body relative to the other.
- Generalizing, the number of degrees of freedom of a rigid body (three for planar bodies and six for spatial bodies) minus the number of constraints provided by a joint must equal the number of freedoms provided by that joint.

Joint type	dof f	Constraints c between two planar rigid bodies	Constraints c between two spatial rigid bodies
Revolute (R)	1	2	5
Prismatic (P)	1	2	5
Helical (H)	1	N/A	5
Cylindrical (C)	2	N/A	4
Universal (U)	2	N/A	4
Spherical (S)	3	N/A	3

The number of degrees of freedom *f* and constraints *c* provided by common joints.

Grubler’s Formula

- The number of degrees of freedom of a mechanism with links and joints can be calculated using the Grubler's formula.
- Consider a mechanism consisting of **N links**, where ground is also regarded as a link. Let **J** be the **number of joints**, **m** be the **number of degrees of freedom of a rigid body (m = 3 for planar mechanisms and m = 6 for spatial mechanisms)**, $\boldsymbol{f_i}$ be the **number of freedoms provided by joint $\boldsymbol{i}$**, and $\boldsymbol{c_i}$ be the **number of constraints provided by joint $\boldsymbol{i}$**, where $\boldsymbol{f_i} + \boldsymbol{c_i} = \boldsymbol{m}$ **for all $\boldsymbol{i}$**.

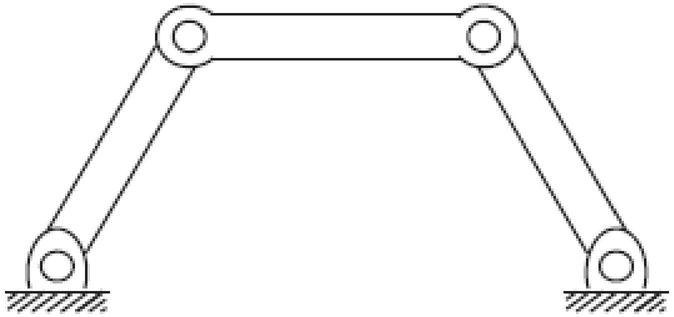
- Then Grubler's formula for the number of degrees of freedom of the robot is

$$\mathrm{dof} = \underbrace{m(N-1)}_{\text{rigid body freedoms}} - \underbrace{\sum_{i=1}^{J} c_i}_{\text{joint constraints}}$$

$$dof = m(N-1) - \sum_{i=1}^{J}(m - f_i)$$

$$= \boldsymbol{m(N-1-J)} + \sum_{\boldsymbol{i=1}}^{\boldsymbol{J}}(\boldsymbol{f_i})$$

- There are two types of mechanisms: open-chain mechanisms (also known as serial mechanisms) and closed-chain mechanisms.
- A closed-chain mechanism is any mechanism that has a closed loop.
- A person standing with both feet on the ground is an example of a closed-chain mechanism, since a closed loop can be traced from the ground, through the right leg, through the waist, through the left leg, and back to ground (recall that the ground itself is a link).
- An open-chain mechanism is any mechanism without a closed loop; an example is your arm when your hand is allowed to move freely in space.



Four-bar Linkage

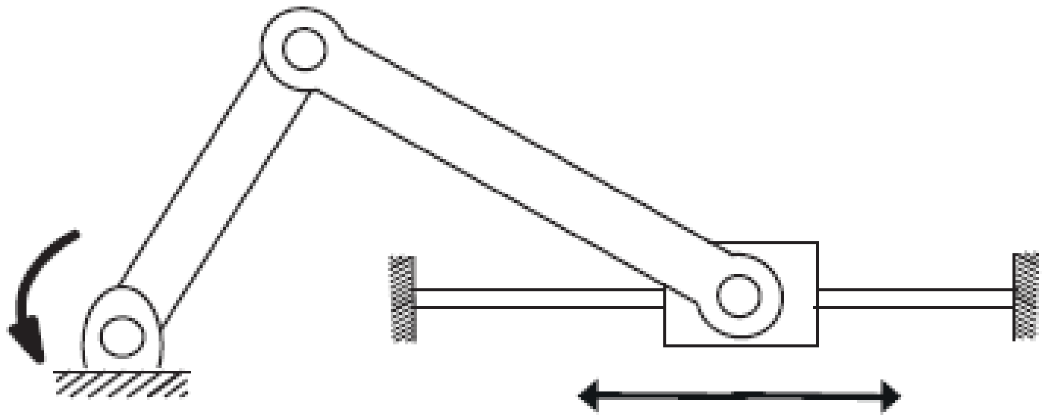
- The planar four-bar linkage consists of four links (one of them ground) arranged in a single closed loop and connected by four revolute joints.
- Since all the links are confined to move in the same plane, we have **m = 3.**

- N = number of links = 4
- J = the number of joints = 4
- m = the number of degrees of freedom of a rigid body (m = 3 for planar mechanisms and m = 6 for spatial mechanisms) = 3
- fi = number of freedoms provided by joint i
- ci = number of constraints provided by joint i, where fi+ci=m for all i.
- i = 1, 2, 3, 4

$$dof = \boldsymbol{m}(\boldsymbol{N} - \mathbf{1} - \boldsymbol{J}) + \sum_{\boldsymbol{i}=\mathbf{1}}^{\boldsymbol{J}}(\boldsymbol{f}_{\boldsymbol{i}})$$

$$dof = 3(4 - 1 - 4) + (1 + 1 + 1 + 1) = -3 + 4 = 1$$

- Using the Gruber’s formula, determine the degrees of freedom of the planar slider-crank mechanism below



Slider-Crank Mechanism Solution (a)

- 3 revolute joints and 1 prismatic joint

$dof = \boldsymbol{m}(\boldsymbol{N} - \mathbf{1} - \boldsymbol{J}) + \sum_{\boldsymbol{i}=\mathbf{1}}^{\boldsymbol{J}}(\boldsymbol{f}_{\boldsymbol{i}})$

$$J = 4; each\ f_i = 1; N = 4, m = 3$$

$$dof = 3(4 - 1 - 4) + 1 + 1 + 1 + 1 = 3(-1) + 4 = 1$$

Slider-Crank Mechanism Solution (b)

- 2 revolute joints and 1 revolute and prismatic (RP) joint

$$dof = \boldsymbol{m}(\boldsymbol{N} - \mathbf{1} - \boldsymbol{J}) + \sum_{\boldsymbol{i}=\mathbf{1}}^{\boldsymbol{J}}(\boldsymbol{f}_{\boldsymbol{i}})$$

$$J = 3;\ each\, revolute\ f_i = 1;\ RP\ f_i = 2; N = 3$$

$$dof = 3(3 - 1 - 3) + 1 + 1 + 2 = 3(-1) + 4 = 1$$

CONFIGURATION SPACE

A robot is mechanically constructed by connecting a set of bodies, called **links**, to each other using various types of **joints**.

Actuators, such as electric motors, deliver forces or torques that cause the robot's links to move.

An **end-effector**, such as a gripper or hand for grasping and manipulating objects, is attached to a specific link.

The location , in space, of a robot is derived from its **configuration**.

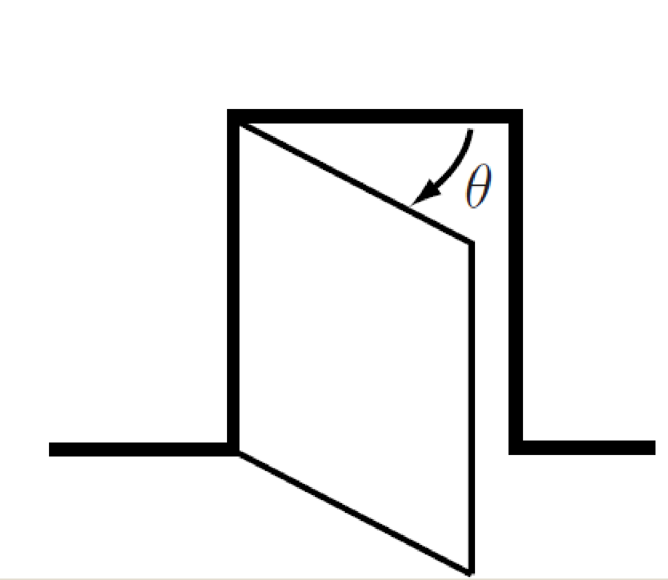
The configuration of a robot is the specification of the positions of all points of the robot.

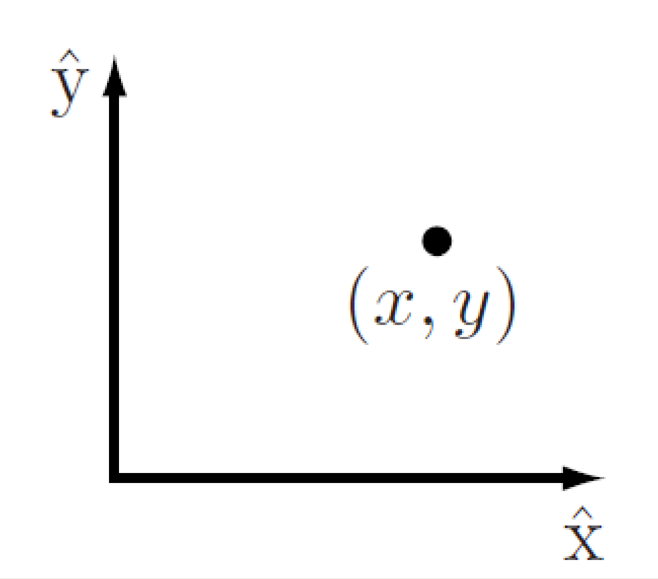
The robot's links are rigid and of a known shape,therefore only a few numbers are needed to represent its configuration.

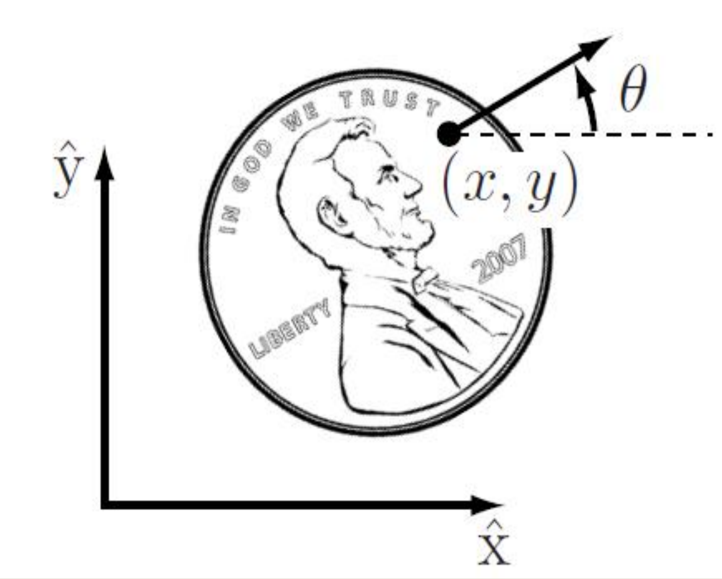
- Examples:
  - The configuration of a door can be represented by a single number, the angle θ about its hinge.
  - The configuration of a point on a plane can be described by two coordinates, (x, y).
  - The configuration of a coin lying heads up on a table can be described by three coordinates: two coordinates (x, y) that specify the location of a particular point on the coin, and one coordinate (θ) that specifies the coin's orientation.

The values of these coordinates are real, continuous numbers.

- The number of **degrees of freedom (dof)** of a robot is the smallest number of real-valued coordinates needed to represent its configuration.





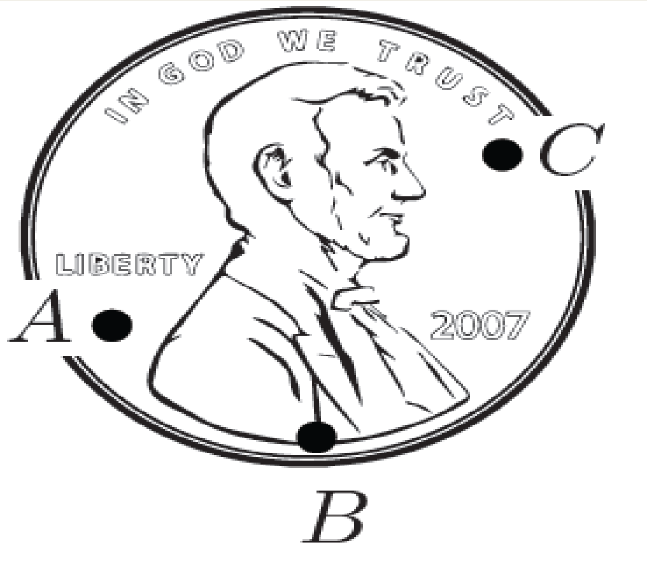


  - The door has one degree of freedom
  - The coin lying heads up, on a table has three degrees of freedom.

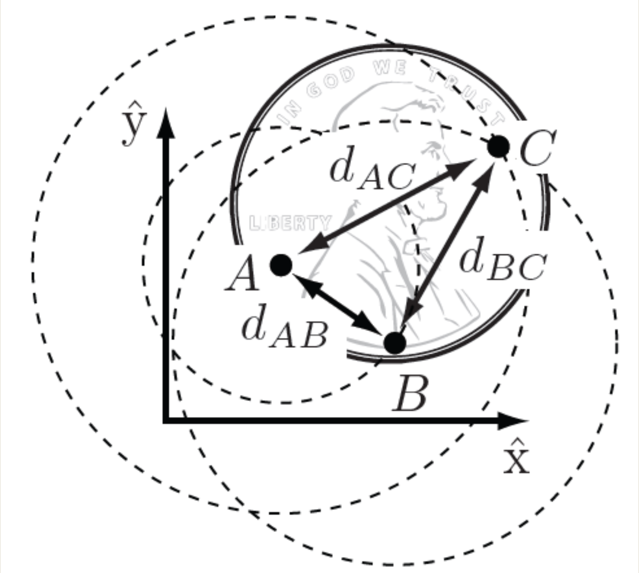
- The **configuration** of a robot is a complete specification of the position of every point of the robot.
- The minimum number, **n** of real-valued coordinates needed to represent the configuration is the number of **degrees of freedom (dof)** of the robot.
- The n-dimensional space containing all possible configurations of the robot is called the **configuration space (C-space).**
- The configuration of a robot is represented by a point in its C-space.

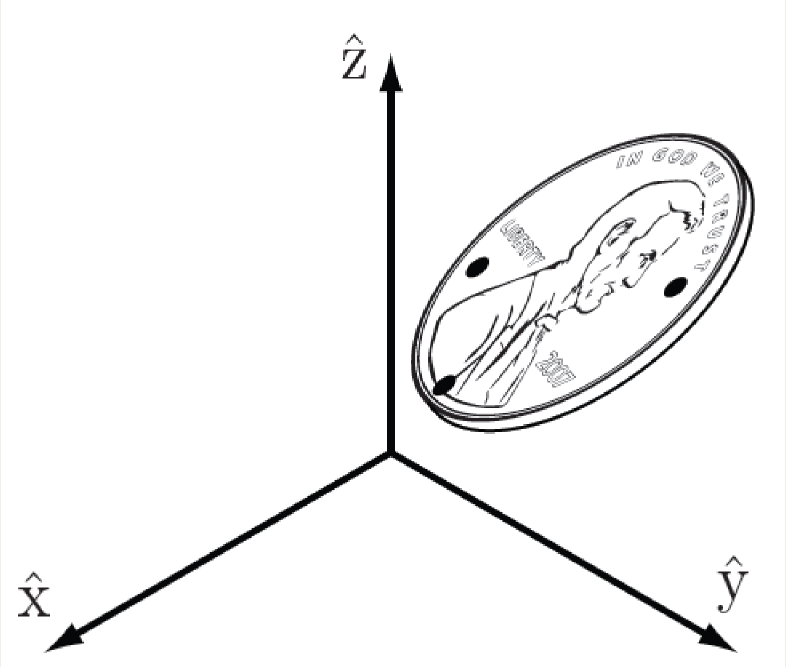
Degrees of Freedom of a Rigid Body

- Choose three fixed points A, B, and C on a coin lying on a table



- Once the location of A is chosen, B must lie on a circle of radius, $\mathbf{d_{AB}}$ centered at A. Once the location of B is chosen, C must lie at the intersection of circles centered at A and B. Only one of these two intersections corresponds to the "heads up" configuration.





- The configuration of a coin in three-dimensional space is given by the three coordinates of A, two angles to the point B on the sphere of radius, $\mathbf{d_{AB}}$ centered at A, and one angle to the point C on the circle defined by the intersection of the sphere centered at A and a sphere centered at B.

- Once a coordinate frame $\hat{x} - \hat{y}$ is attached to the plane, the positions of these points A, B, and C in the plane are written as $(x_A, y_A), (x_B, y_B), and\ (x_C, y_C)$
- If the points could be placed independently anywhere in the plane, the coin would have six degrees of freedom - two for each of the three points.
- But, according to the definition of a rigid body, the distance between point A and point B, denoted d(A,B), is always constant regardless of where the coin is.

- Similarly, the distances d(B,C) and d(A,C) must be constant.
- The following equality constraints on the coordinates $(x_A, y_A), (x_B, y_B), and\ (x_C, y_C)$ must therefore always be satisfied:

$$d(A,B) = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2} = d_{AB},$$

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$$d(A,C) = \sqrt{(x_A - x_C)^2 + (y_A - y_C)^2} = d_{AC}.$$

- To determine the number of degrees of freedom of the coin on the table, first choose the position of point A in the plane.
- We may choose it to be anything we want, so we have two degrees of freedom to specify, namely (x_A, y_A).
- Once (x_A, y_A) is specified, the constraint d(A,B) = $\mathbf{d_{AB}}$ restricts the choice of (x_B, y_B) to those points on the circle of radius $\mathbf{d_{AB}}$ centered at A.
- A point on this circle can be specified by a single parameter, e.g., the angle specifying the location of B on the circle centered at A.

- Let's call this angle $\emptyset_{AB}$ and define it to be the angle that the vector $\overrightarrow{AB}$ makes with the $\hat{x}$-axis.
- Once we have chosen the location of point B, there are only two possible locations for C: at the intersections of the circle of radius $\mathbf{d_{AC}}$ centered at A and the circle of radius $\mathbf{d_{BC}}$ centered at B.
- These two solutions correspond to heads or tails.
- This implies once we have placed A and B and chosen heads or tails, the two constraints d(A,C) = $\mathbf{d_{AC}}$ and d(B,C) = $\mathbf{d_{BC}}$ eliminate the two apparent freedoms provided by (x_C, y_C), and the location of C is fixed.
- The coin has exactly three degrees of freedom in the plane, which can be specified by $(x_A, y_A, \emptyset_{AB})$.

- If we choose to specify the position of an additional point D on the coin.
- This introduces three additional constraints: d(A,D) = $\mathbf{d_{AD}}$, d(B,D) = $\mathbf{d_{BD}}$, and d(C,D) = $\mathbf{d_{CD}}$.
- One of these constraints is redundant, i.e., it provides no new information; only two of the three constraints are independent.
- The two freedoms apparently introduced by the coordinates (x_D, y_D) are then immediately eliminated by these two independent constraints.
- The same would hold for any other newly chosen point on the coin, so that there is no need to consider additional points.

- The general rule for determining the number of degrees of freedom of a system is:

$$\boldsymbol{degrees\ of\ freedom = (sum\ of\ freedoms\ of\ the\ points) - (number\ of\ independent\ constraints)}$$

- Expressing this rule in terms of the number of variables and independent equation that describe the system:

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- A rigid body moving in three-dimensional space is called a **spatial rigid body**, and it has **six degrees of freedom**.

- A rigid body moving in a two-dimensional plane is called a **planar rigid body**, and it has **three degrees of freedom**.

- For robots with rigid bodies,

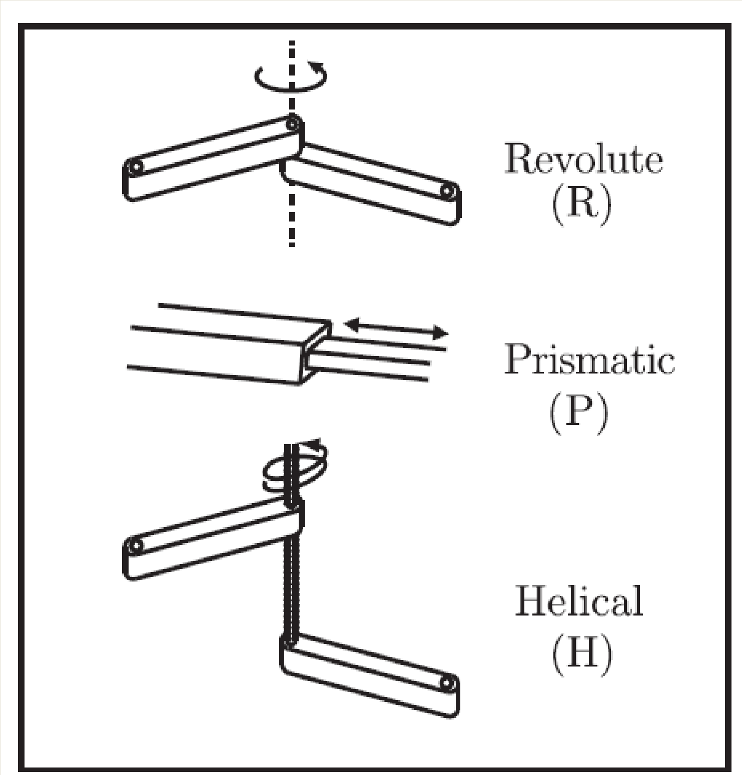
$$\boldsymbol{degrees\ of\ freedom = (sum\ of\ freedoms\ of\ the\ bodies) - (number\ of\ independent\ constraints)}$$

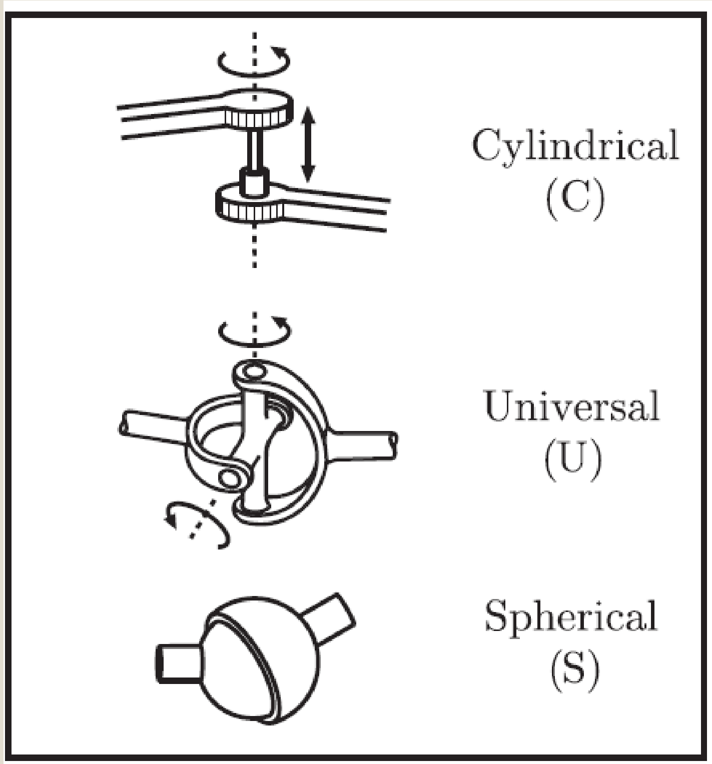
Degrees of Freedom of a Robot

- The previous example of a door, which is a single rigid body connected to a wall by a hinge joint; has only one degree of freedom, represented by the hinge joint angle, θ.
  - Without the hinge joint, the door would be free to move in three-dimensional space and would have six degrees of freedom.
  - By connecting the door to the wall via the hinge joint, five independent constraints are imposed on the motion of the door, leaving only one independent coordinate (θ).
  - Alternatively, the door can be viewed from above and regarded as a planar body, which has three degrees of freedom.
  - The hinge joint then imposes two independent constraints, again leaving only one independent coordinate (θ).

- The door's C-space is represented by some range in the interval $[0, 2\pi)$ over which θ is allowed to vary.
- In both cases the joints constrain the motion of the rigid body, thus reducing the overall degrees of freedom.
- This suggests a formula for determining the number of degrees of freedom of a robot, simply by counting the number of rigid bodies and joints.
- The Grubler's formula is used to determine the number of degrees of freedom of planar and spatial robots.

Robot Joints





Typical Robot Joints

- Every joint connects exactly two links; joints that simultaneously connect three or more links are not allowed.
- The **revolute joint (R**), also called a **hinge joint**, allows rotational motion about the joint axis.
- The **prismatic joint (P**), also called a **sliding or linear joint**, allows translational (or rectilinear) motion along the direction of the joint axis.
- The **helical joint (H**), also called a **screw joint**, allows simultaneous rotation and translation about a screw axis.
- Revolute, prismatic, and helical joints all have one degree of freedom.

- Joints can also have multiple degrees of freedom.
- The **cylindrical joint (C**) has two degrees of freedom and allows independent translations and rotations about a single fixed joint axis.
- The **universal joint (U)** is another two-degree-of-freedom joint that consists of a pair of revolute joints arranged so that their joint axes are orthogonal.
- **The spherical joint (S**), also called a **ball-and-socket joint**, has three degrees of freedom and functions much like our shoulder joint.

- A joint can be viewed as providing freedoms to allow one rigid body to move relative to another.
- It can also be viewed as providing constraints on the possible motions of the two rigid bodies it connects.
- For example, a revolute joint can be viewed as allowing one freedom of motion between two rigid bodies in space, or it can be viewed as providing five constraints on the motion of one rigid body relative to the other.
- Generalizing, the number of degrees of freedom of a rigid body (three for planar bodies and six for spatial bodies) minus the number of constraints provided by a joint must equal the number of freedoms provided by that joint.

Joint type	dof f	Constraints c between two planar rigid bodies	Constraints c between two spatial rigid bodies
Revolute (R)	1	2	5
Prismatic (P)	1	2	5
Helical (H)	1	N/A	5
Cylindrical (C)	2	N/A	4
Universal (U)	2	N/A	4
Spherical (S)	3	N/A	3

The number of degrees of freedom *f* and constraints *c* provided by common joints.

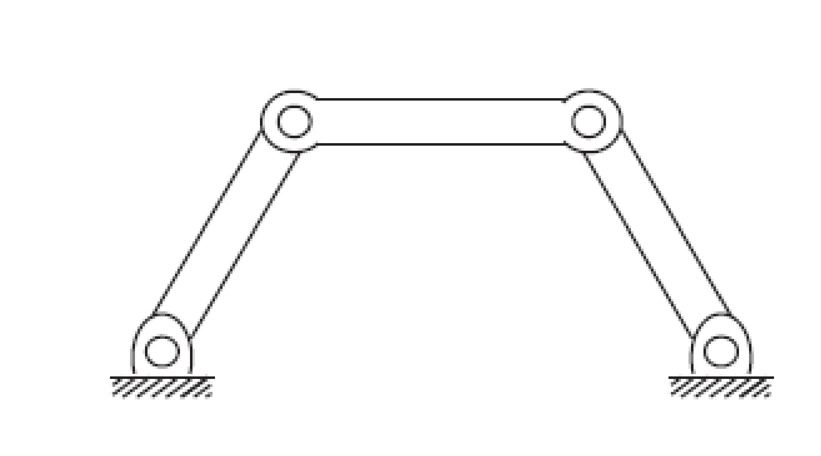
Grubler's Formula

- The number of degrees of freedom of a mechanism with links and joints can be calculated using the Grubler's formula.
- Consider a mechanism consisting of **N links**, where ground is also regarded as a link. Let **J** be the **number of joints**, **m** be the **number of degrees of freedom of a rigid body (m = 3 for planar mechanisms and m = 6 for spatial mechanisms)**, $\boldsymbol{f_i}$ be the **number of freedoms provided by joint** $\boldsymbol{i}$, and $\boldsymbol{c_i}$ be the **number of constraints provided by joint** $\boldsymbol{i}$, where $\boldsymbol{f_i} + \boldsymbol{c_i} = \boldsymbol{m}$ **for all** $\boldsymbol{i}$.
- Then Grubler's formula for the number of degrees of freedom of the robot is

$$\mathrm{dof} = \underbrace{m(N-1)}_{\text{rigid body freedoms}} - \underbrace{\sum_{i=1}^{J} c_i}_{\text{joint constraints}}$$

$$dof = m(N-1) - \sum_{i=1}^{J}(m - f_i)$$
$$= \boldsymbol{m(N-1-J)} + \sum_{\boldsymbol{i=1}}^{\boldsymbol{J}}(\boldsymbol{f_i})$$

- There are two types of mechanisms: open-chain mechanisms (also known as serial mechanisms) and closed-chain mechanisms.
- A closed-chain mechanism is any mechanism that has a closed loop.
- A person standing with both feet on the ground is an example of a closed-chain mechanism, since a closed loop can be traced from the ground, through the right leg, through the waist, through the left leg, and back to ground (recall that the ground itself is a link).
- An open-chain mechanism is any mechanism without a closed loop; an example is your arm when your hand is allowed to move freely in space.



Four-bar Linkage

- The planar four-bar linkage consists of four links (one of them ground) arranged in a single closed loop and connected by four revolute joints.
- Since all the links are confined to move in the same plane, we have **m = 3.**

◦ N = number of links = 4

◦ J = the number of joints = 4

◦ m = the number of degrees of freedom of a rigid body (m = 3 for planar mechanisms and m = 6 for spatial mechanisms) = 3

◦ fi = number of freedoms provided by joint i

◦ ci = number of constraints provided by joint i, where fi+ci=m for all i.

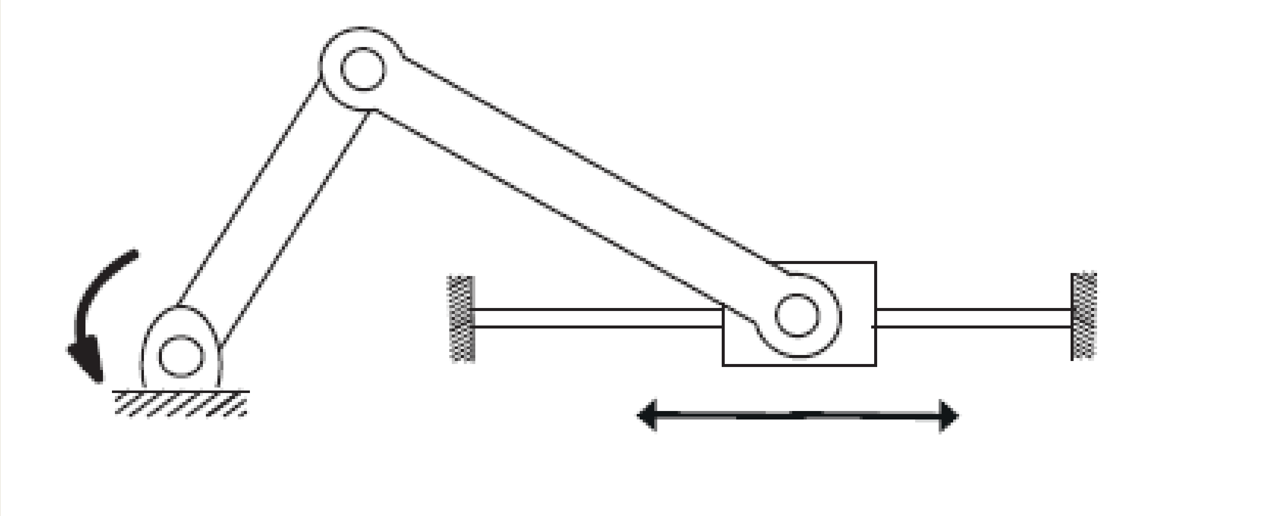
◦ i = 1, 2, 3, 4

$$dof = \boldsymbol{m}(\boldsymbol{N} - \boldsymbol{1} - \boldsymbol{J}) + \sum_{\boldsymbol{i=1}}^{\boldsymbol{J}}(\boldsymbol{f_i})$$

$$dof = 3(4-1-4) + (1+1+1+1)$$

$$= -3 + 4 = 1$$

- Using the Gruber's formula, determine the degrees of freedom of the planar slider-crank mechanism below



Slider-Crank Mechanism Solution (a)

- 3 revolute joints and 1 prismatic joint

$$dof = \boldsymbol{m}(\boldsymbol{N} - \boldsymbol{1} - \boldsymbol{J}) + \sum_{\boldsymbol{i=1}}^{\boldsymbol{J}} (\boldsymbol{f_i})$$

$$J = 4; each\ f_i = 1; N = 4, m = 3$$

$$dof = 3(4 - 1 - 4) + 1 + 1 + 1 + 1 = 3(-1) + 4 = 1$$

Slider-Crank Mechanism Solution (b)

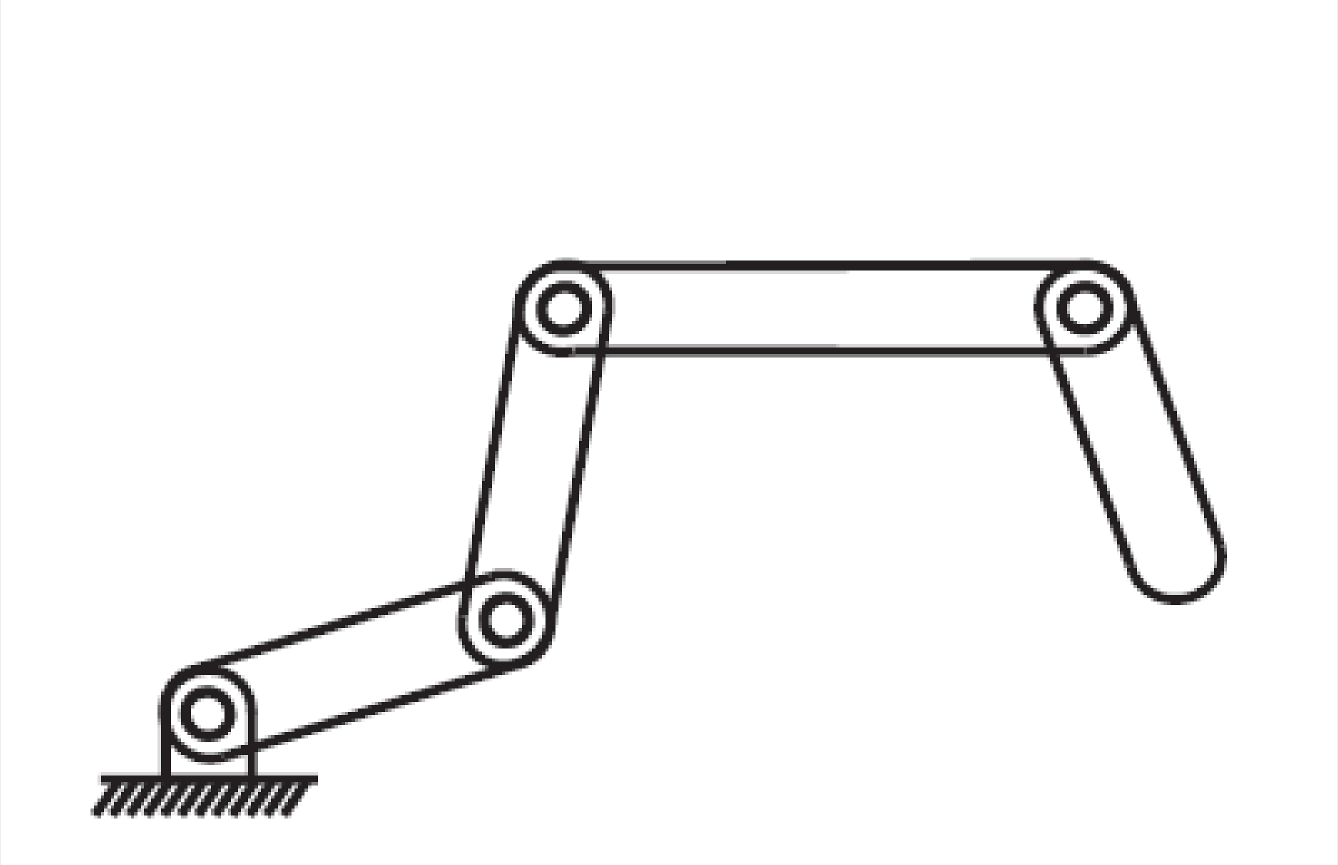
- 2 revolute joints and 1 revolute and prismatic (RP) joint

$$dof = \boldsymbol{m}(\boldsymbol{N} - \mathbf{1} - \boldsymbol{J}) + \sum_{\boldsymbol{i}=\mathbf{1}}^{\boldsymbol{J}} (\boldsymbol{f}_{\boldsymbol{i}})$$

$$J = 3;\ each\, revolute\ f_i = 1;\ RP\ f_i = 2; N = 3$$

$$dof = 3(3 - 1 - 3) + 1 + 1 + 2 = 3(-1) + 4 = 1$$

K-link Planar Serial Chain



K-link Planar Serial Chain Solution

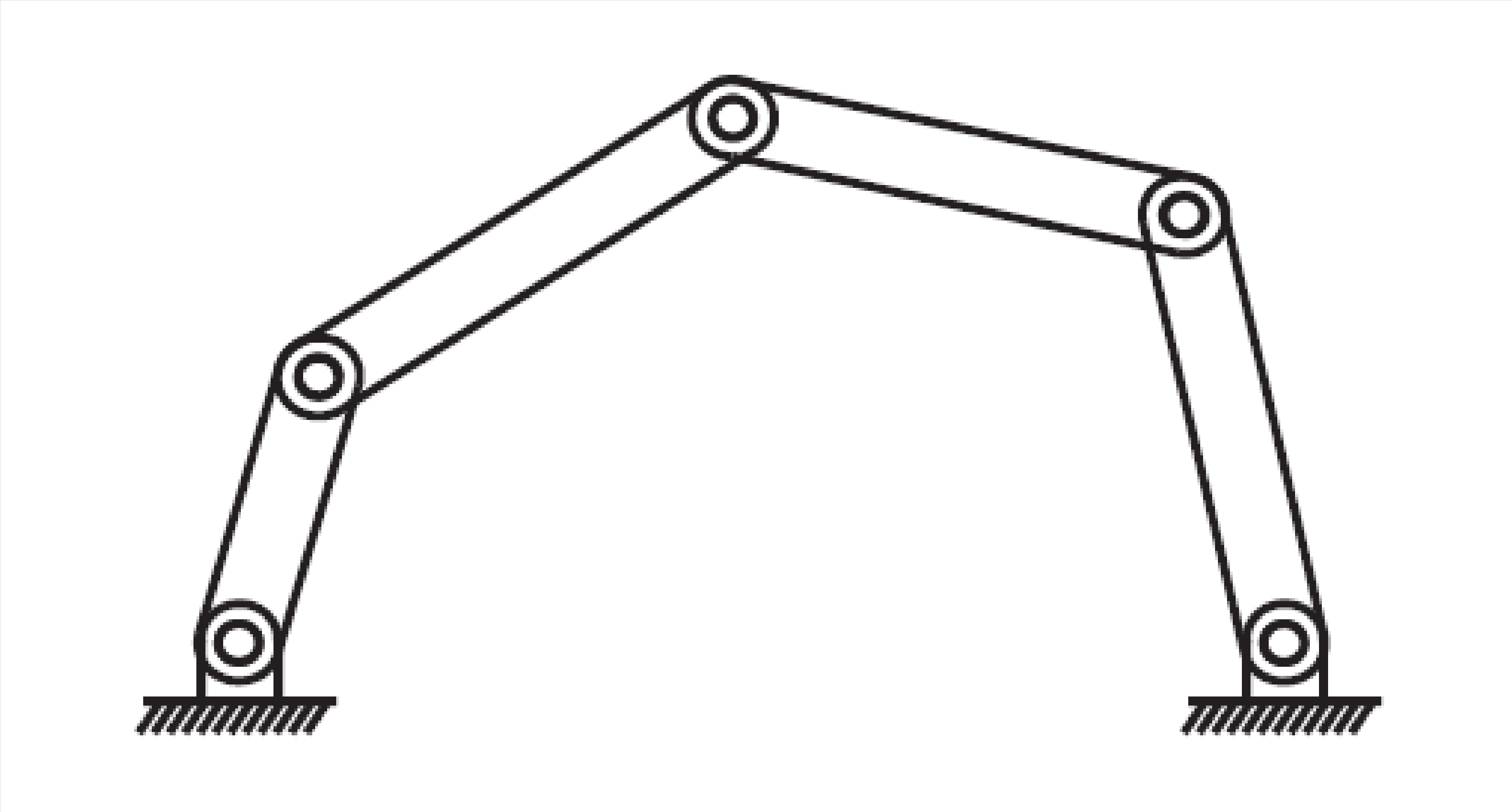
- K-link planar serial chain of revolute joints
- Called a kR robot because of its k revolute joints

$$dof = \boldsymbol{m}(\boldsymbol{N} - \boldsymbol{1} - \boldsymbol{J}) + \sum_{\boldsymbol{i=1}}^{\boldsymbol{J}}(\boldsymbol{f_i})$$

$$J = k\ joints;\ each\ f_i = 1;\ N = k + 1\ links\ (k\ links\ plus\ ground)$$

$$dof = 3[(k + 1) - 1 - k] + k = k$$

Five-bar Planar Linkage



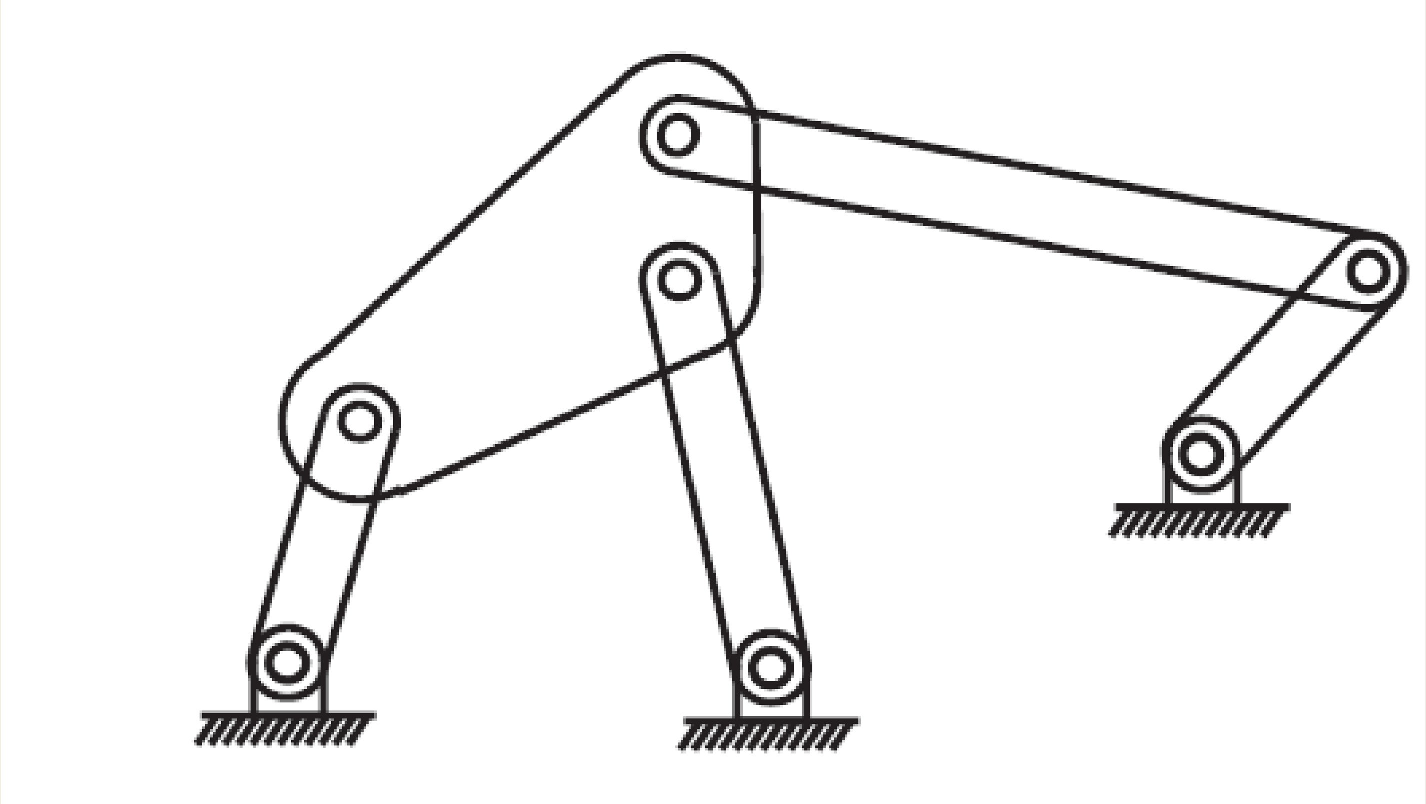
Five-bar Planar Linkage Solution

$$dof = \boldsymbol{m}(\boldsymbol{N} - \mathbf{1} - \boldsymbol{J}) + \sum_{\boldsymbol{i}=\mathbf{1}}^{\boldsymbol{J}} (\boldsymbol{f}_{\boldsymbol{i}})$$

$$J = 5;\ each\ f_i = 1;\ N = 5$$

$$dof = 3(5 - 1 - 5) + 5 = -3 + 5 = 2$$

Stephenson six-bar linkage



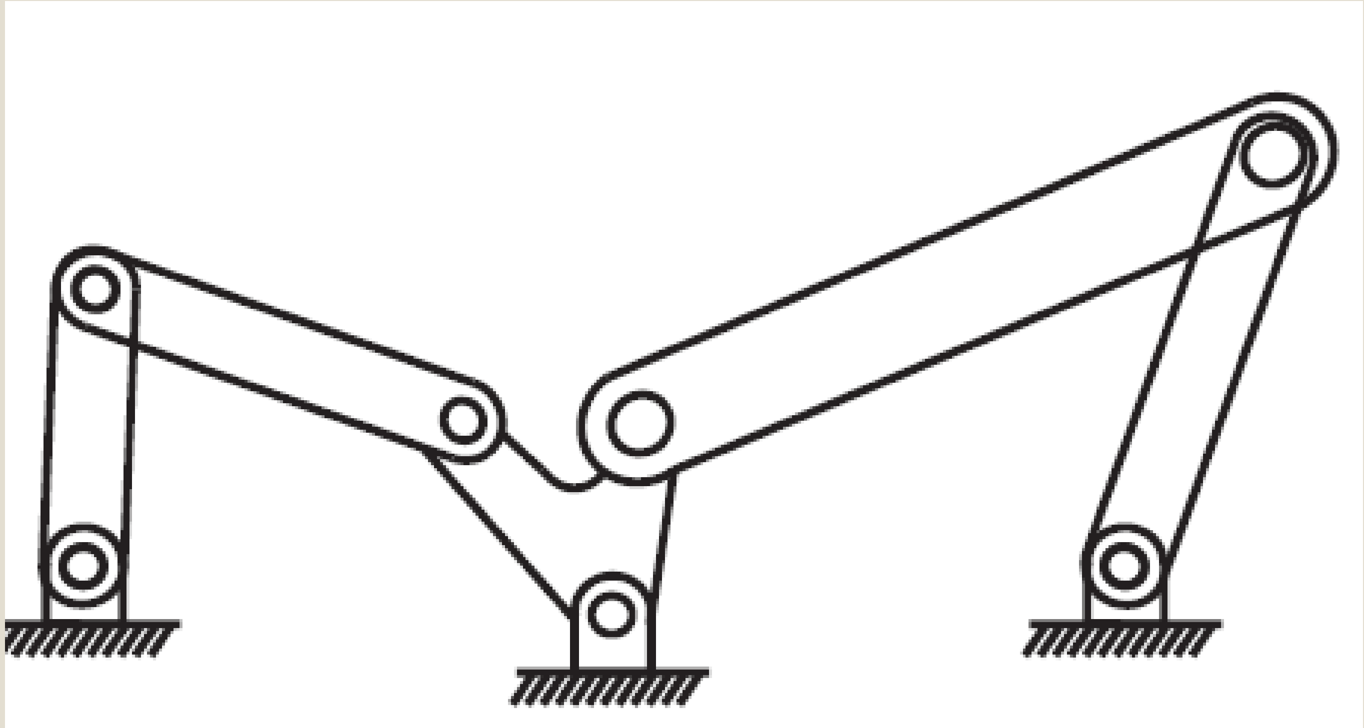
Stephenson six-bar linkage Solution

$$dof = \boldsymbol{m}(\boldsymbol{N} - \boldsymbol{1} - \boldsymbol{J}) + \sum_{\boldsymbol{i=1}}^{\boldsymbol{J}}(\boldsymbol{f_i})$$

$$J = 7;\ each\ f_i = 1;\ N = 6$$

$$dof = 3(6 - 1 - 7) + 7 = 3(-2) + 7 = 1$$

Watt six-bar linkage



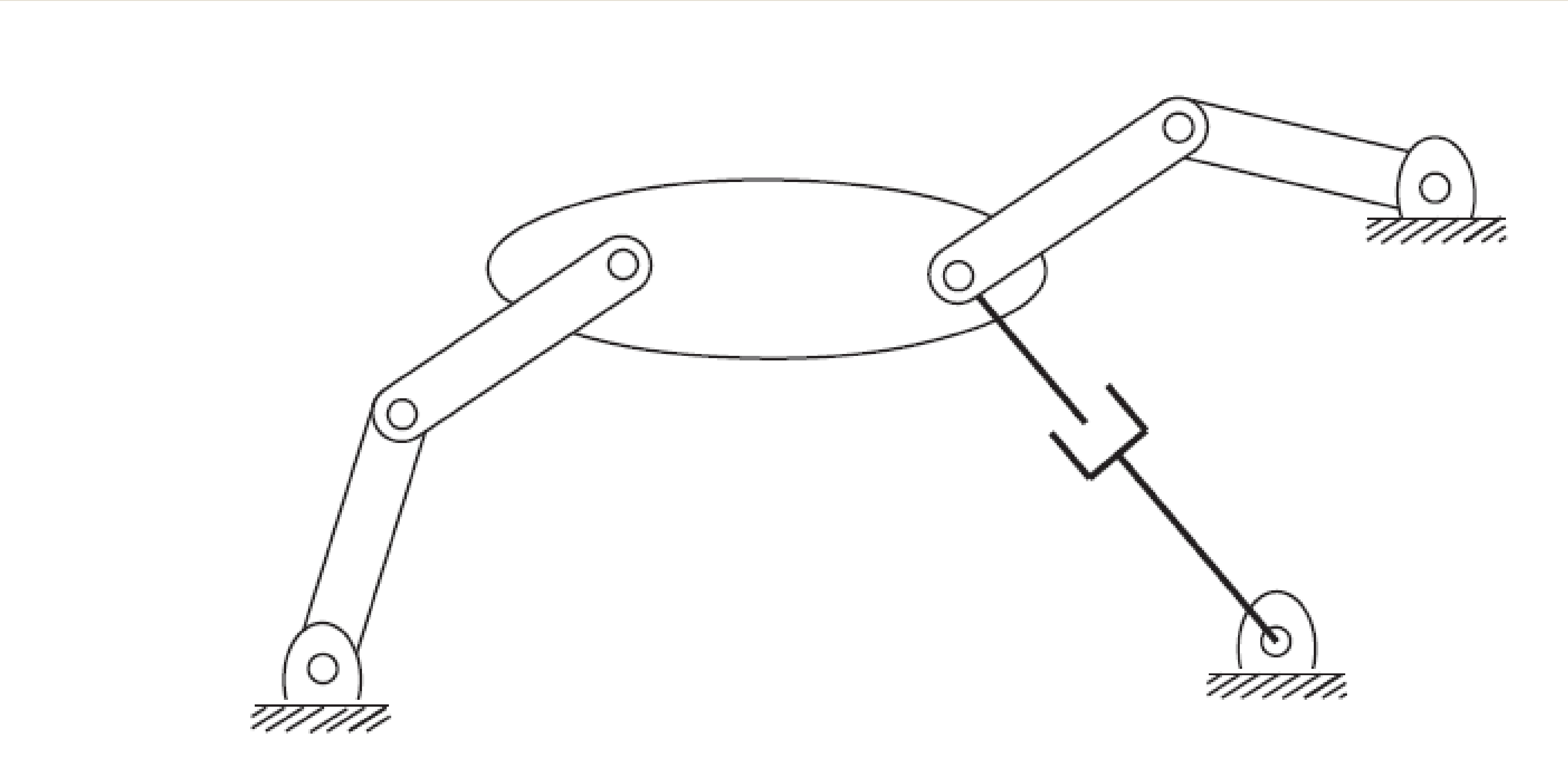
Watt six-bar linkage Solution

$$dof = \boldsymbol{m}(\boldsymbol{N} - \mathbf{1} - \boldsymbol{J}) + \sum_{\boldsymbol{i}=\mathbf{1}}^{\boldsymbol{J}}(\boldsymbol{f}_{\boldsymbol{i}})$$

$$J = 7;\ each\ f_i = 1;\ N = 6$$

$$dof = 3(6 - 1 - 7) + 7 = 3(-2) + 7 = 1$$

Planar mechanism with overlapping joints



Planar mechanism with overlapping joints Solution (a)

- 3 links meet at a single point on the right
- Recall that a joint connects two links
- Two revolute joints overlapping each other

$$dof = \boldsymbol{m}(\boldsymbol{N} - \mathbf{1} - \boldsymbol{J}) + \sum_{\boldsymbol{i}=\mathbf{1}}^{\boldsymbol{J}}(\boldsymbol{f_i})$$

$$J = 9\ (8\ revolute\ joints\ and\ 1\ Prismatic\ joint);$$

$$each\ f_i = 1;\ N = 8$$

$$dof = 3(8 - 1 - 9) + 9 = 3(-2) + 9 = 3$$

Planar mechanism with overlapping joints Solution (b)

- Alternatively, the lower-right revolute-prismatic (RP) joint pair can be regarded as a single, 2 dof joint

$$dof = \boldsymbol{m}(\boldsymbol{N} - \mathbf{1} - \boldsymbol{J}) + \sum_{\boldsymbol{i}=\mathbf{1}}^{\boldsymbol{J}}(\boldsymbol{f}_{\boldsymbol{i}})$$

$$J = 8;\ each\, revolute\ f_i = 1; RP\ f_i = 2;\ N = 7$$

$$dof\ = 3(7 - 1 - 8) + 9 = 3(-2) + 9 = 3$$